

SickKids

Functional Pediatric Neurosurgery

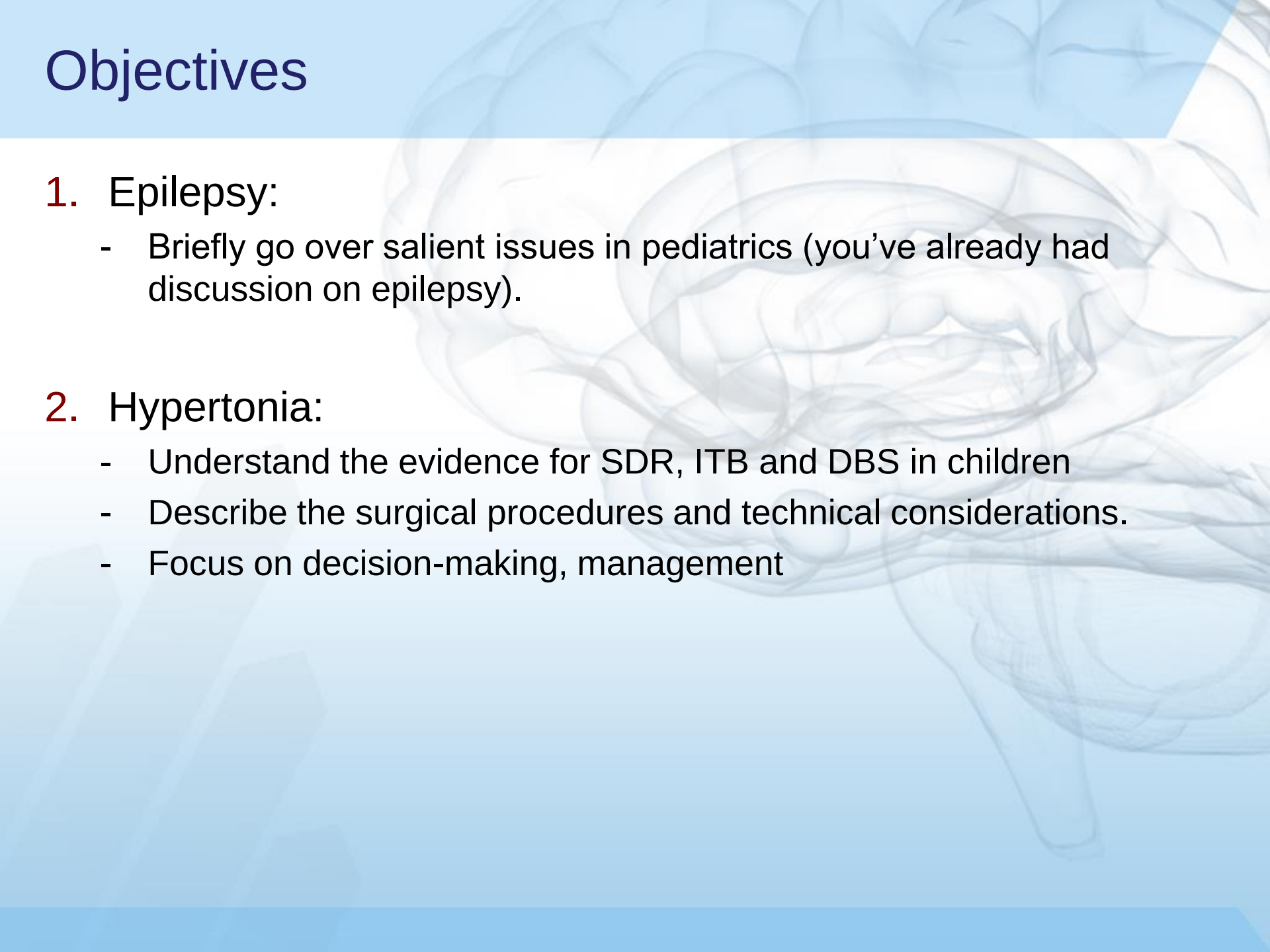


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Hospital for Sick Children
Institute of Biomaterials and Biomedical Engineering
University of Toronto

Disclosures

- None relevant to this talk
- Speaking Fees from LivaNova Inc.
- Investigator Initiated Grant support from LivaNova Inc.

Objectives



1. Epilepsy:

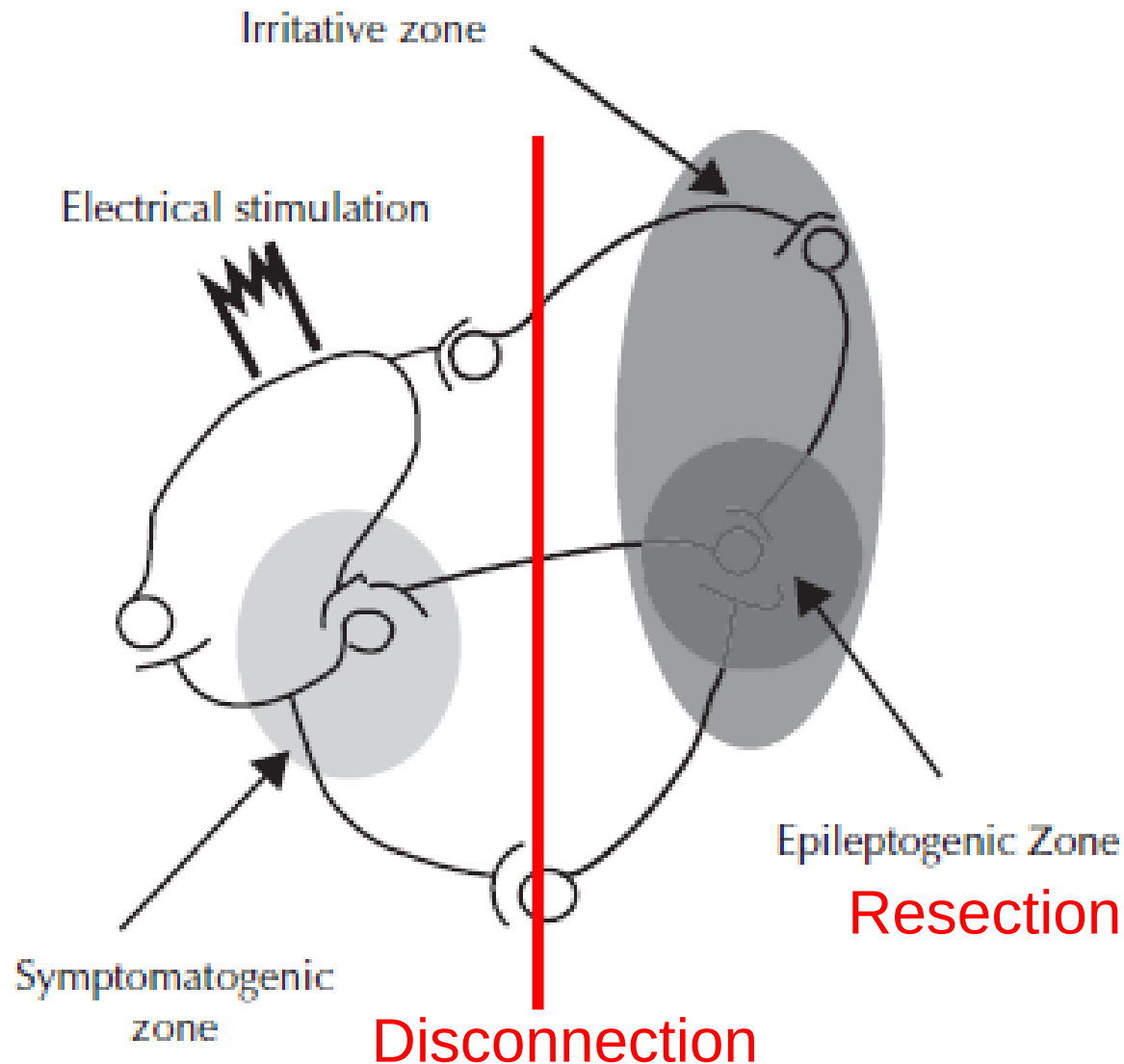
- Briefly go over salient issues in pediatrics (you've already had discussion on epilepsy).

2. Hypertonia:

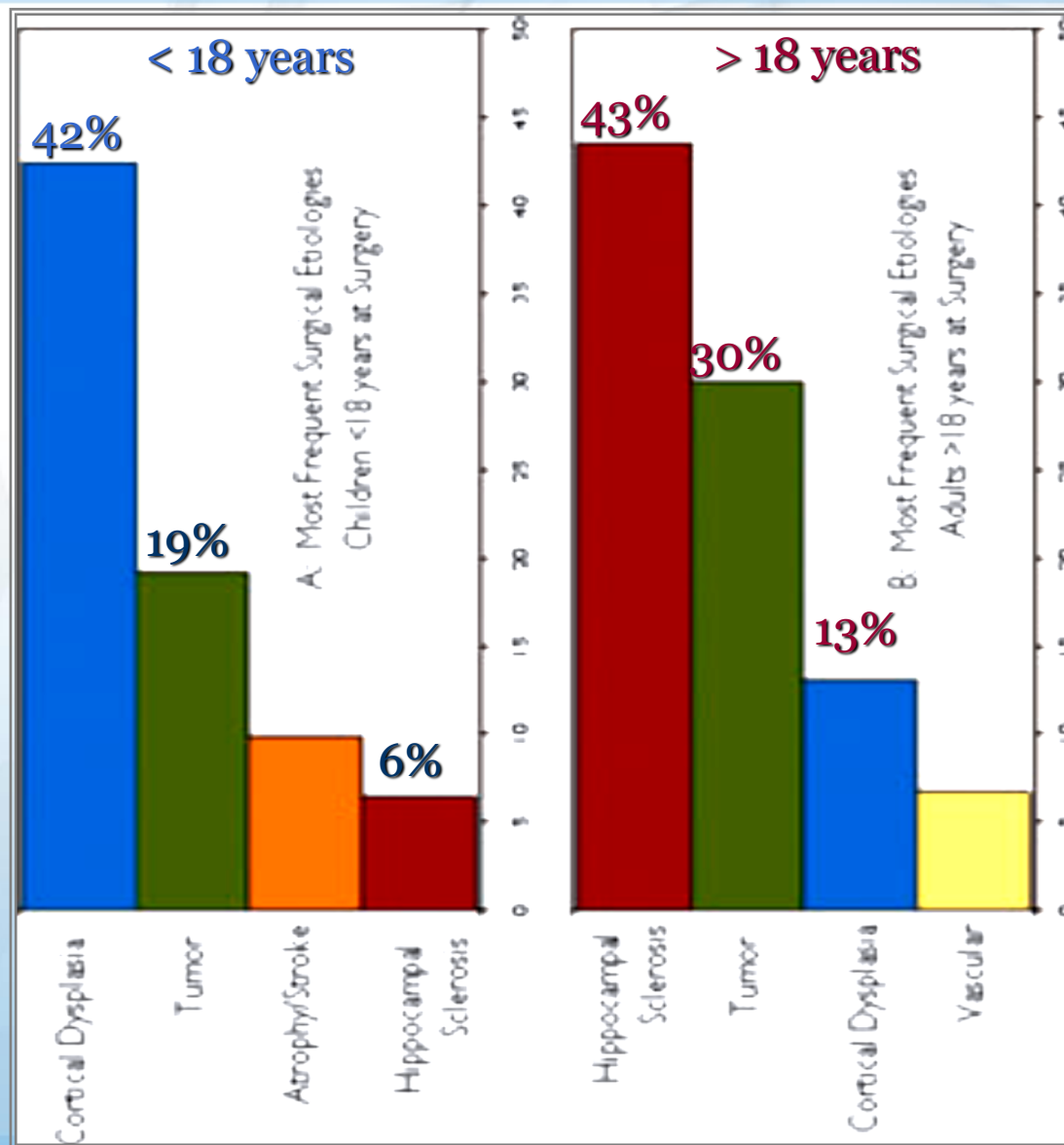
- Understand the evidence for SDR, ITB and DBS in children
- Describe the surgical procedures and technical considerations.
- Focus on decision-making, management

Childhood Epilepsy: Treatment Options

VNS
DBS
RNS

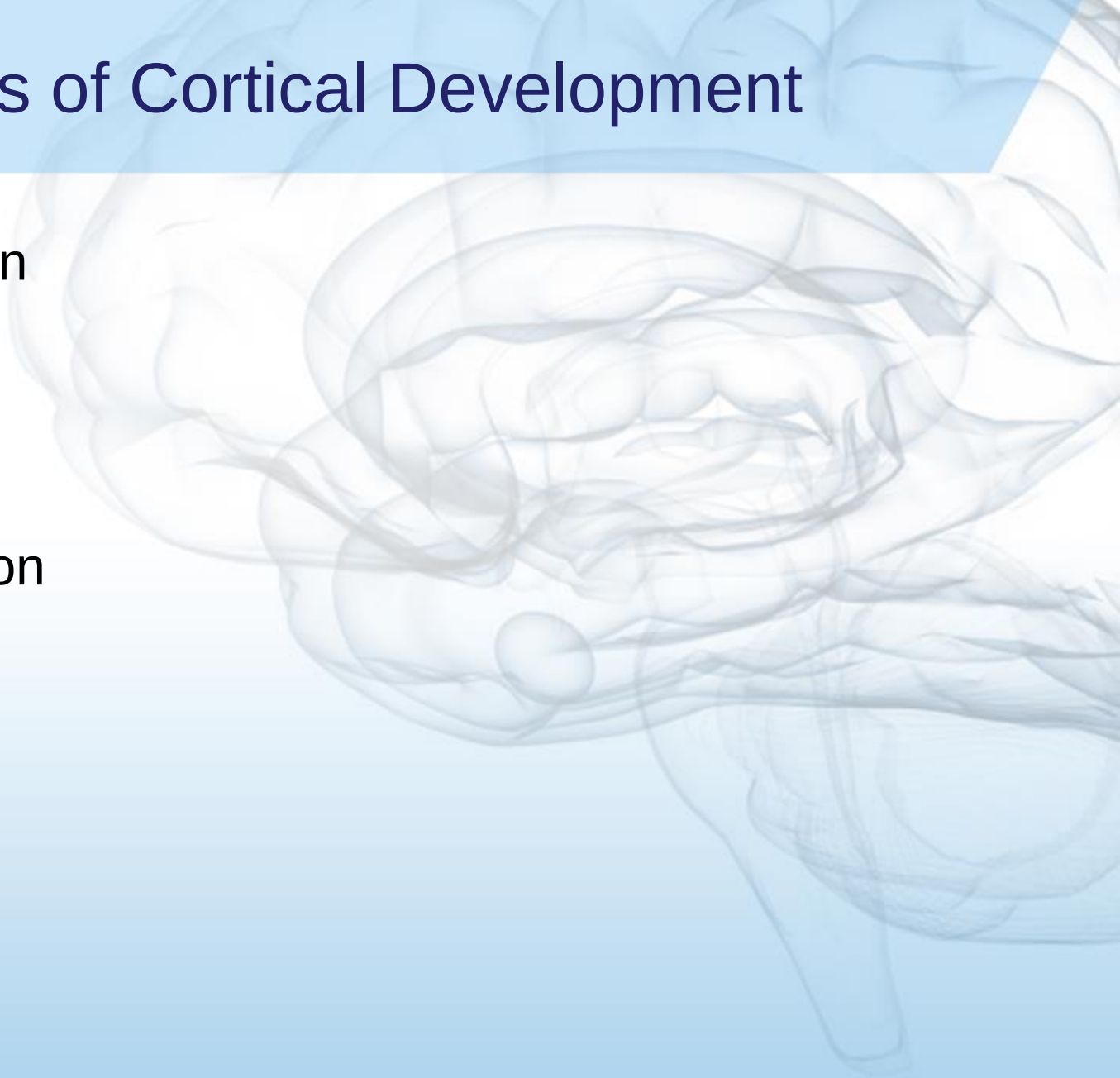


Surgically amenable syndromes in children



Malformations of Cortical Development

- Cell Proliferation
- Cell Migration
- Cell Organization



Focal Cortical Dysplasia

Rectangular Snip

Epilepsia, 52(1):158–174, 2011
doi: 10.1111/j.1528-1167.2010.02777.x

SPECIAL REPORT

The clinicopathologic spectrum of focal cortical dysplasias: A consensus classification proposed by an ad hoc Task Force of the ILAE Diagnostic Methods Commission¹

²Ingmar Blümcke, †Maria Thom, ‡Eleonora Aronica, §Dawna D. Armstrong, ¶Harry V. Vinters, #Andre Palmi, **Thomas S. Jacques, ††Giuliano Avanzini, ‡‡A. James Barkovich, §§Giorgio Battaglia, ¶¶Albert Becker, ##Carlos Cepeda, ***³Fernando Cendes, †††Nadia Colombo, ‡‡‡Peter Crino, §§§J. Helen Cross, ¶¶¶Olivier Delalande, ####François Dubeau, *John Duncan, ††††Renzo Guerrini, ‡‡‡‡Philippe Kahane, §§§§Gary Mathern, ¶¶¶¶Imad Najm, #####Çiğdem Özkara, *****Charles Raynaud, †††††Alfonso Represa, ‡‡‡‡‡Steven N. Roper, §§§§§Noriko Salamon, ¶¶¶¶¶Andreas Schulze-Bonhage, #####Laura Tassi, *****Annamaria Vezzani, and ††Roberto Spreafico**

Focal Cortical Dysplasia

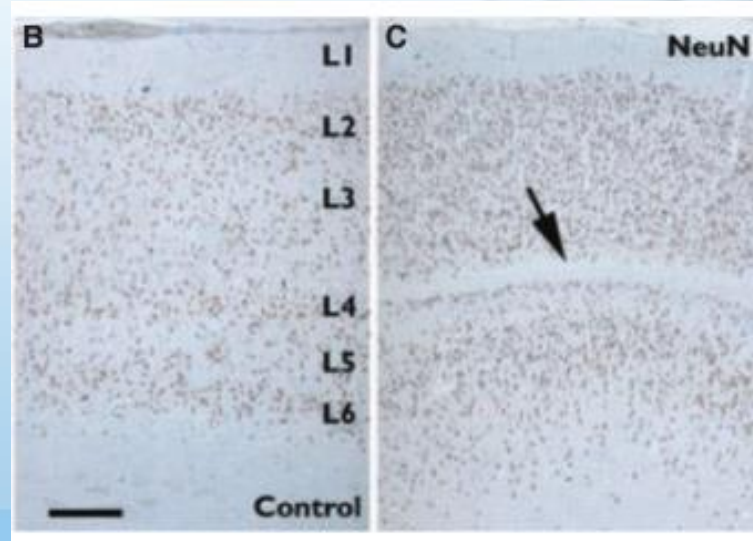
Table 1. The three-tiered ILAE classification system of focal cortical dysplasia (FCD) distinguishes isolated forms (FCD Types I and II) from those associated with another principal lesion (FCD Type III).

FCD Type I (isolated)			
FCD Type II (isolated)			
FCD Type III (associated with principal lesion)			

Focal Cortical Dysplasia

Table 1. The three-tiered ILAE classification system of focal cortical dysplasia (FCD) distinguishes isolated forms (FCD Types I and II) from those associated with another principal lesion (FCD Type III).

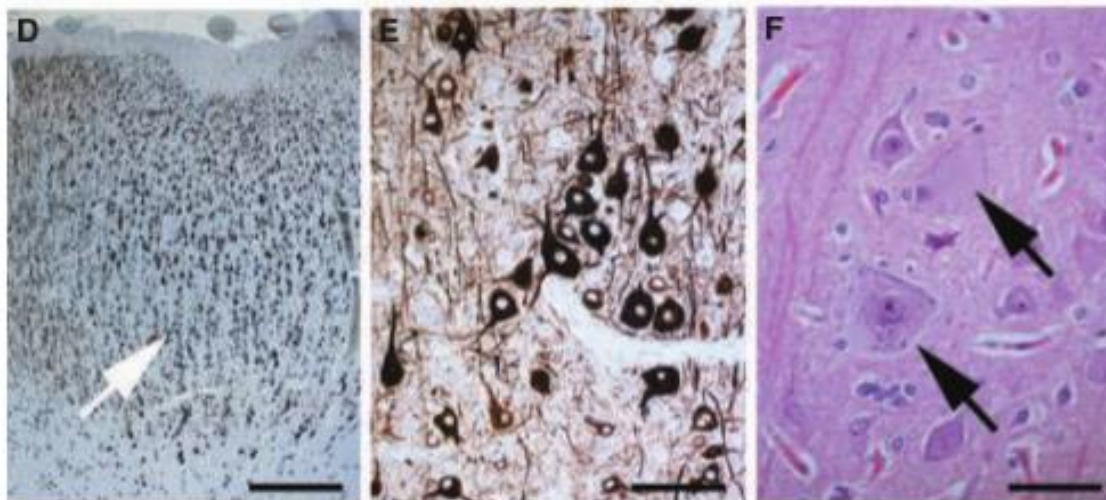
FCD Type I (isolated)	Focal cortical dysplasia with abnormal radial cortical lamination (FCD Type Ia)	Focal cortical dysplasia with abnormal tangential cortical lamination (FCD Type Ib)	Focal cortical dysplasia with abnormal radial and tangential cortical lamination (FCD Type Ic)	
FCD Type II (isolated)				
FCD Type III (associated with principal lesion)				



Focal Cortical Dysplasia

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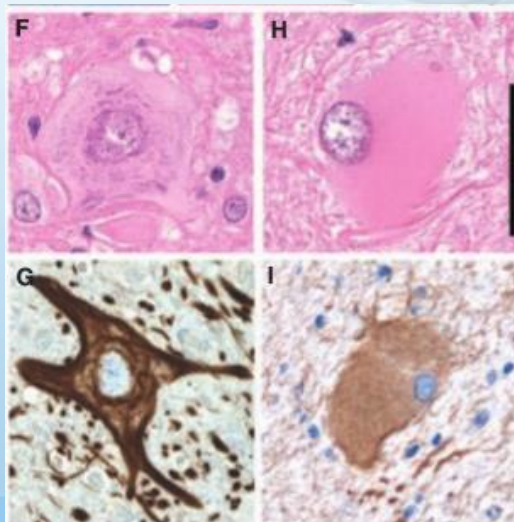
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FCD Type II (isolated)	Focal cortical dysplasia with dysmorphic neurons (FCD Type IIa)			
FCD Type III (associated with principal lesion)				



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FCD Type III (associated with principal lesion)				



Focal Cortical Dysplasia

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FCD Type III (associated with principal lesion)	Hippocampal Sclerosis	Glioneuronal tumour	Vascular malformation	Lesion acquired during early life (i.e. trauma, ischemia, encephalitis)

Phase 1
Investigations required in all patients

High-resolution MRI

Rectangular Snip

Video scalp EEG monitoring

Neuropsychological assessment

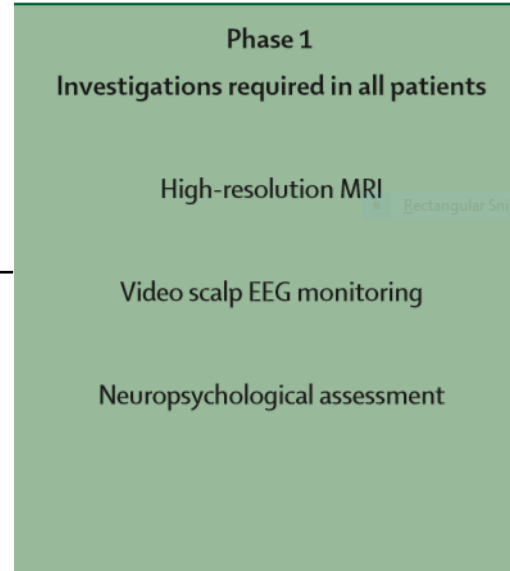
ILAE (2006)

Phase 1
Investigations required in all patients

ilar Snip

ILAE (2006)

Localize EZ



Localize
Function

**Non-systematic complementary investigations
to better localise EZ**

Improve detection of structural lesion on MRI
Morphometric analysis of T1-weighted images

Map source of interictal epileptiform discharges
MEG, EEG-fMRI, high-resolution EEG

Detect focal interictal brain dysfunction
PET imaging with FDG-PET or other radiotracers

Detect ictal focal brain hyperperfusion
Ictal and inter-ictal SPECT with SISCOM and SPM analysis

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Neuropsychological assessment

**Non-systematic complementary investigations
to assess risk of postoperative deficits**

Determine hemisphere dominant for language
fMRI, Wada test, MEG

Predict risk of postoperative memory decline
fMRI, Wada test

Reduce risk of visual field deficit
Tractography of Meyer's loop

Reduce risk of motor deficit
fMRI, tractography of pyramidal tract

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Assessment of patient's eligibility for epilepsy surgery

Epilepsy surgery

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Assessment of patient's eligibility for epilepsy surgery

Epilepsy surgery

No reasonable hypothesis about localisation of EZ
or unfavourable ratio between risk of
postoperative deficit and expected seizure outcome

Not eligible for
epilepsy surgery

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to better localise EZ**

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Morphometric analysis of T1-weighted images

Map source of interictal epileptiform discharges
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Investigations required in all patients

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Video scalp EEG monitoring

Neuropsychological assessment

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fMRI, Wada test, MEG

Predict risk of postoperative memory decline
fMRI, Wada test

Reduce risk of visual field deficit
Tractography of Meyer's loop

Reduce risk of motor deficit
fMRI, tractography of pyramidal tract

Assessment of patient's eligibility for epilepsy surgery

No formal conclusion about EZ
localisation/extension

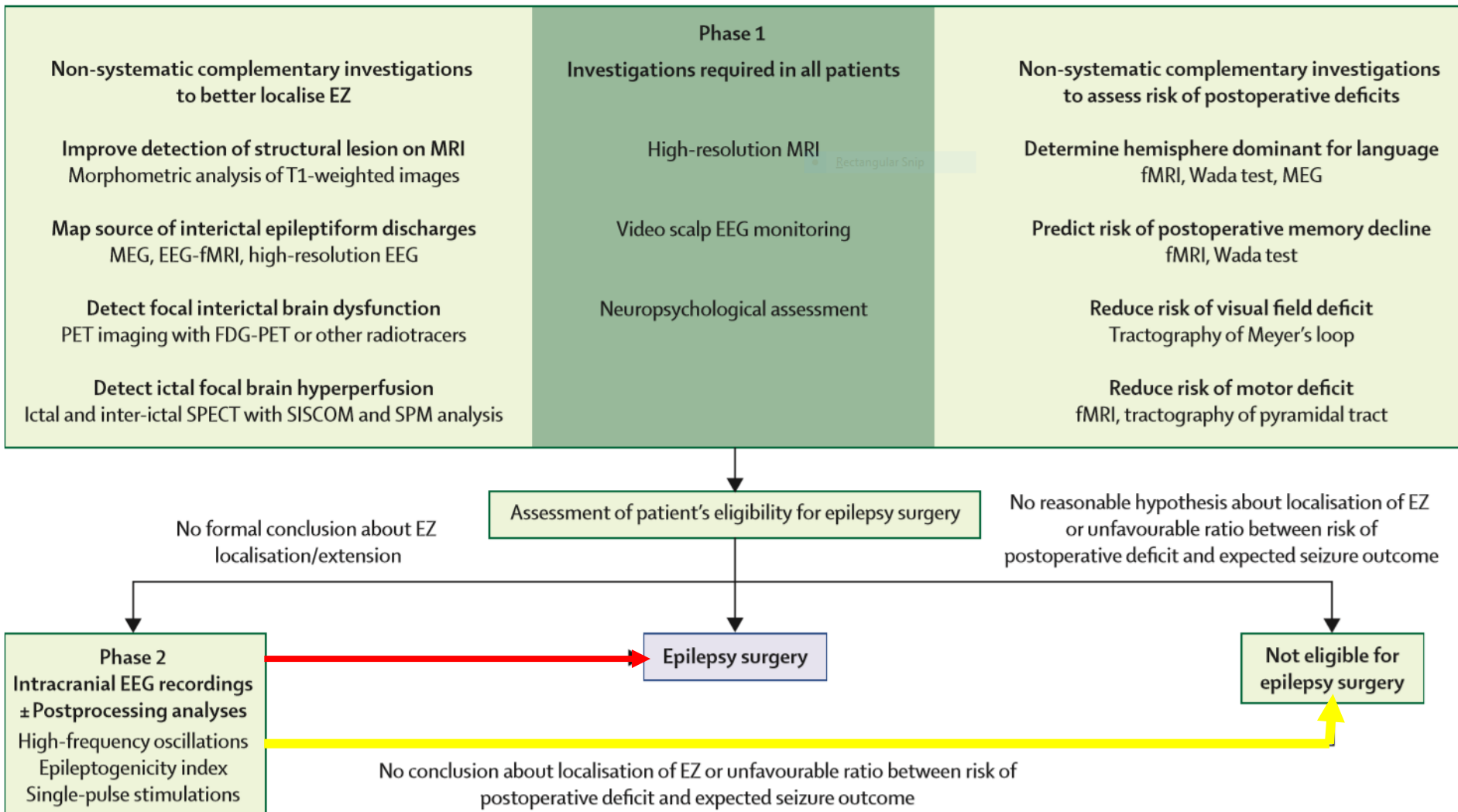
No reasonable hypothesis about localisation of EZ
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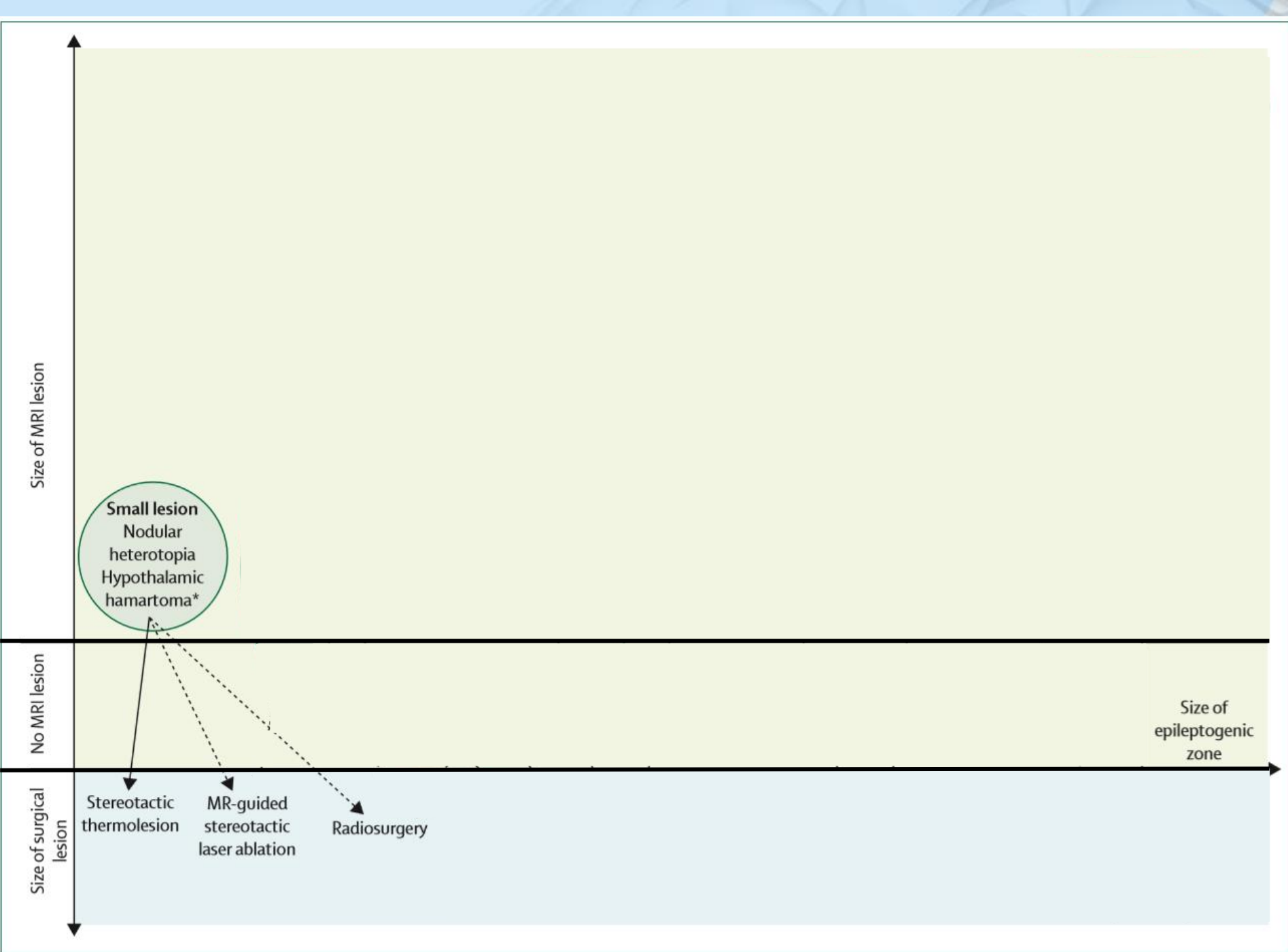
Phase 2

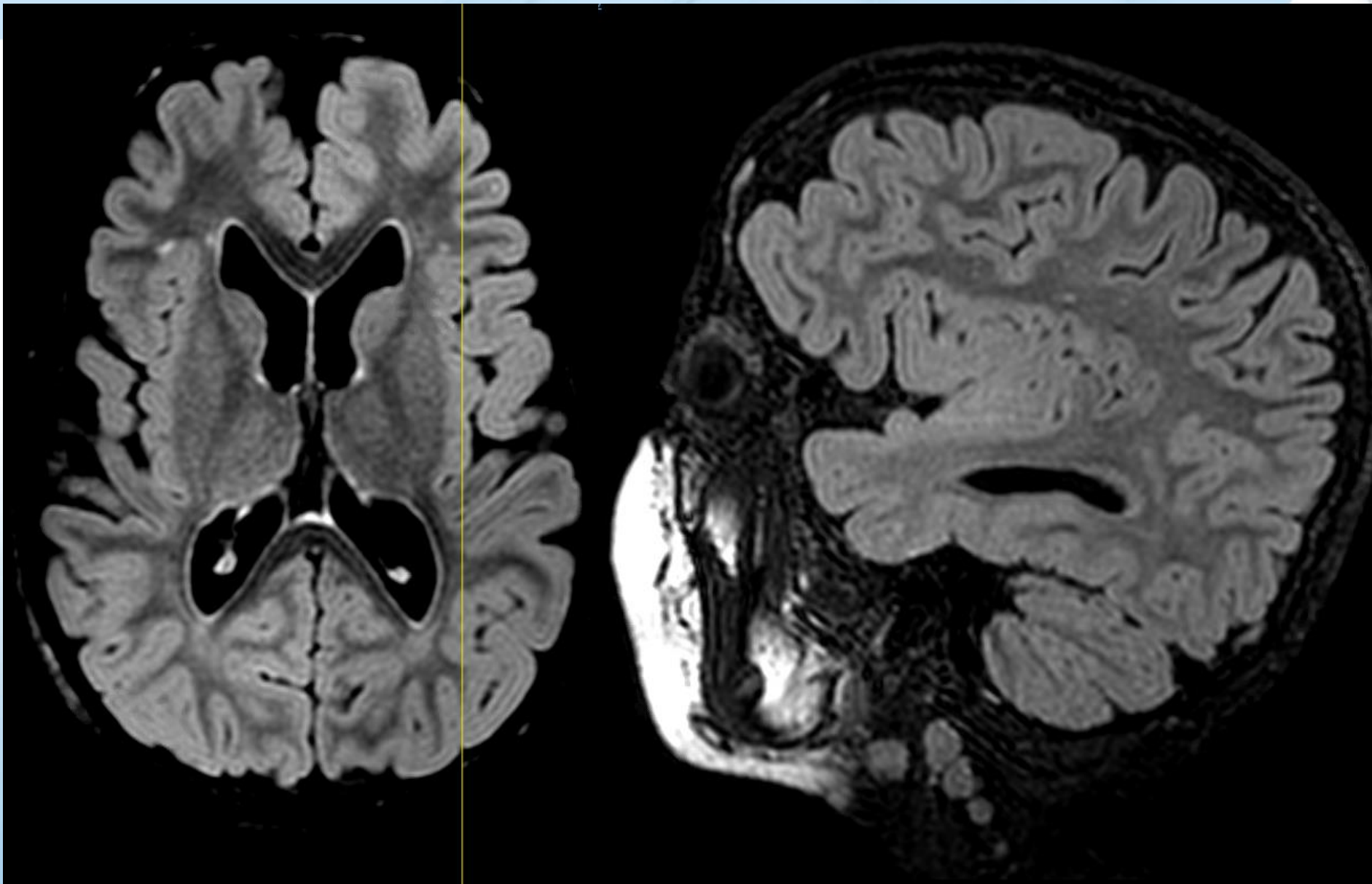
Intracranial EEG recordings
± Postprocessing analyses
High-frequency oscillations
Epileptogenicity index
Single-pulse stimulations

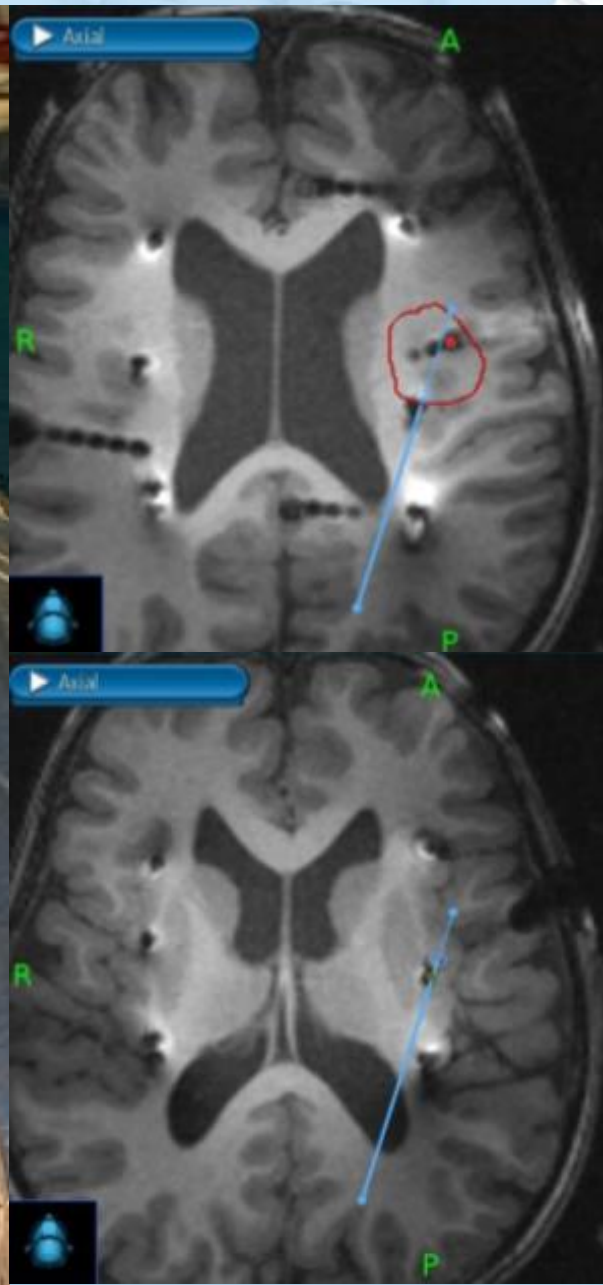
Epilepsy surgery

**Not eligible for
epilepsy surgery**

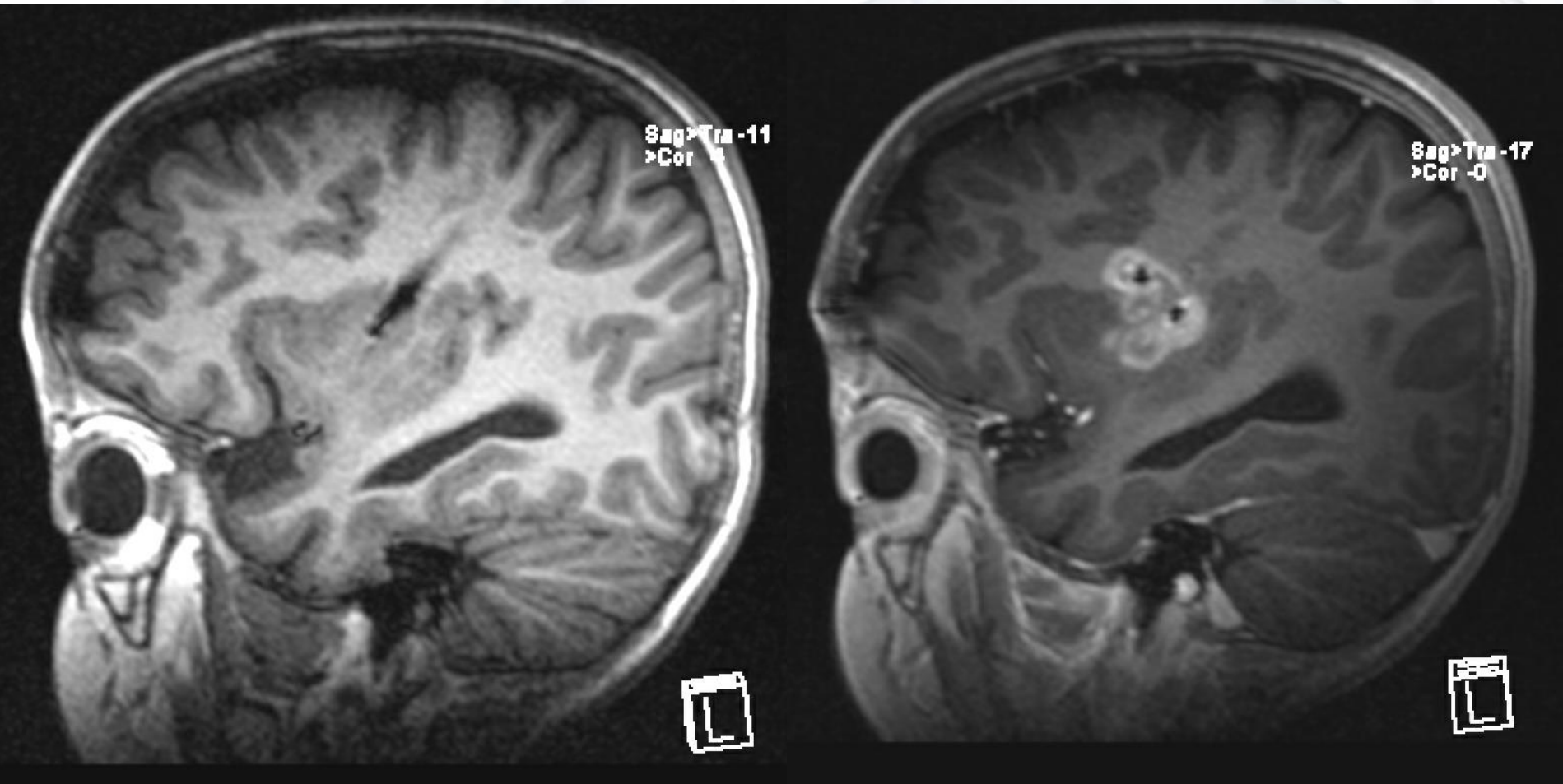




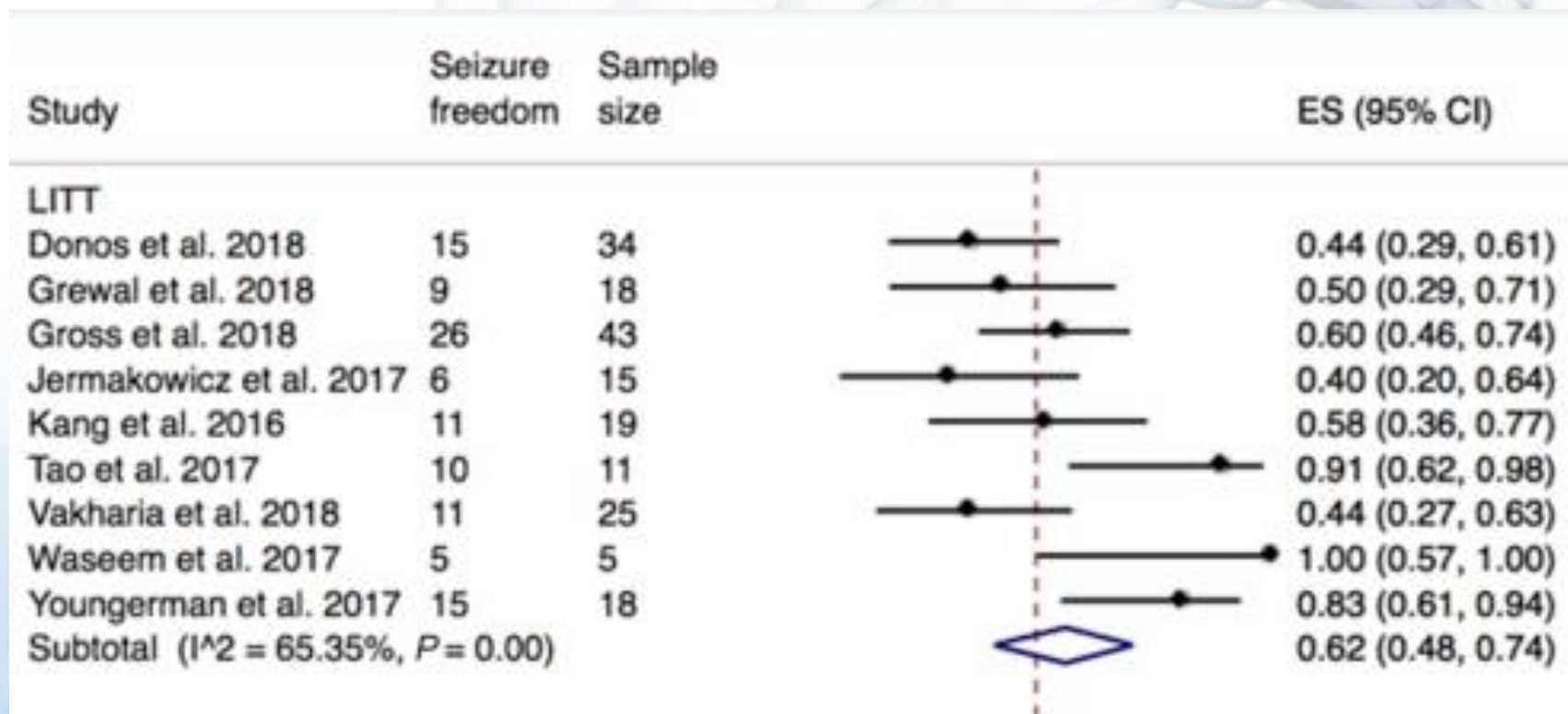


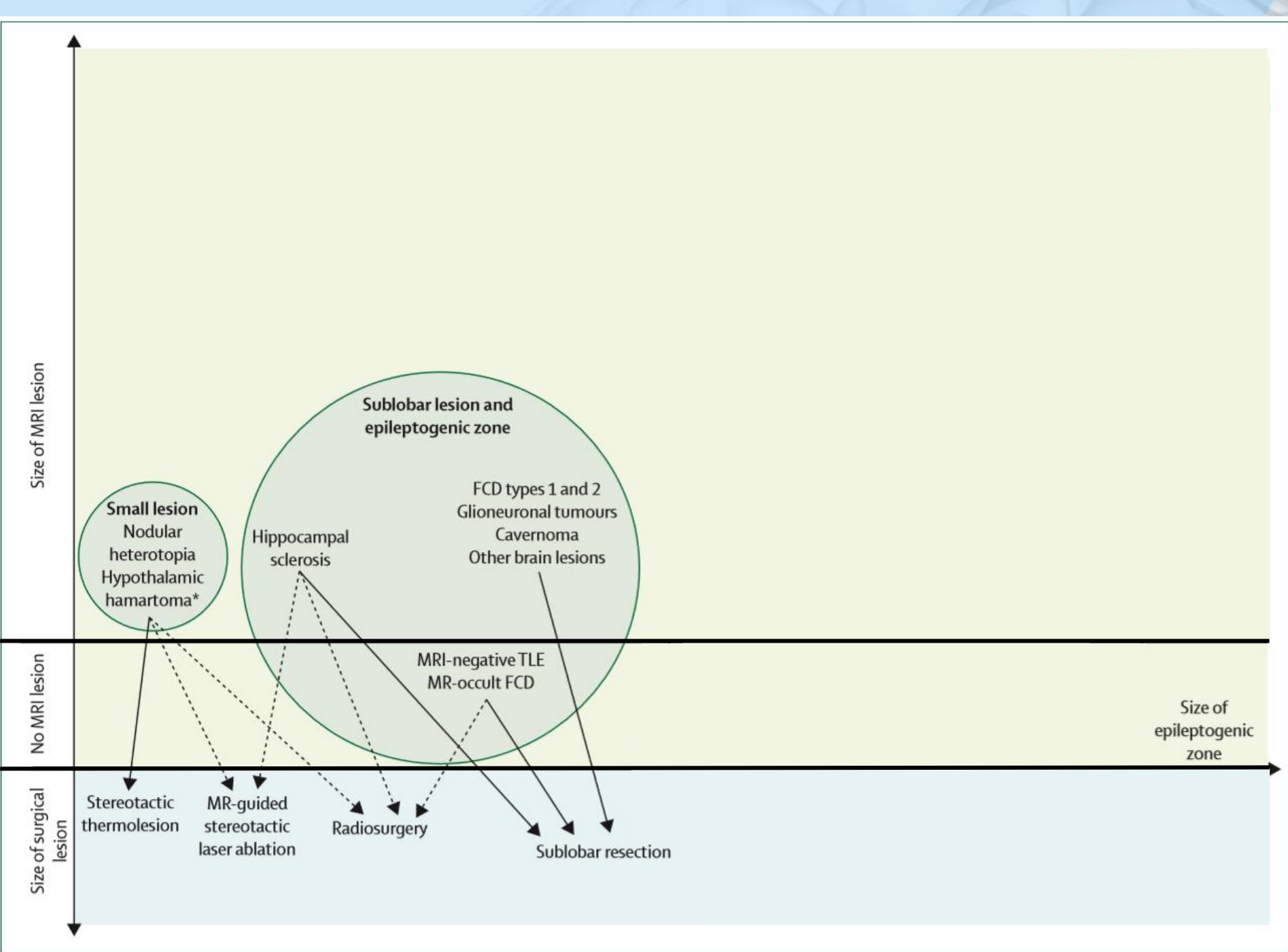


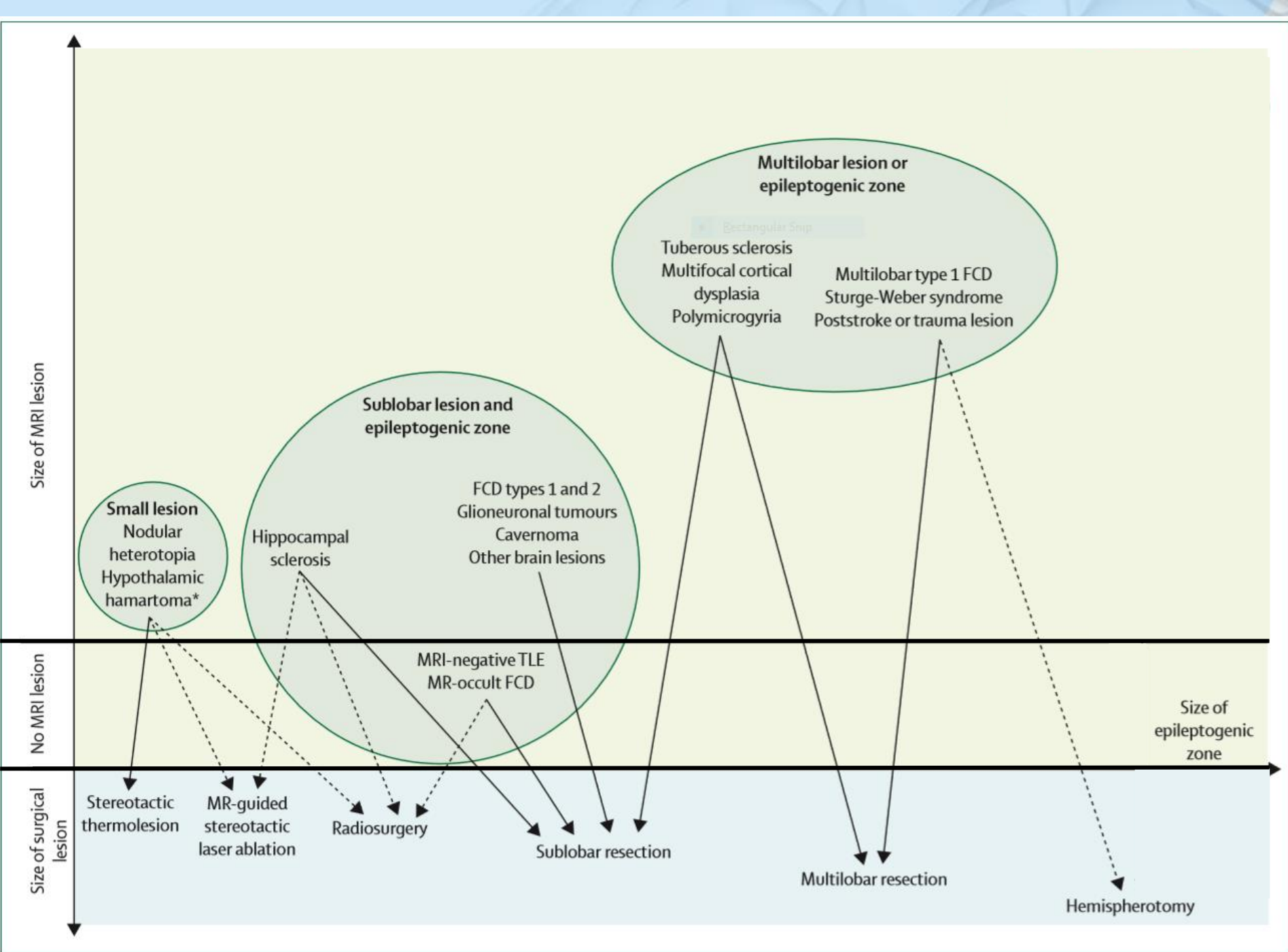
MRgLITT



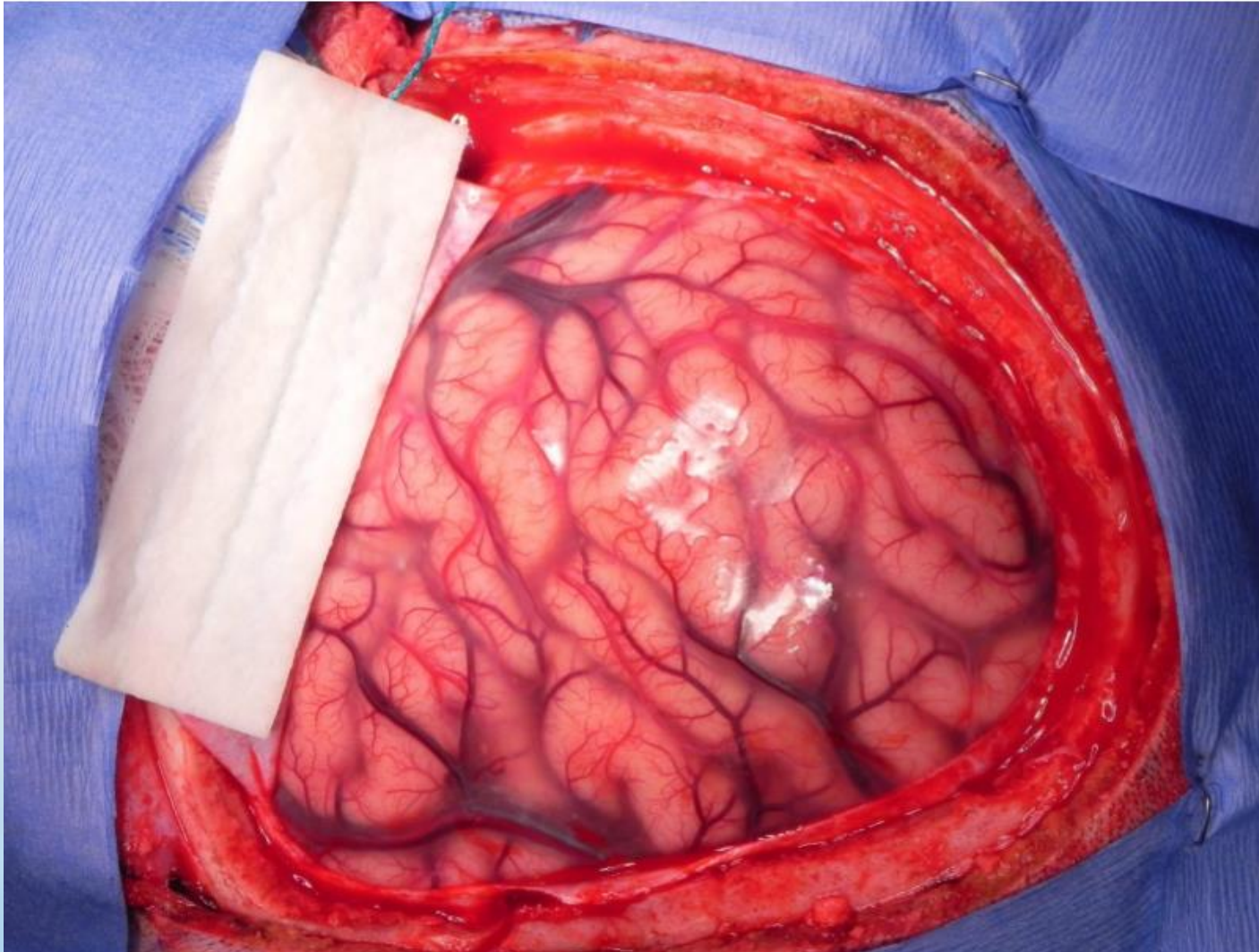
MRgLITT



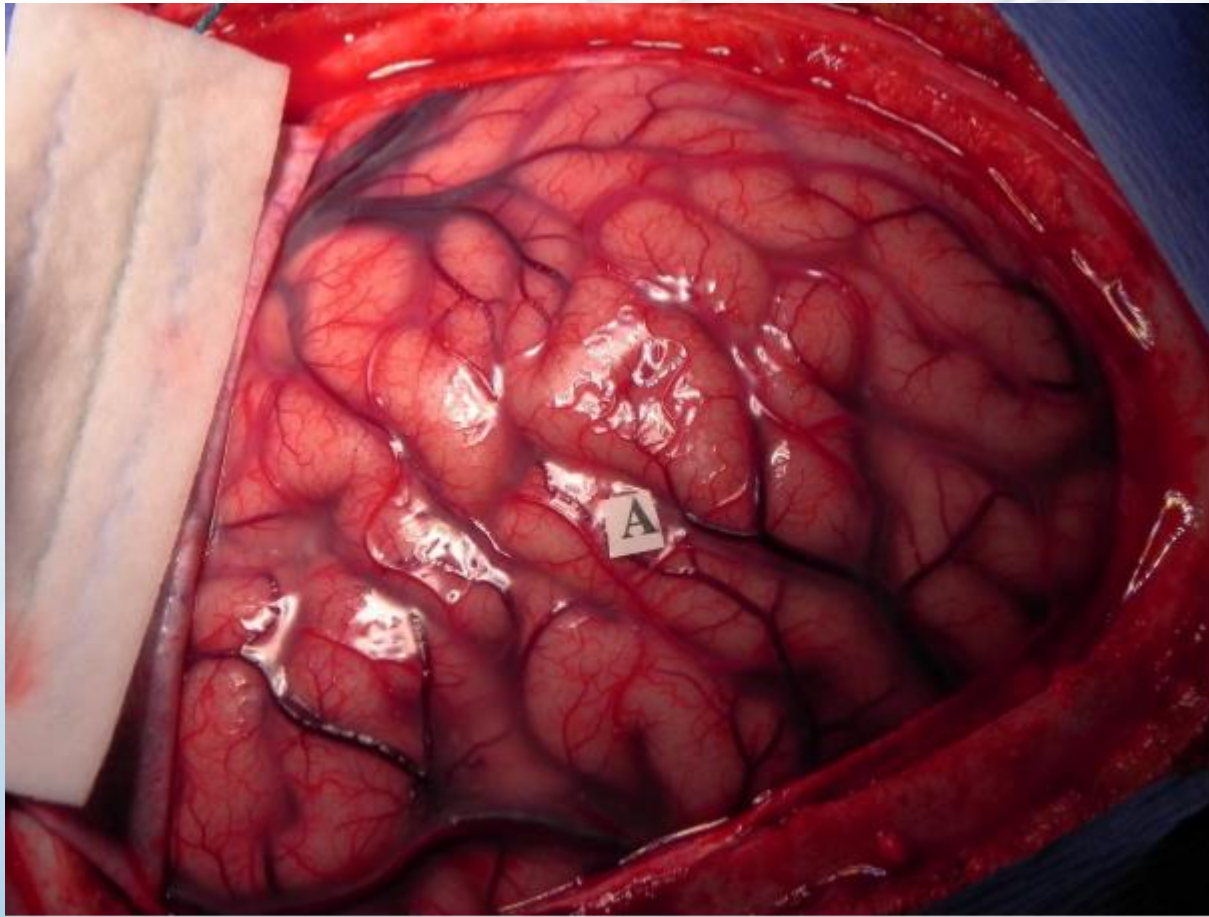




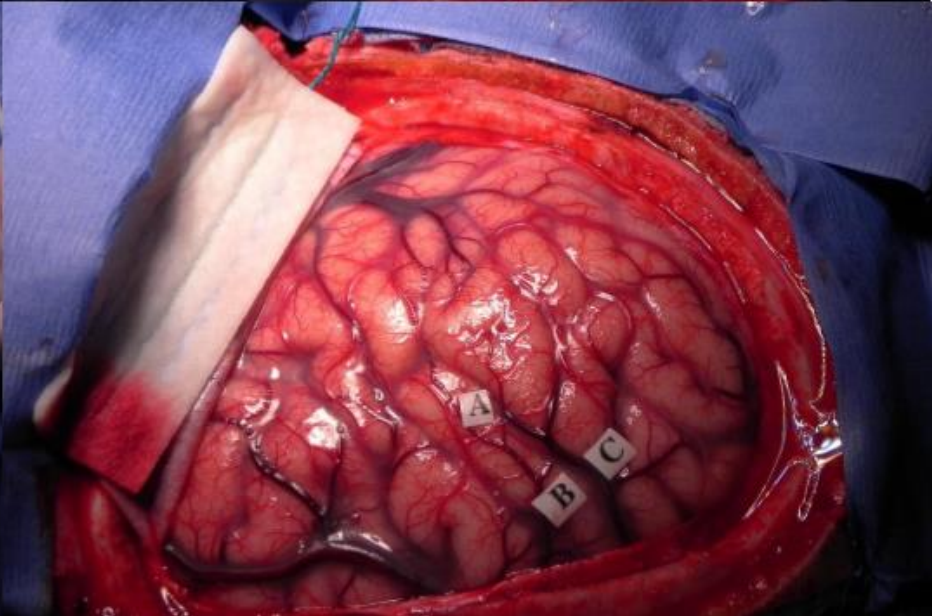
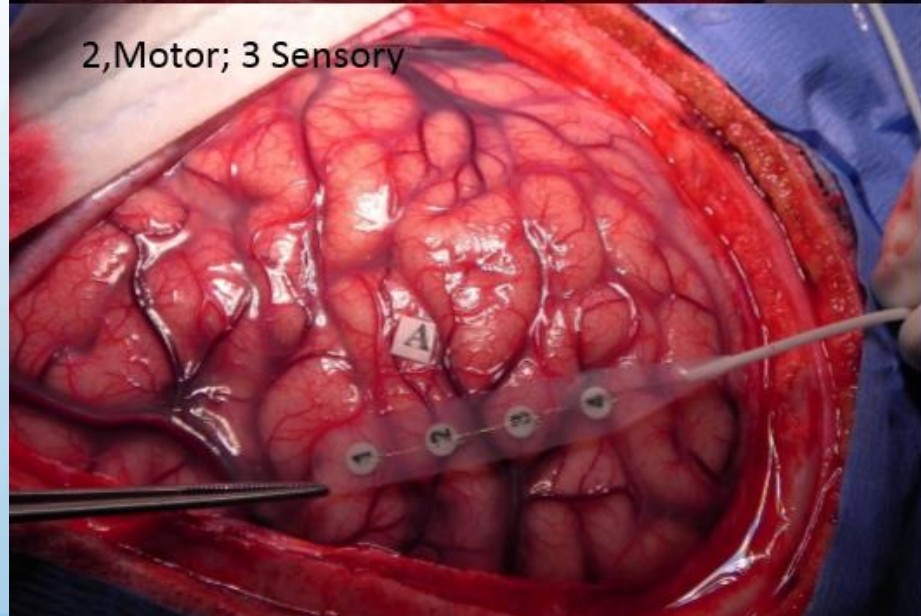
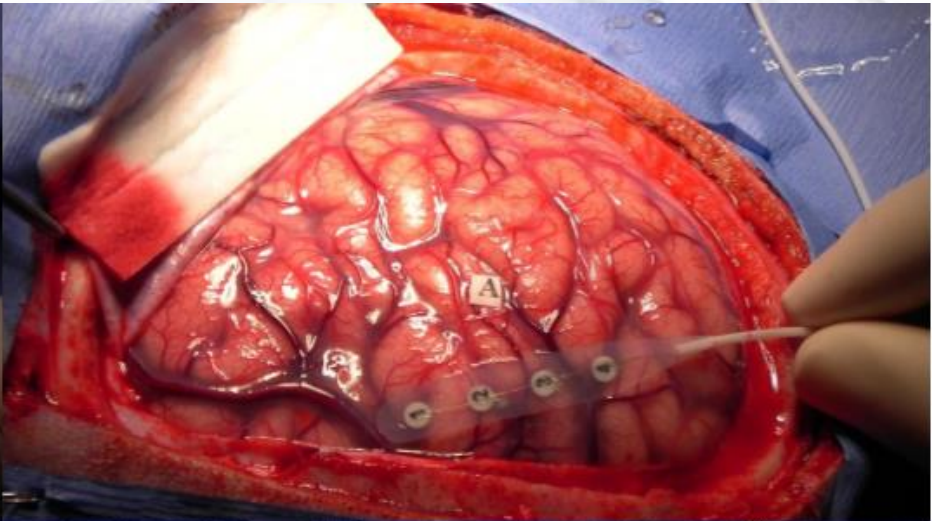
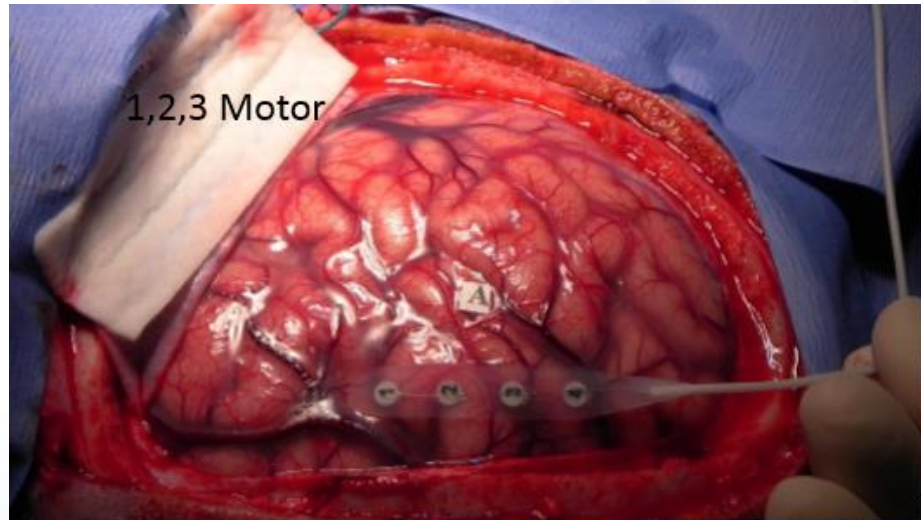
Resective surgery in eloquent cortex



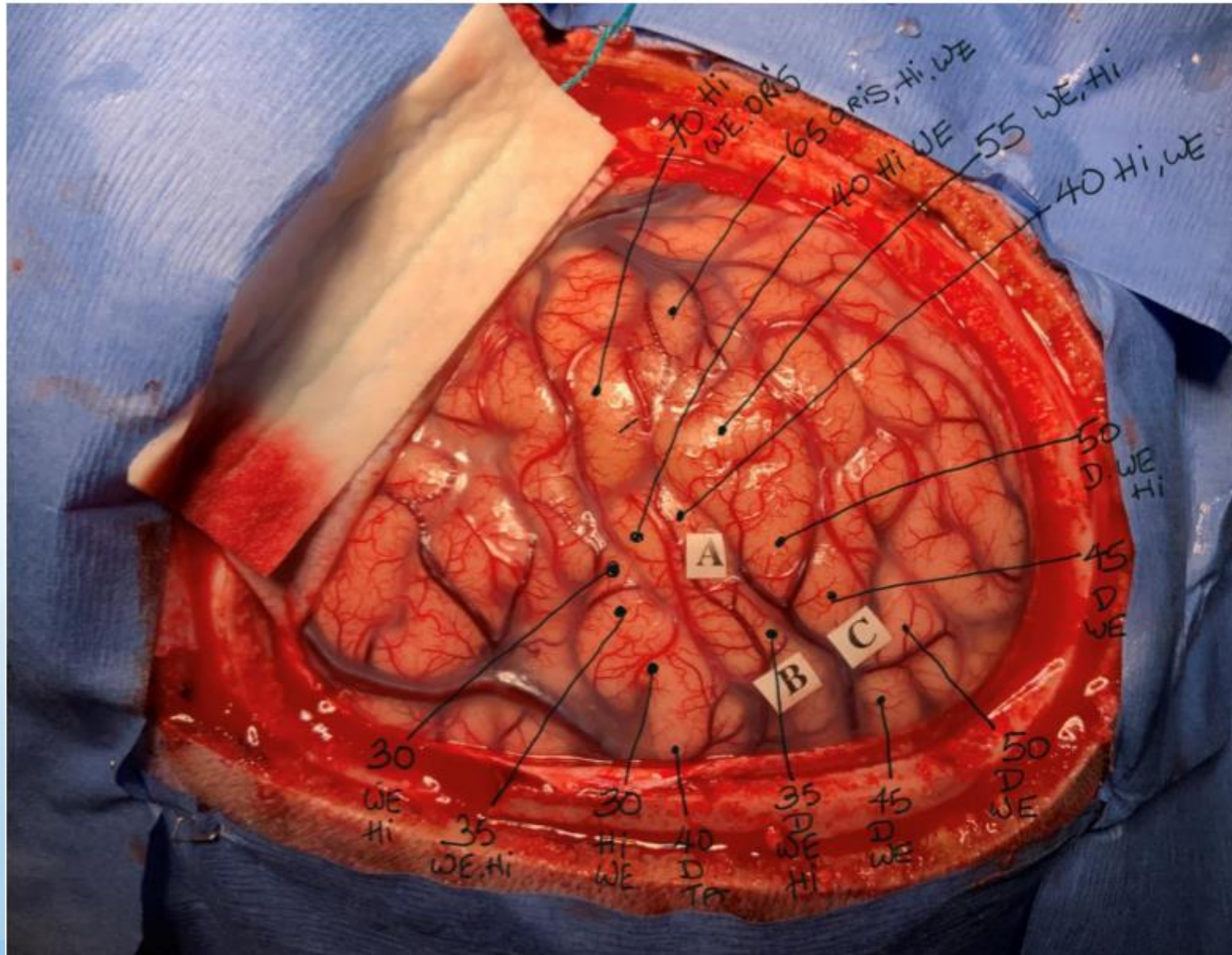
Preoperative functional localization SEF (MEG)



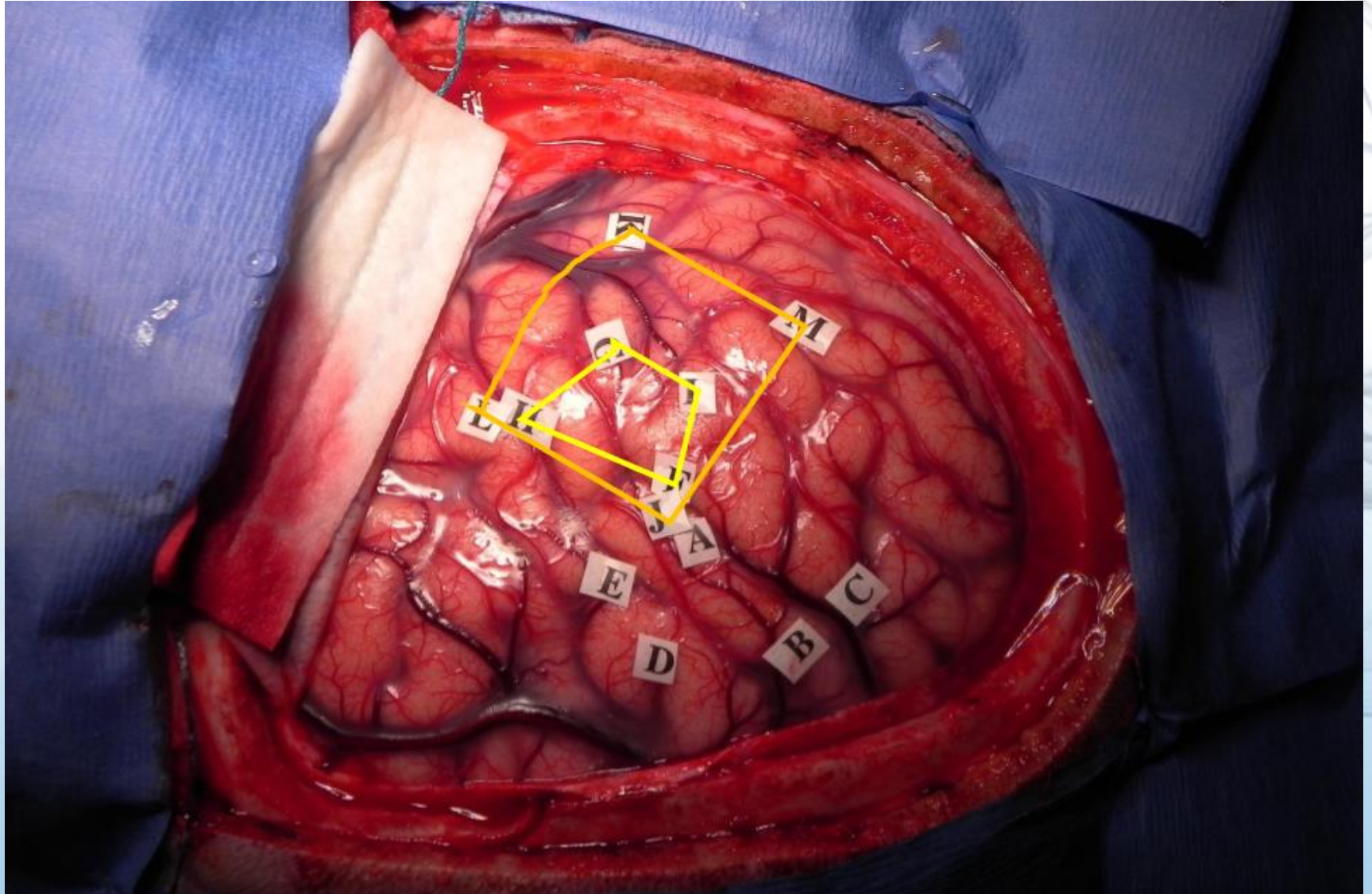
Intraoperative functional localization (phase reversal)



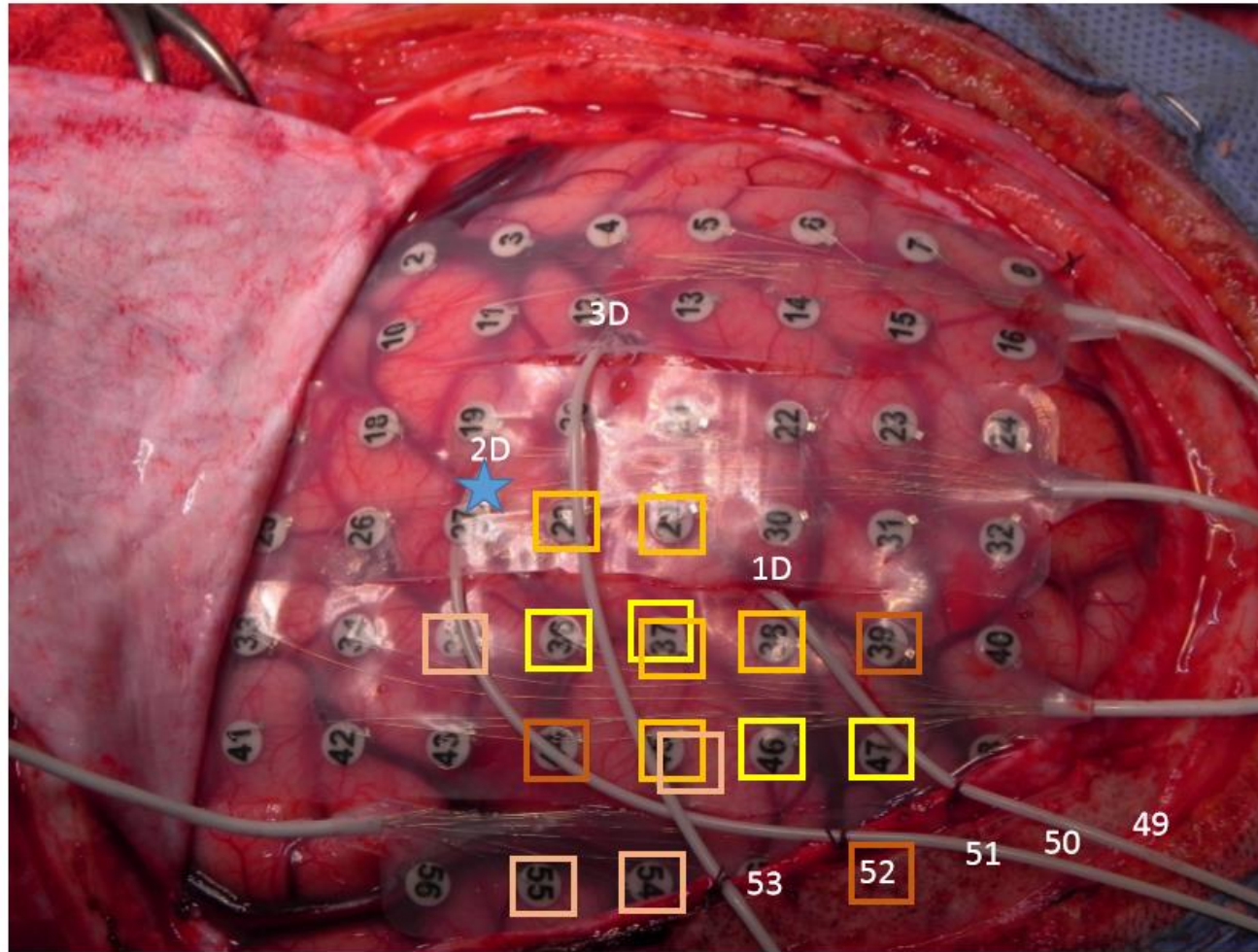
Intraoperative functional localization (direct train of five stimulation)



Lesion/MEG cluster mapping



Extraoperative mapping



WE, wrist extensor

HI, hand intrinsic

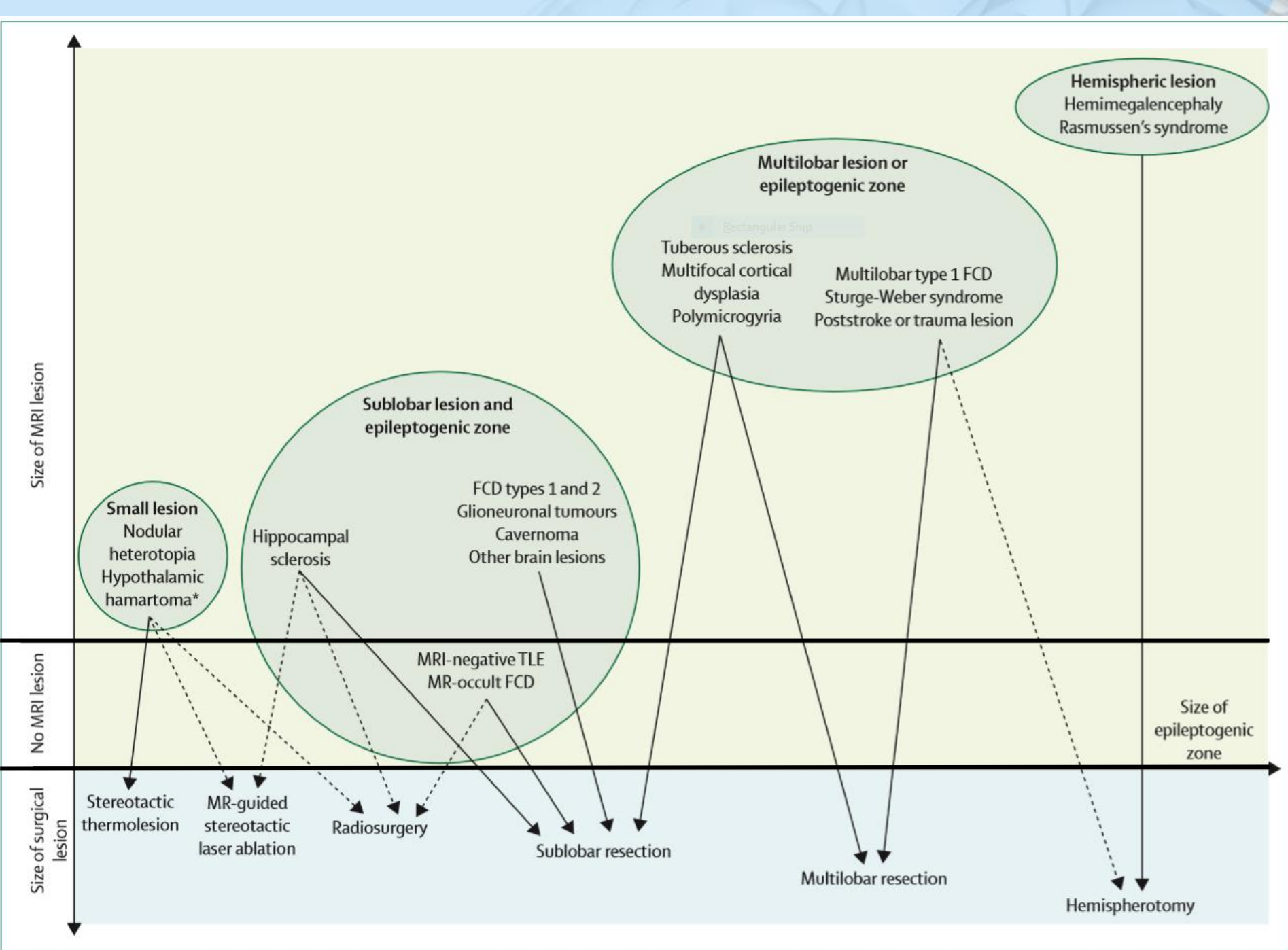
25V=2D1(everything), 2D2 (o, we, hi), 2D3(o), pyramidal tract

30V= 1D45(we, hi), 2D1 (we, hi), 36,37,46,47

35V= 1D6, 2D4 (o), 3D1, 28,29, 38, 45

40V=1D2, 2D5,39, 44, 52 (ta)

45V=1D1 (o), 2D6, 3D3, 35, 45, 54-55 (ta)



Objectives

1. Epilepsy:

- Briefly go over salient issues in pediatrics (you've already had discussion on epilepsy).

2. Hypertonia:

- Understand the evidence for SDR, ITB and DBS in children
- Describe the surgical procedures and technical considerations.
- Focus on decision-making, management

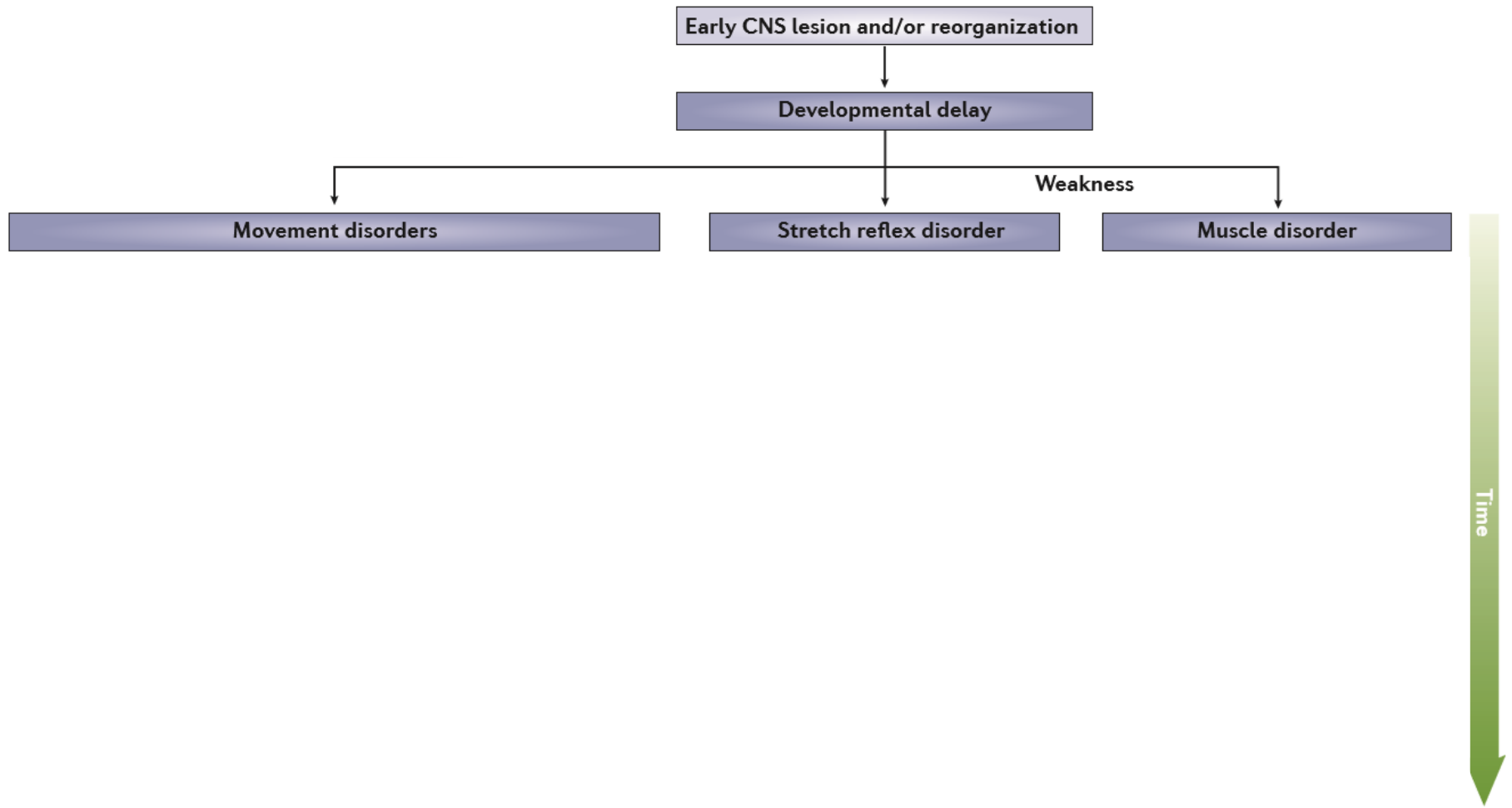
Early CNS lesion and/or reorganization

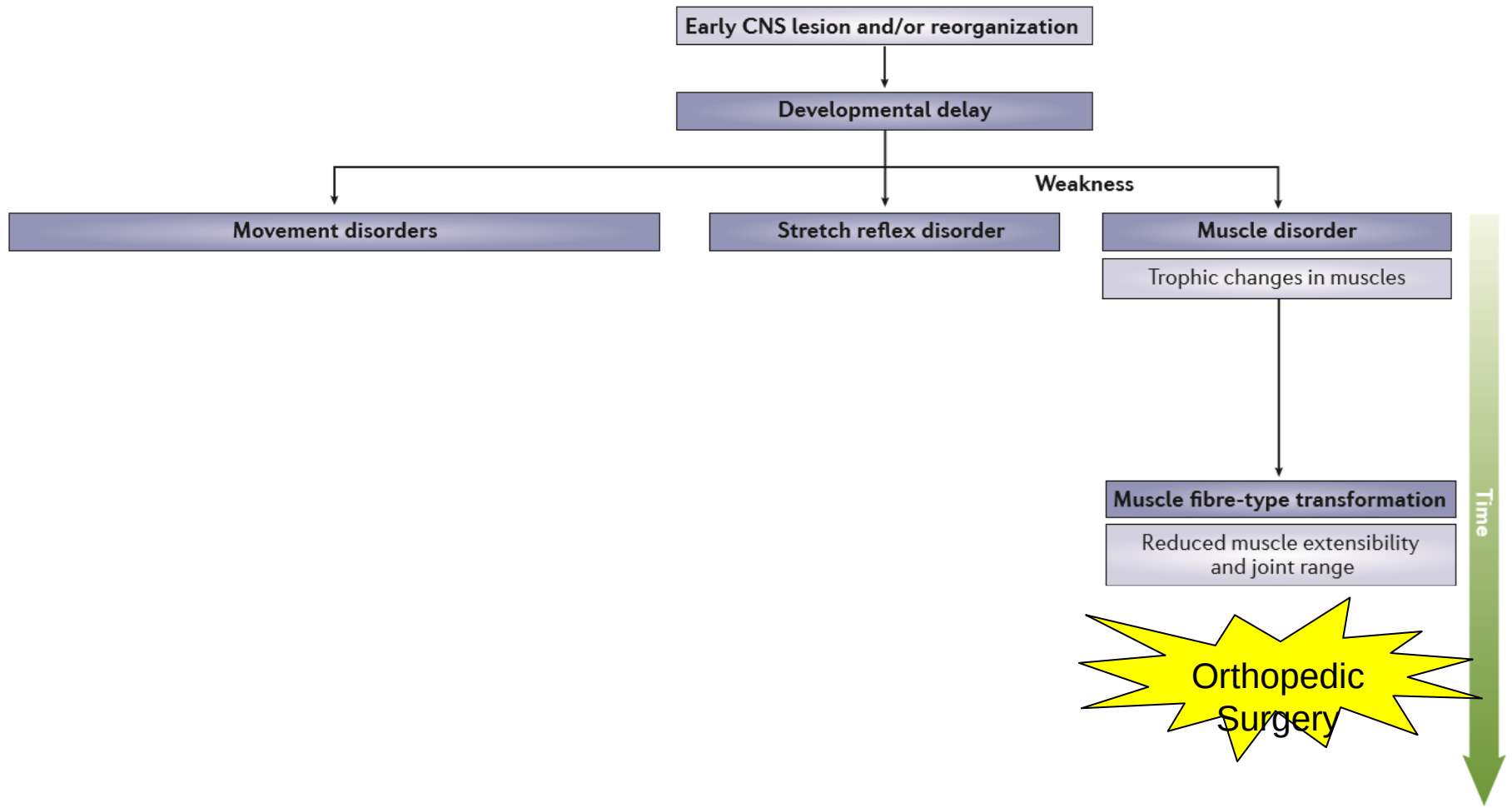


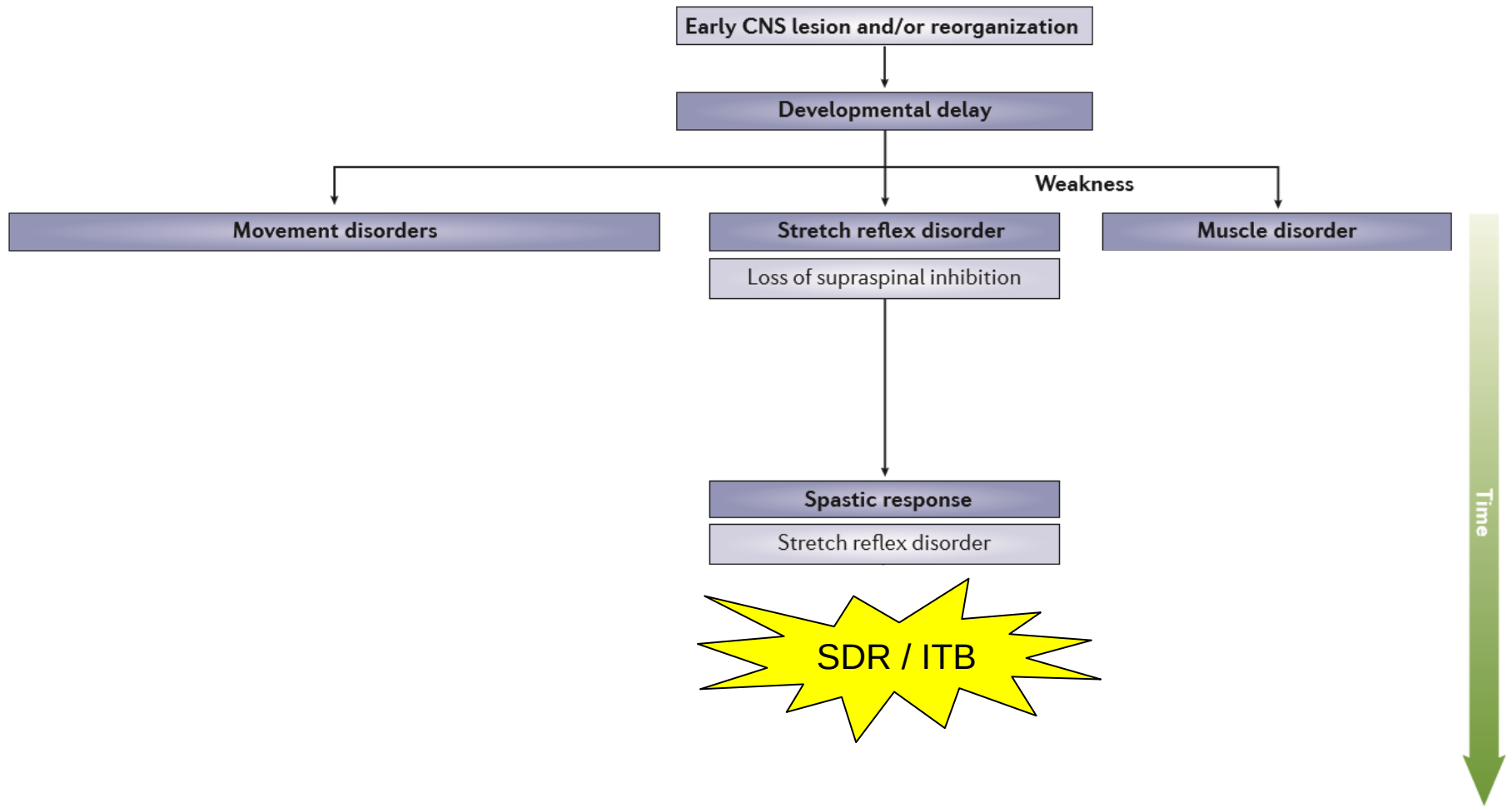
Developmental delay

Time



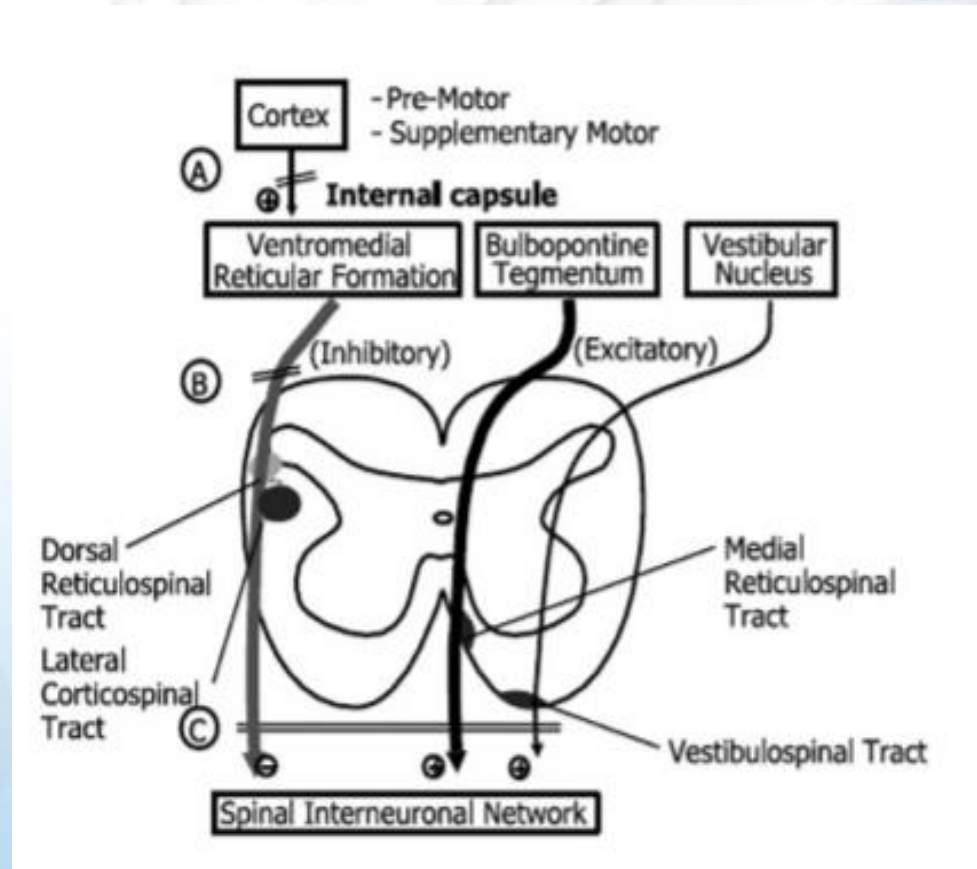






Spasticity

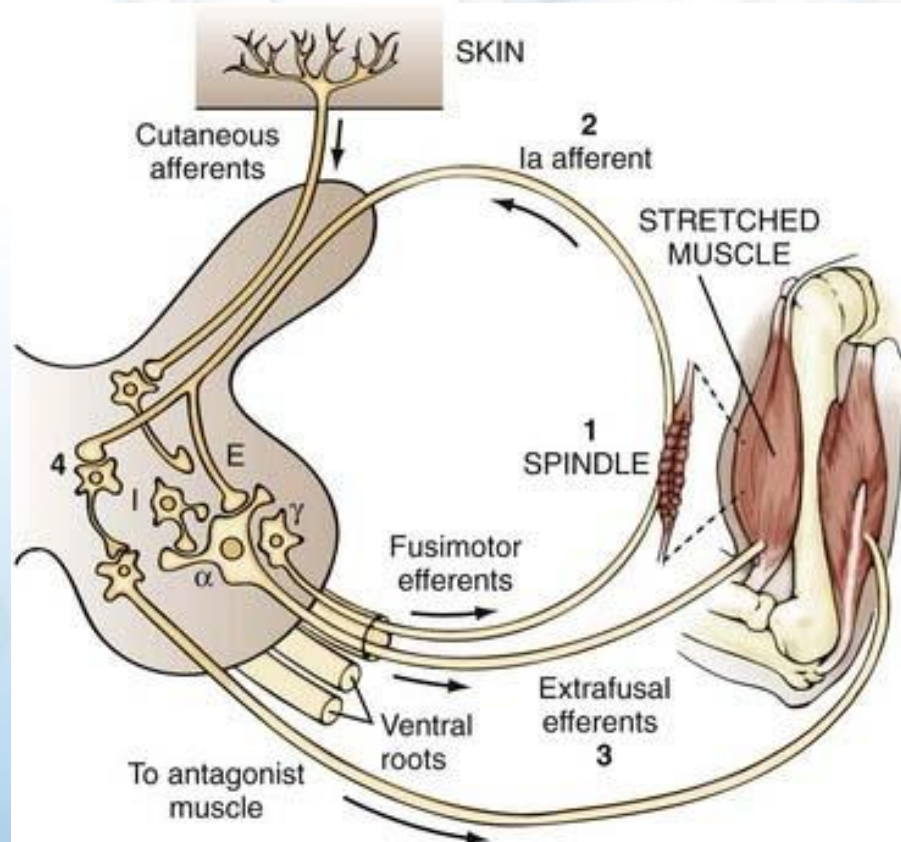
Top-Down Control (basal ganglia, reticular formation, cerebellum)



Spasticity

Top-Down Control (basal ganglia, reticular formation, cerebellum)

Reflex Arc: Afferent excitatory impulses that arise from muscle spindles and travel in Ia fibres in the posterior root



Extracted from Sheean G, The pathophysiology of spasticity. Eur J Neurol, Wiley-Blackwell (2002).

Modified Ashworth Scale

0	No increase in muscle tone
1	Slight increase in muscle tone, manifested by a catch or by minimal resistance at the end of the range of motion (ROM) when the affected part(s) is moved in flexion or extension
1+	Slight increase in muscle tone, manifested by a catch, followed by minimal resistance throughout the remainder (less than half) of the ROM
2	More marked increase in muscle tone through most of the ROM, but affected part(s) easily moved
3	Considerable increase in muscle tone, passive movement difficult
4	Affected part(s) rigid in flexion or extension

Tardieu Scale

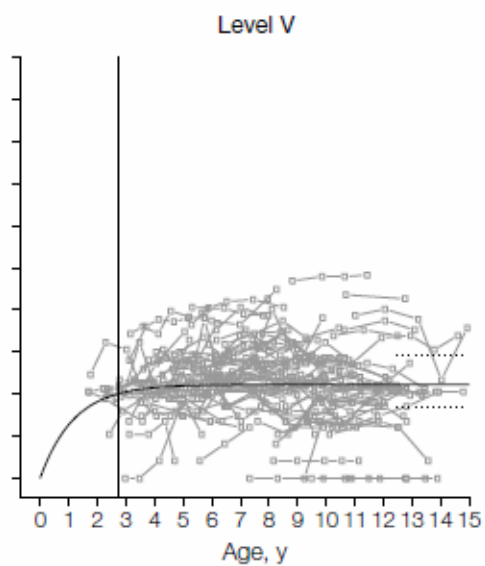
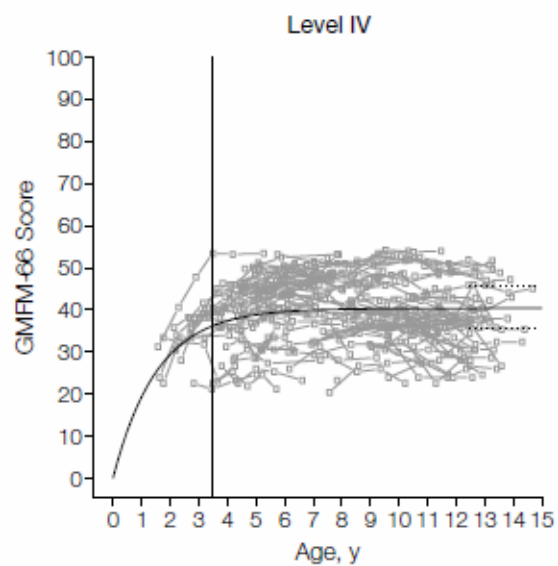
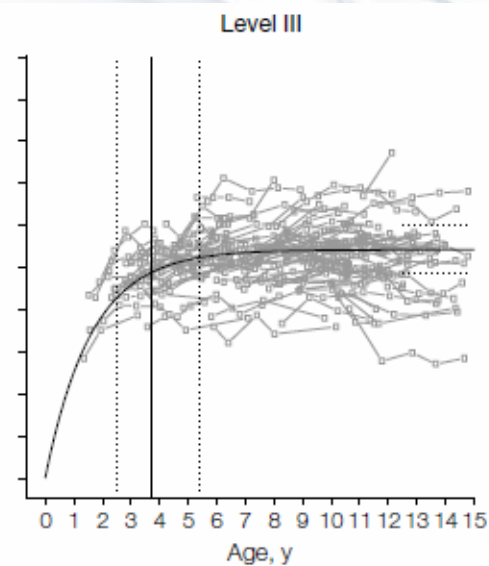
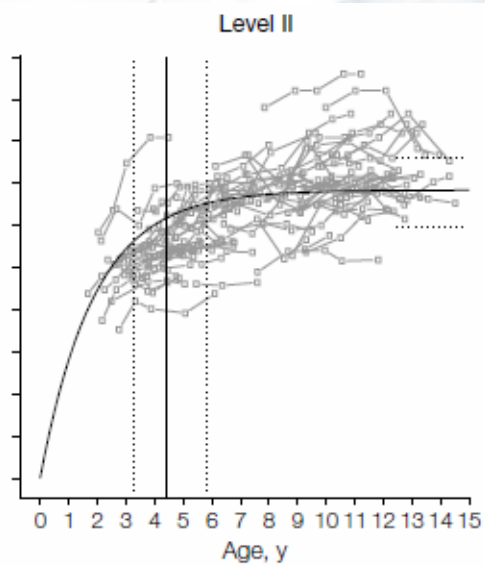
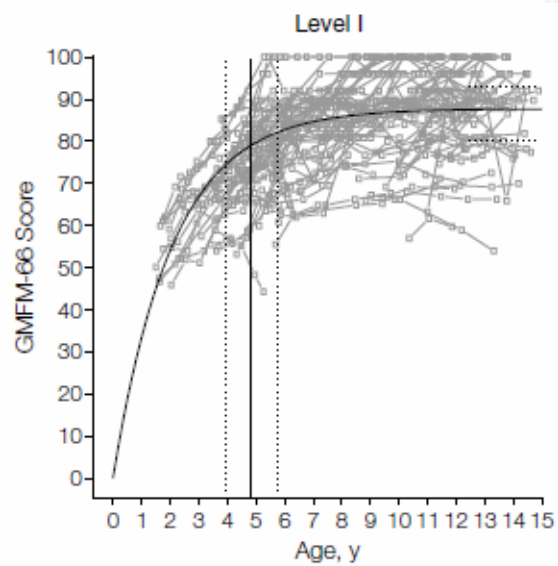
Velocity to Stretch

V1	As slow as possible
V2	Speed of the limb segment falling (with gravitational pull)
V3	At a fast rate (>gravitational pull)

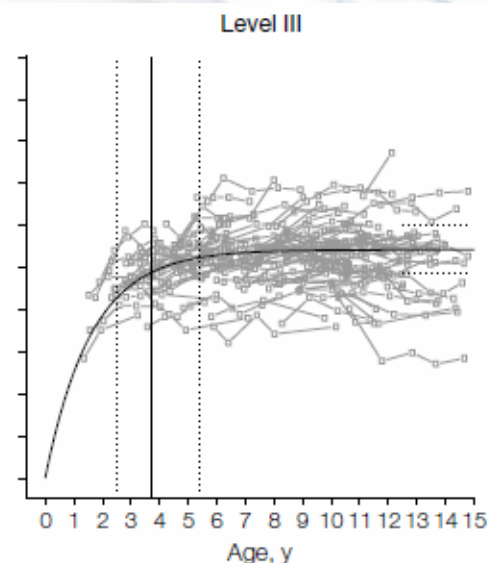
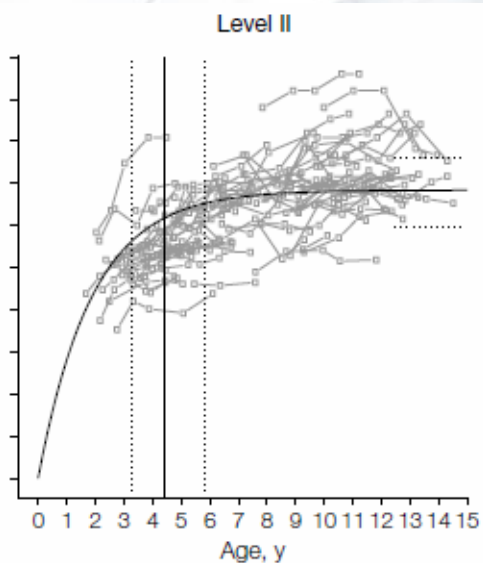
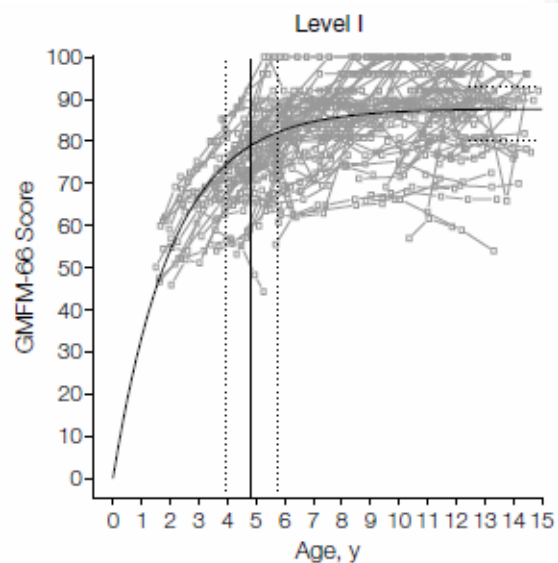
Spasticity Angle

R1	Angle of catch seen at Velocity V2 or V3
R2	Full range of motion achieved when muscle is at rest and tested at V1 velocity

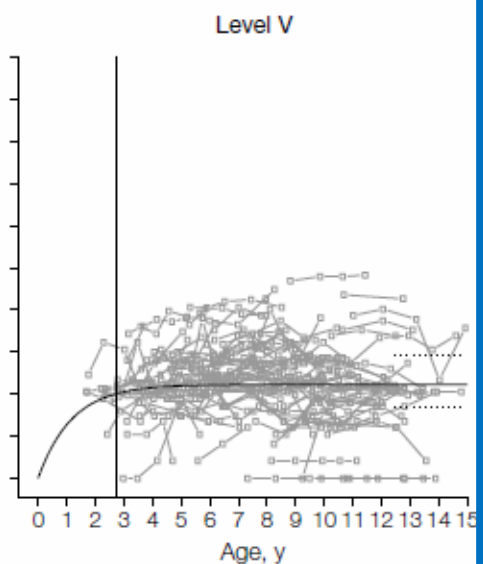
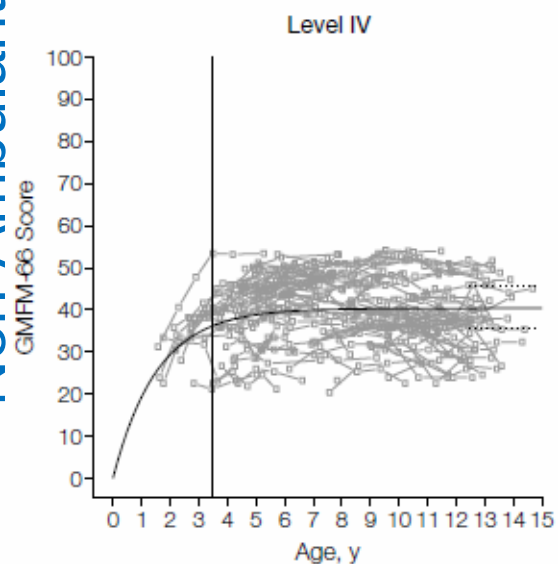
GMFCS Grade	Function	
I	Walk all the time	
II	Walk most of the time, but need rails	
III	Walker, wheelchair rarely	
IV	Wheelchair usually, walker sometimes	
V	Wheelchair, no independence	



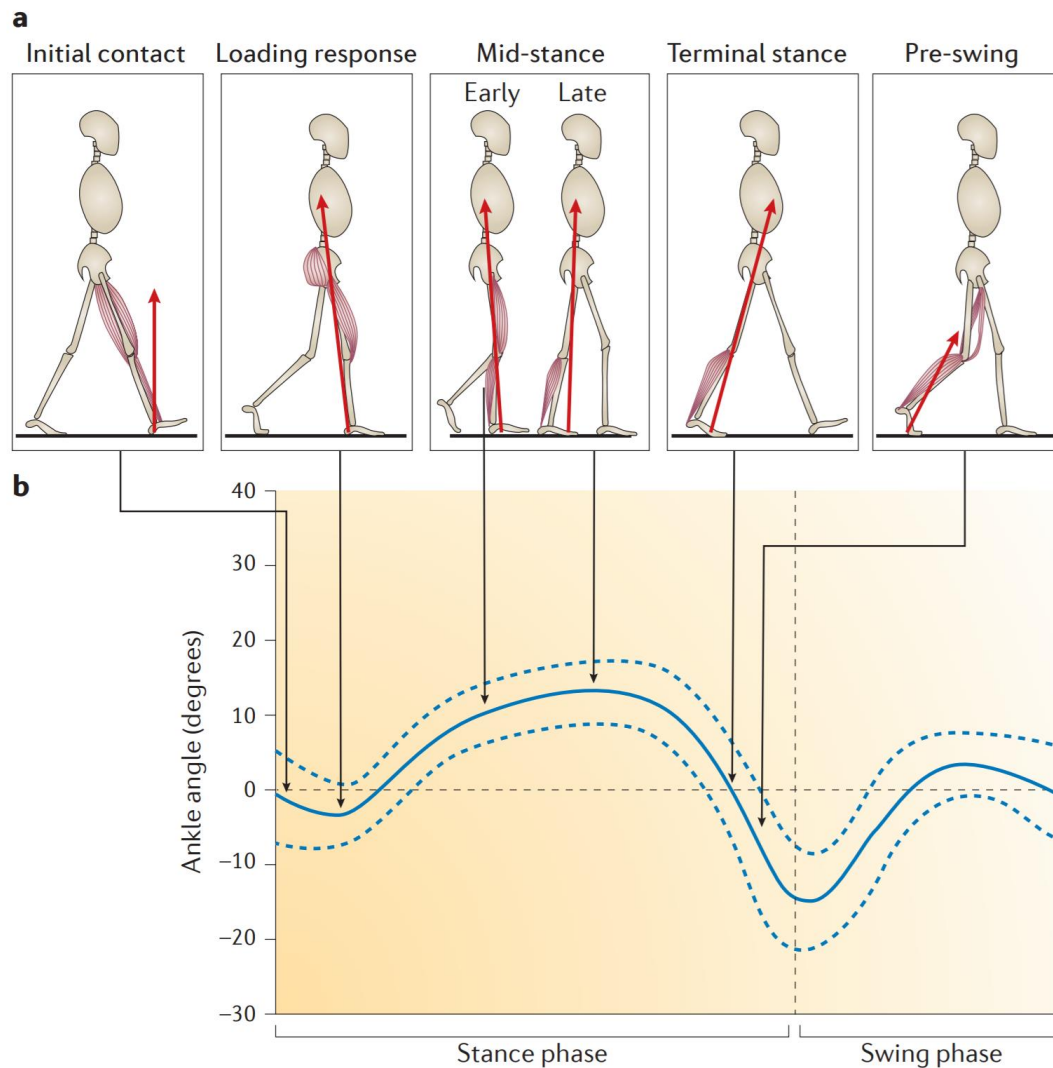
Ambulant



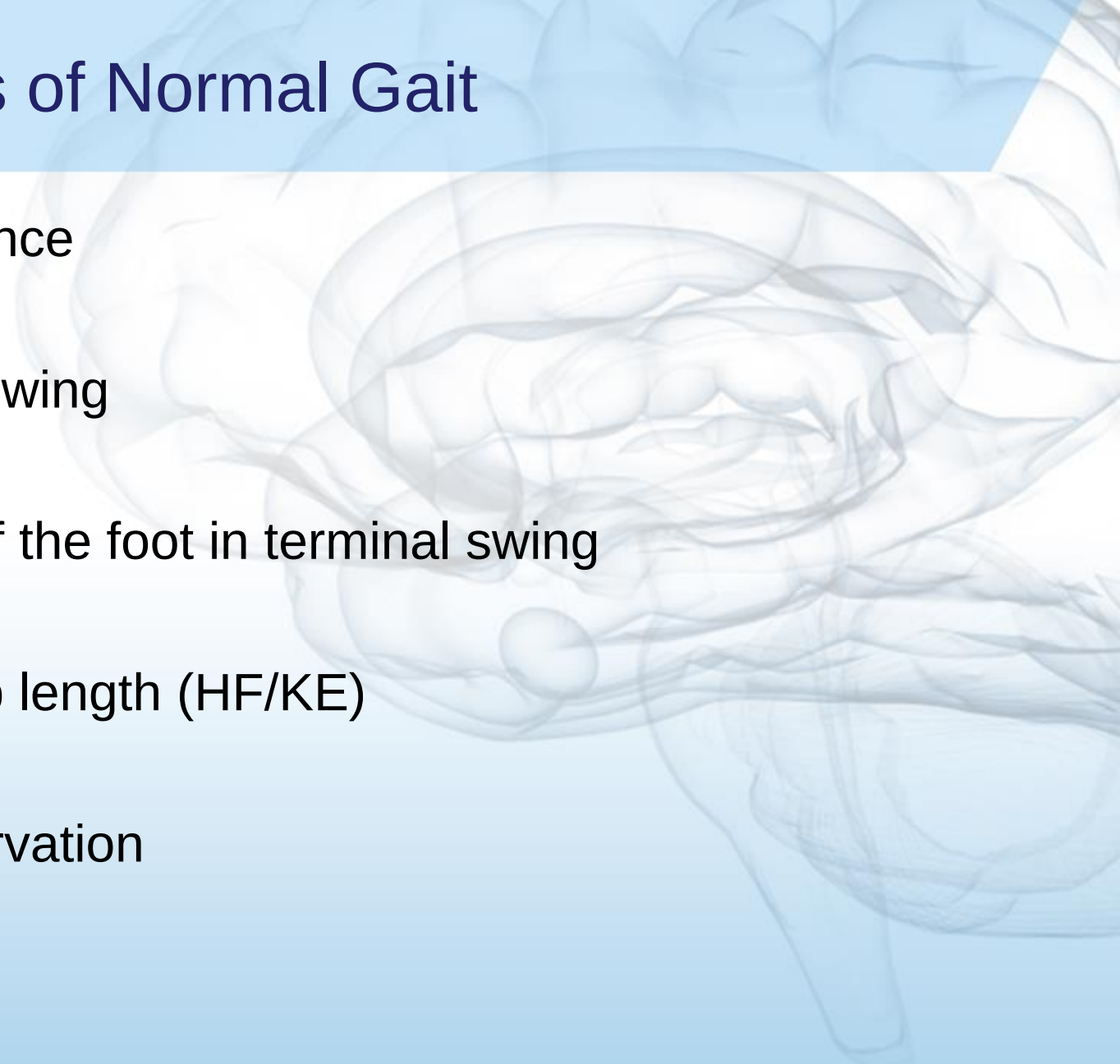
Non-Ambulant



Prerequisites of Normal Gait

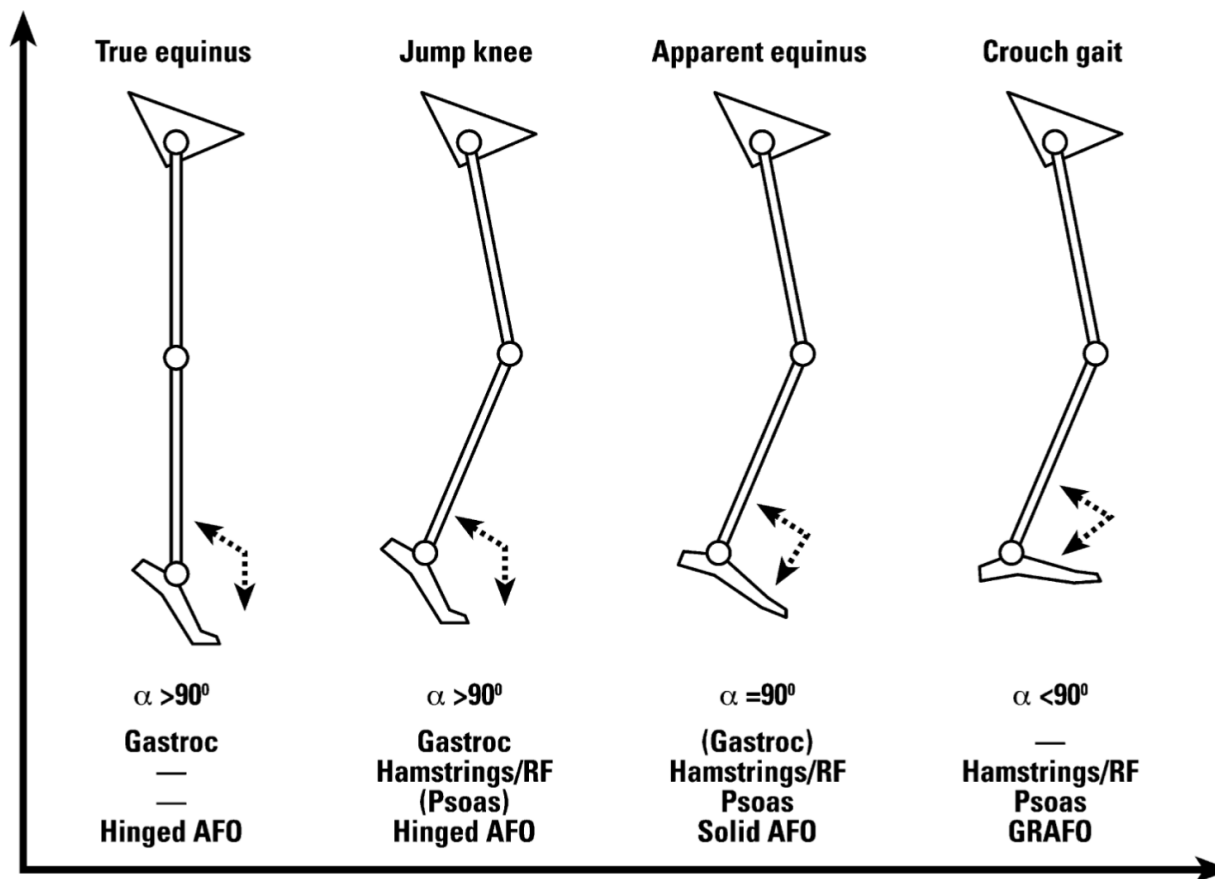


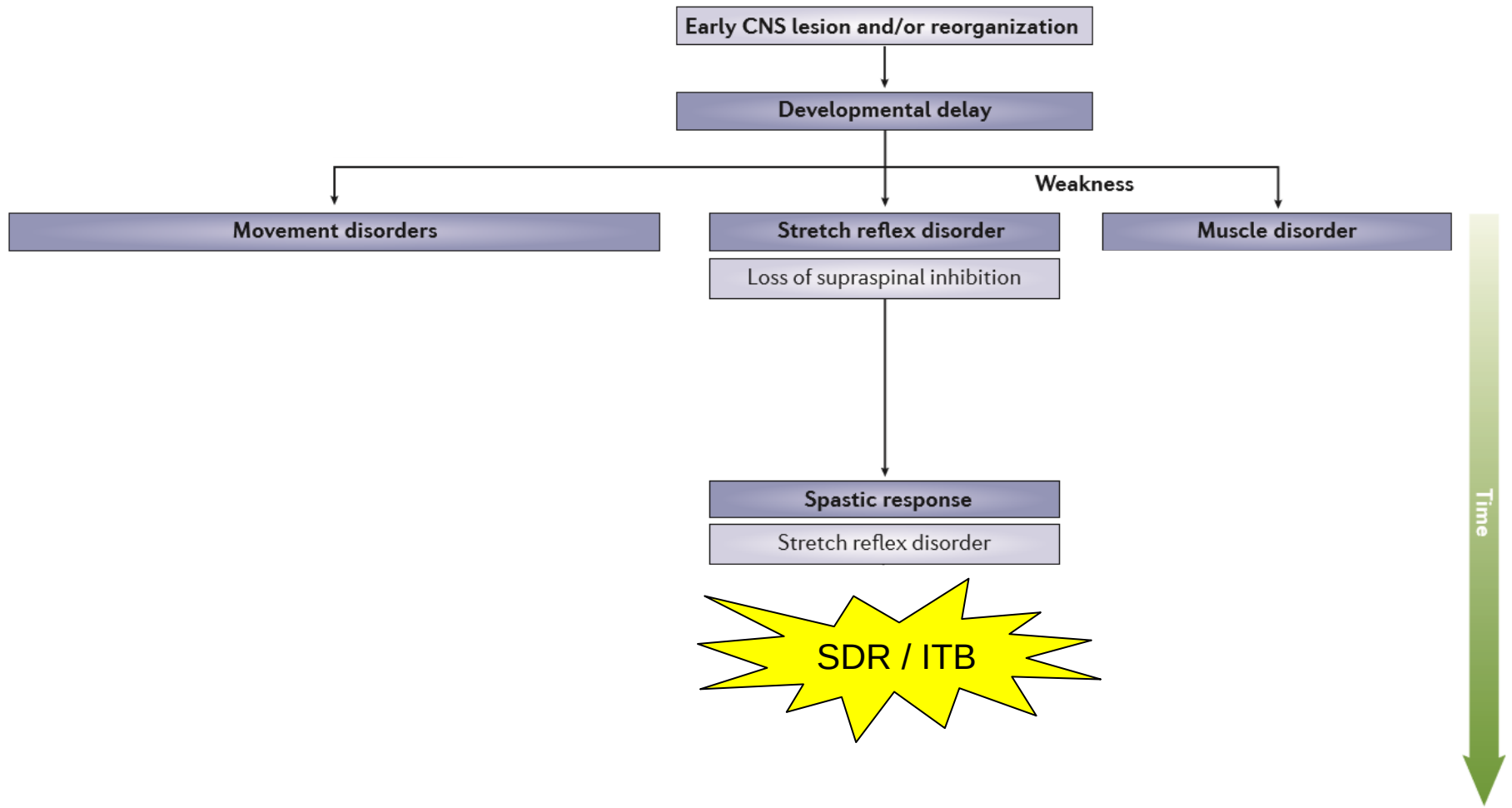
Prerequisites of Normal Gait

1. Stability in stance
 2. Clearance in swing
 3. Pre-position of the foot in terminal swing
 4. Adequate step length (HF/KE)
 5. Energy conservation
- 
- A faint, stylized illustration of a human brain and a hand pointing towards the list of prerequisites. The brain is shown in a light blue, wireframe-like style, and the hand is also in a similar style, with the index finger pointing towards the list. The background is a light blue gradient with some abstract shapes.

“Spastic gait”

Common Gait Patterns: Spastic Diplegia





Who is the ideal surgical candidate?

- Diagnosis:
- Age:
- Function:
- Spasticity:
- Gait:
- Strength:
- Cooperation
- Negatives

Who stands to benefit the most?

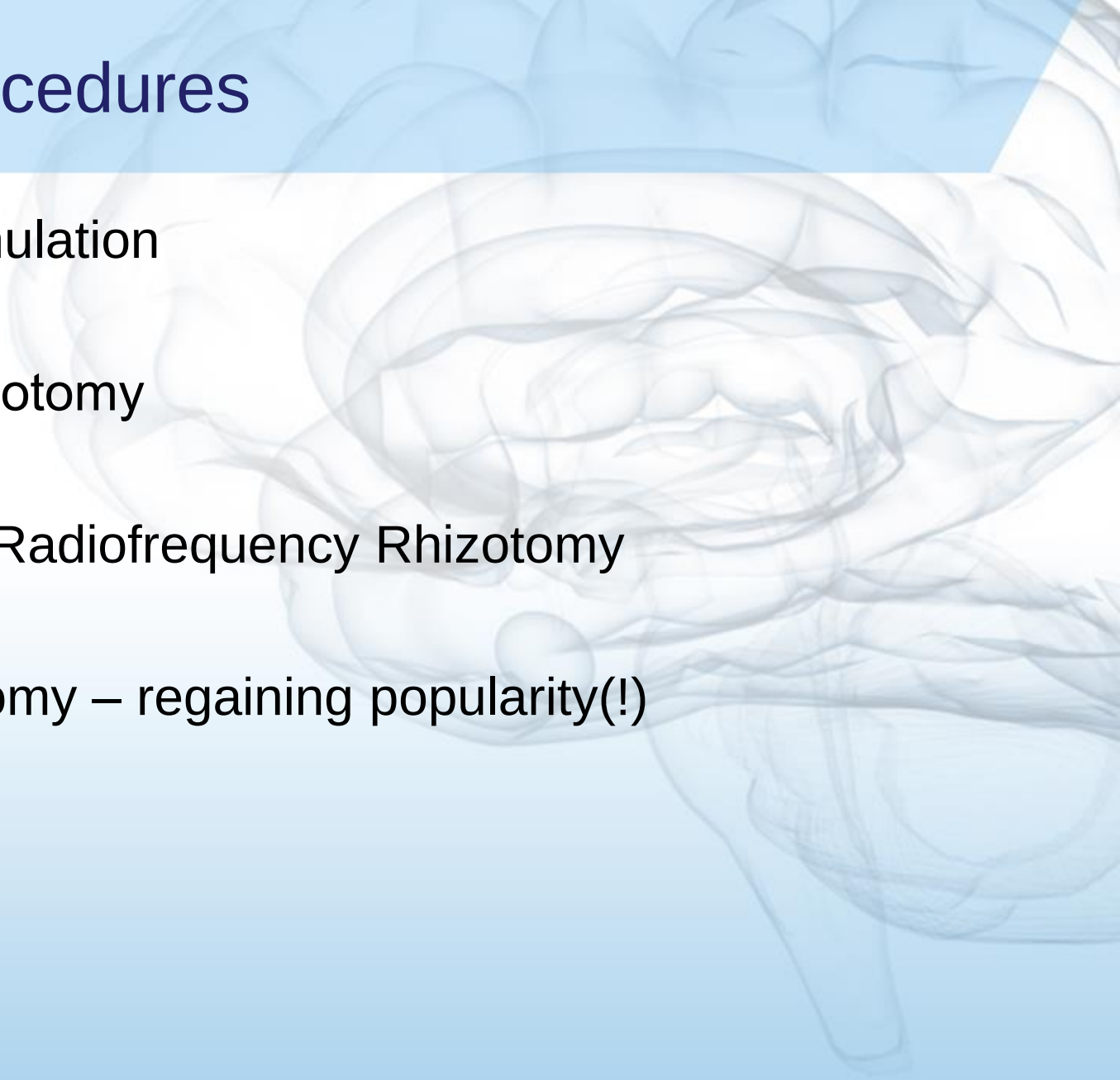
- Diagnosis:
- Age:
- Function:
- Spasticity:
- Gait:
- Strength:
- Cooperation
- Negatives

Who stands to benefit the most?

- Diagnosis: Spastic diplegic sub-type
- Age: not too young, not too old
- Function: GMFCS II-III (vs. ITB)
- Spasticity: At least 4 lower extremity muscles
- Gait: Evidence of spastic gait
- Strength: Adequate proximal strength
- Cooperation
- Negatives
 - No dystonia
 - No contractures

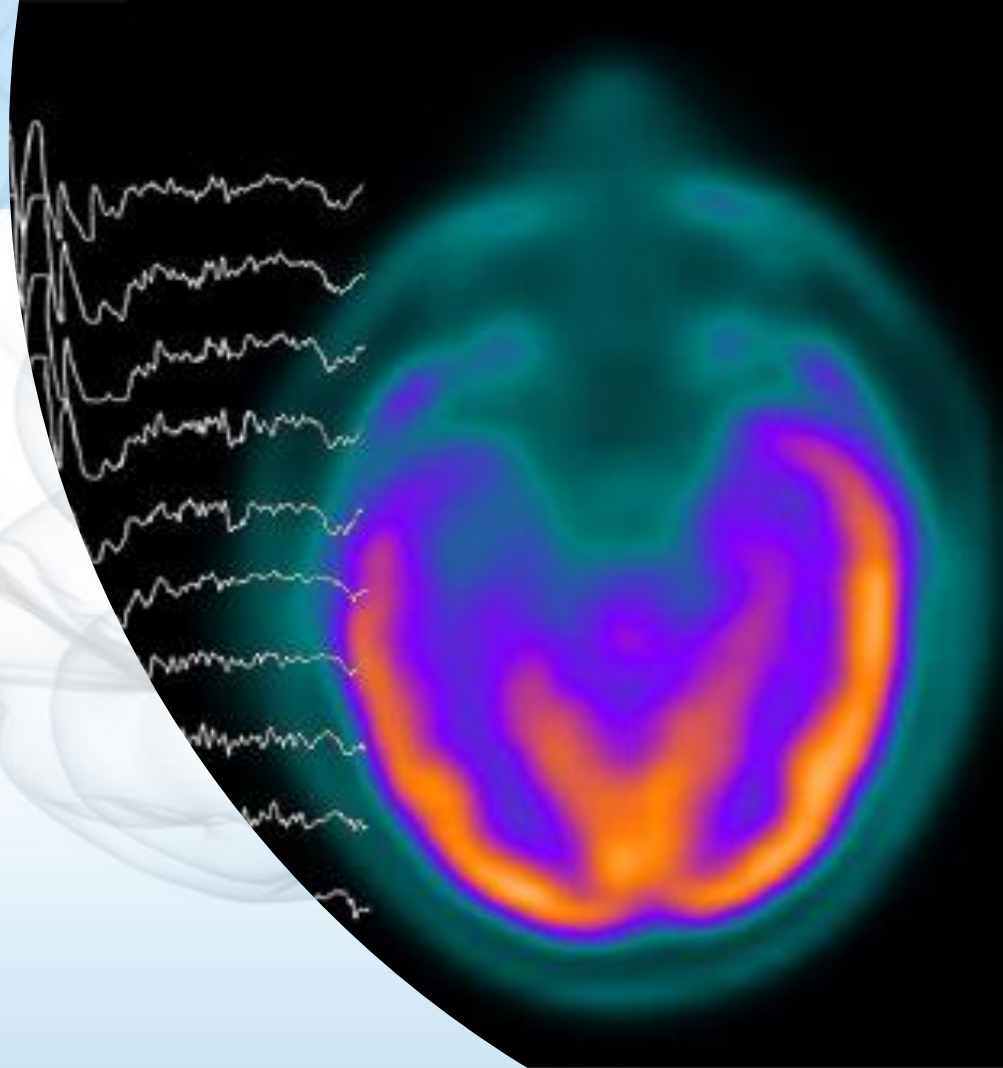
Historical procedures

- Cerebellar stimulation
- Bischoff's myelotomy
- Percutaneous Radiofrequency Rhizotomy
- Anterior rhizotomy – regaining popularity(!)

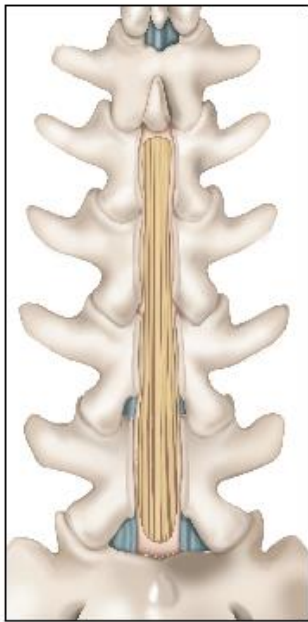


Selective Dorsal Rhizotomy

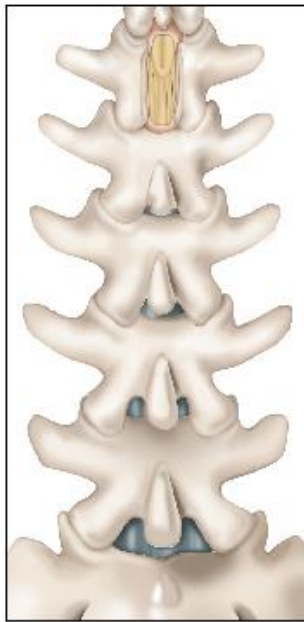
- **1908 (Foerster):** Section the dorsal (sensory) roots.
- **1978 (Fasano):** Partial sectioning on the basis of intraoperative electrophysiology.
- **1982 (Peacock):** modified procedure, identified candidates.



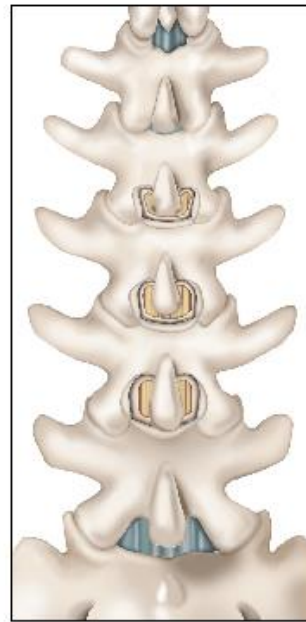




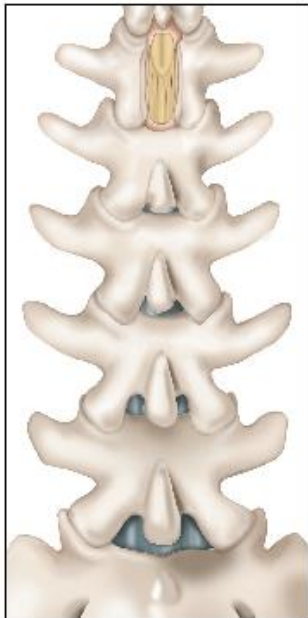
Peacock et al. 1987



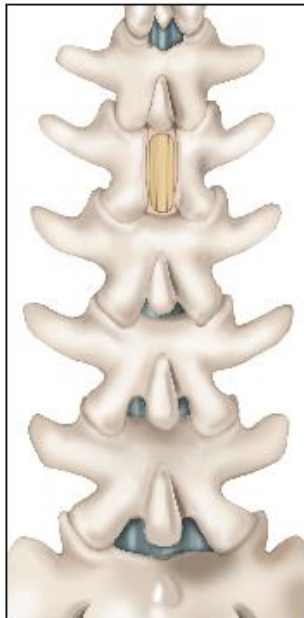
Park et al. 2006



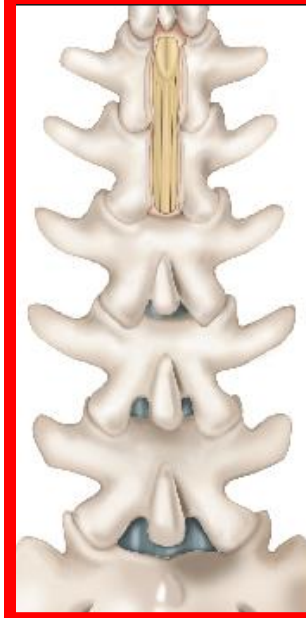
Sindou et al. 2015



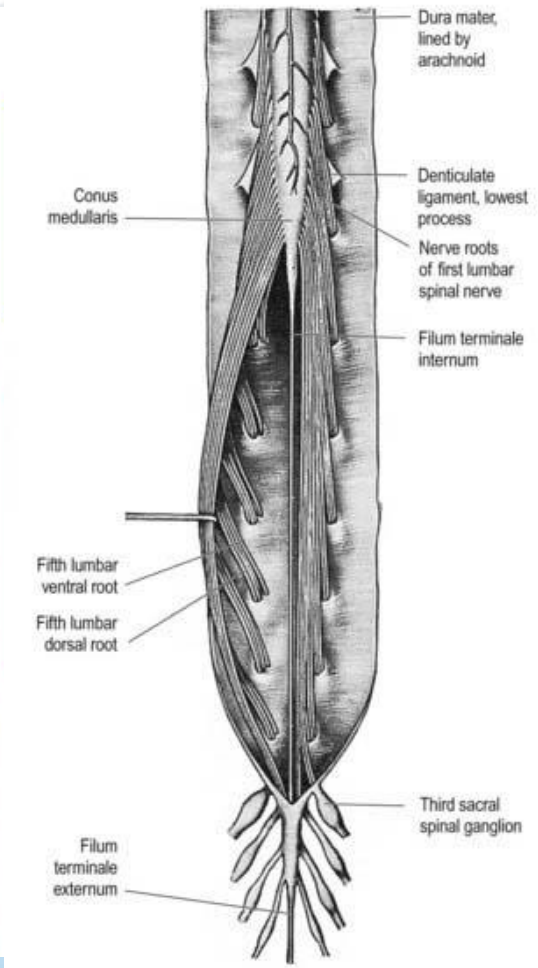
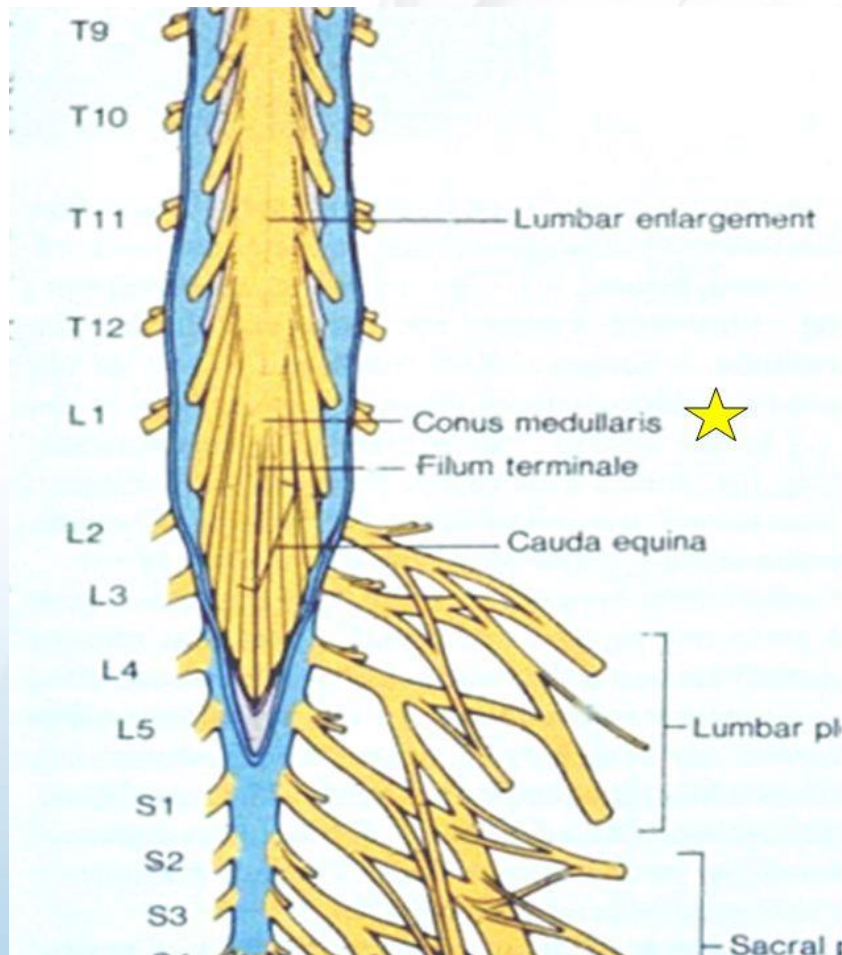
Funk et al. 2016

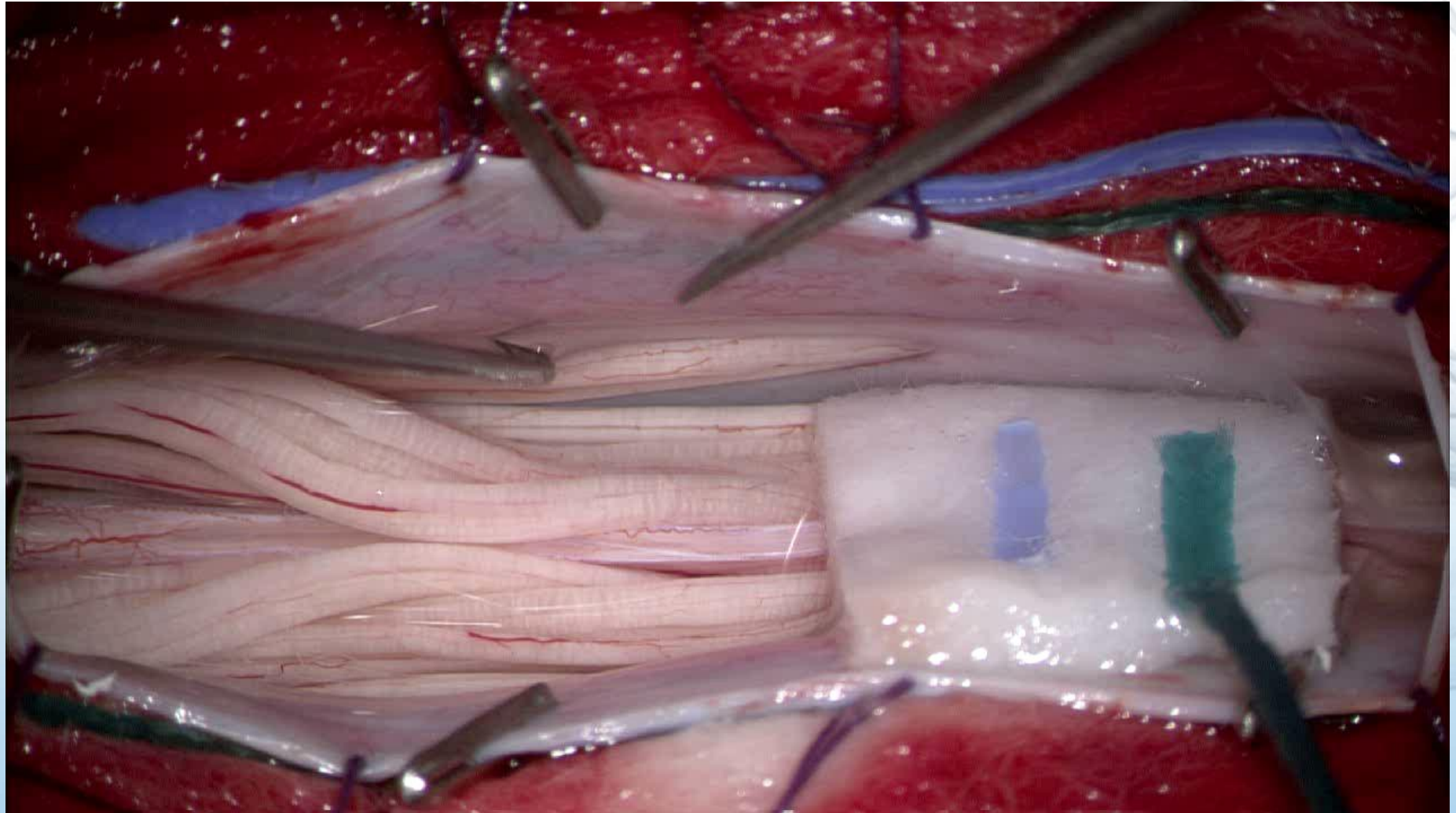


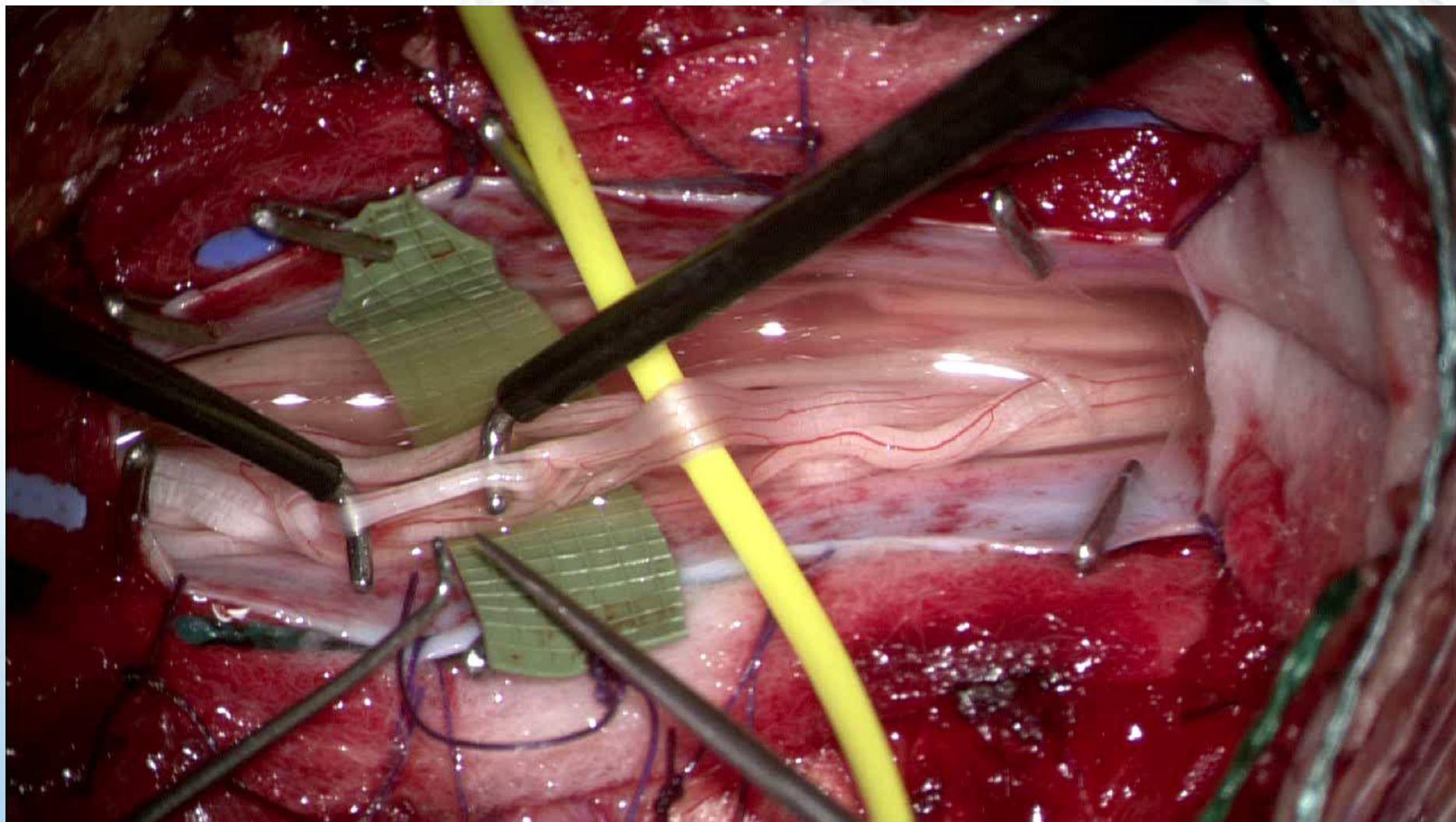
Bales et al. 2016



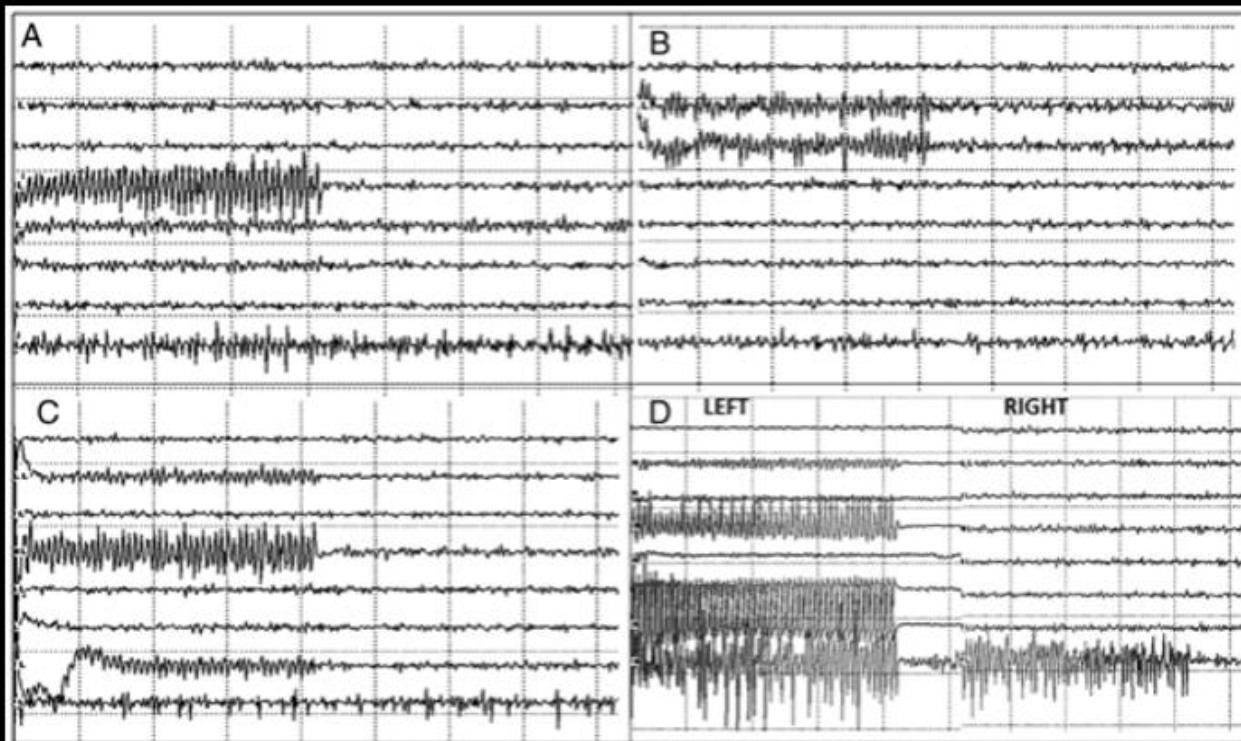
SickKids Method







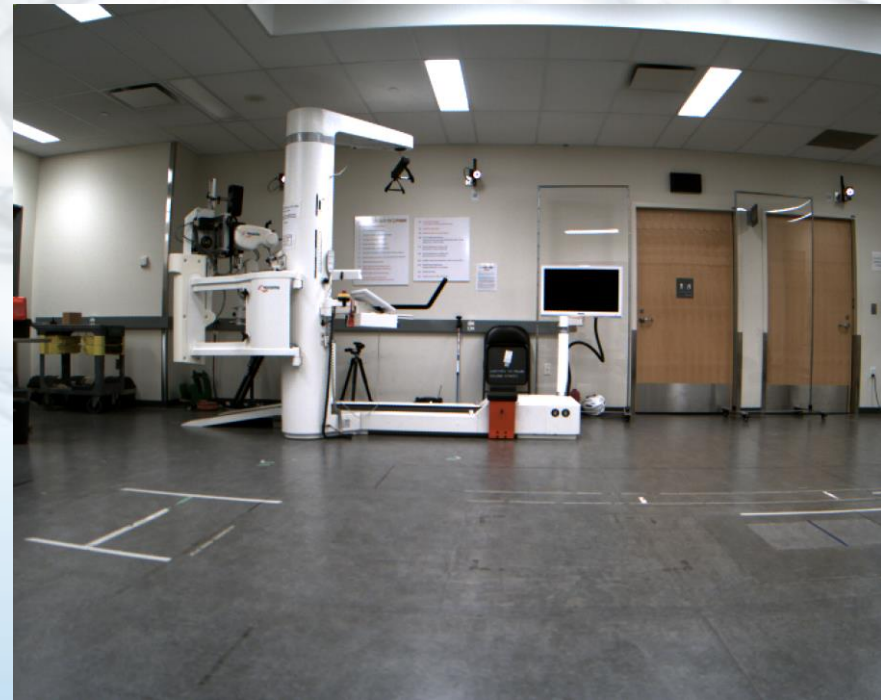
Grade	EMG response
0	Unsustained or single discharge to a train of stimuli
1	Sustained discharges from muscle group being stimulated
2	Sustained discharges from muscle group being stimulated + adjacent segments
3	Sustained discharges from muscle group being stimulated + distant segments
4	Sustained discharges from muscle group being stimulated + contralateral segments







Preoperative Gait



Postoperative Gait

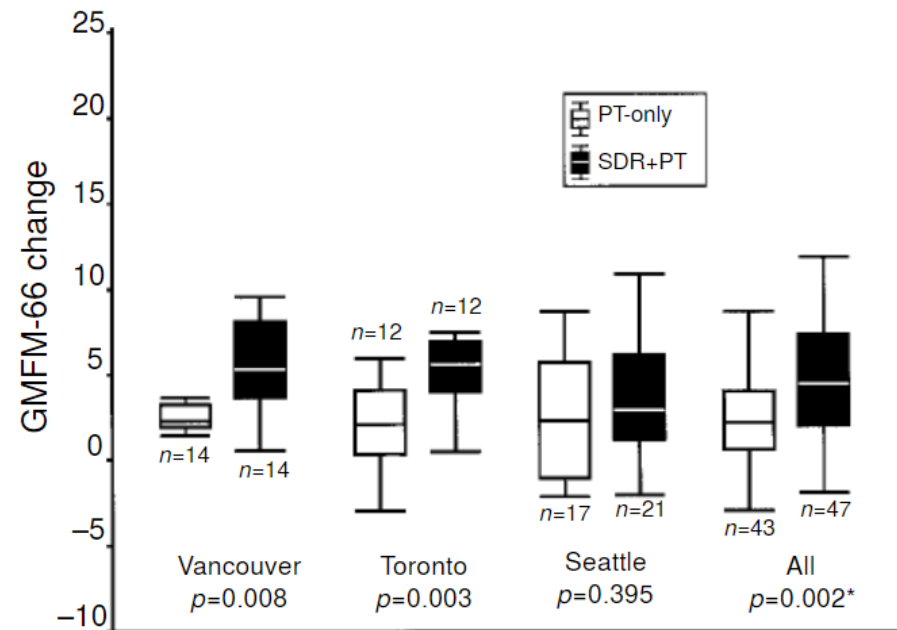
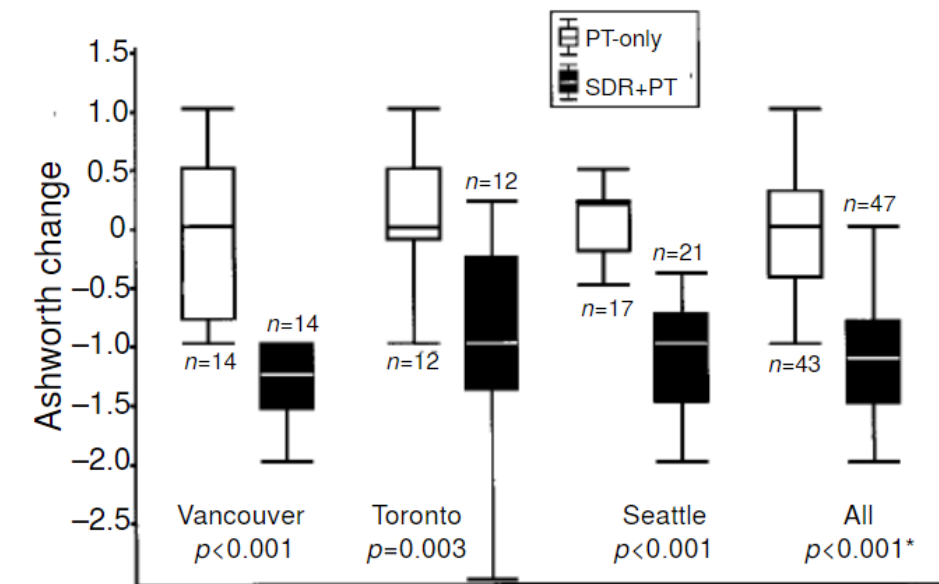
Selective dorsal rhizotomy: meta-analysis of three randomized controlled trials

**Table I: Selective dorsal rhizotomy randomized clinical trials
outcomes summary**

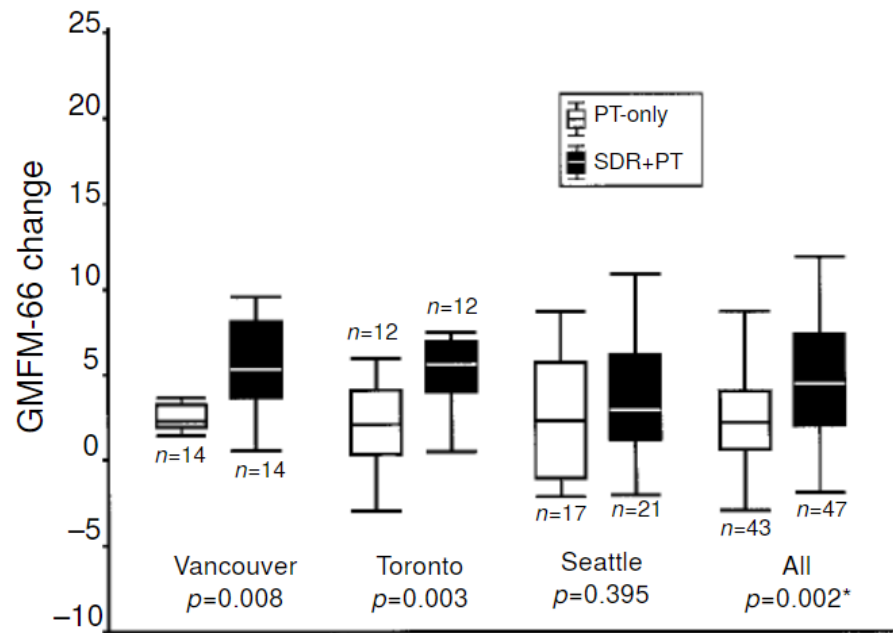
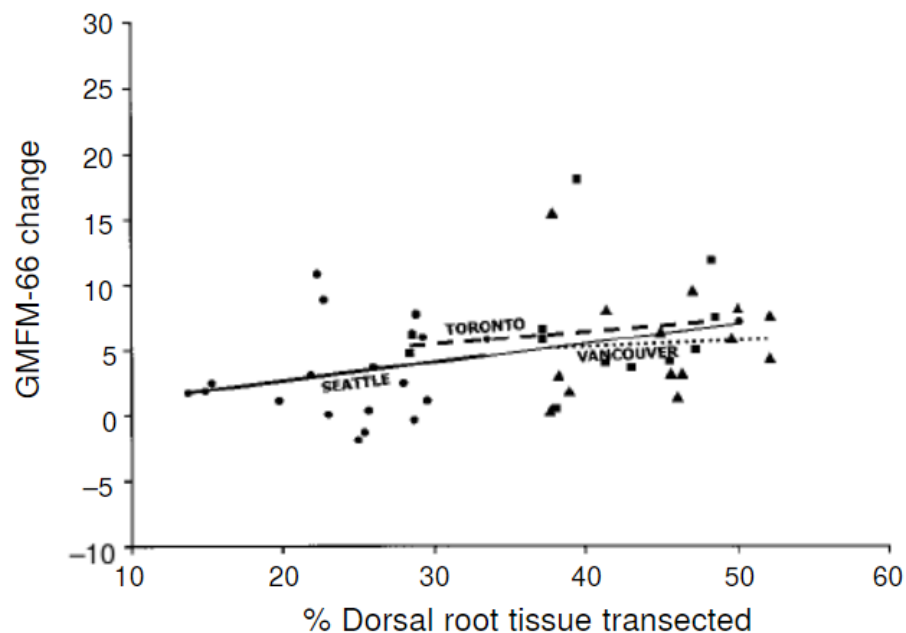
	<i>Study</i>		
	<i>Vancouver</i>	<i>Toronto</i>	<i>Seattle</i>
Children (<i>n</i>)	28	24	38
Interval (mo)	9	12	24
Mean difference in Ashworth change scores	-1.1 (<i>p</i> <0.001)	-1.0 (<i>p</i> =0.002)	-1.0 (<i>p</i> =0.001)
Mean difference in GMFM change scores	6.1% (<i>p</i> =0.007)	7.7% (<i>p</i> =0.02)	0.2% (<i>p</i> =0.94)

Vancouver, Steinbok et al. 1997; Toronto, Wright et al. 1998; Seattle, McLaughlin et al. 1998.

Selective dorsal rhizotomy: meta-analysis of three randomized controlled trials



Selective dorsal rhizotomy: meta-analysis of three randomized controlled trials



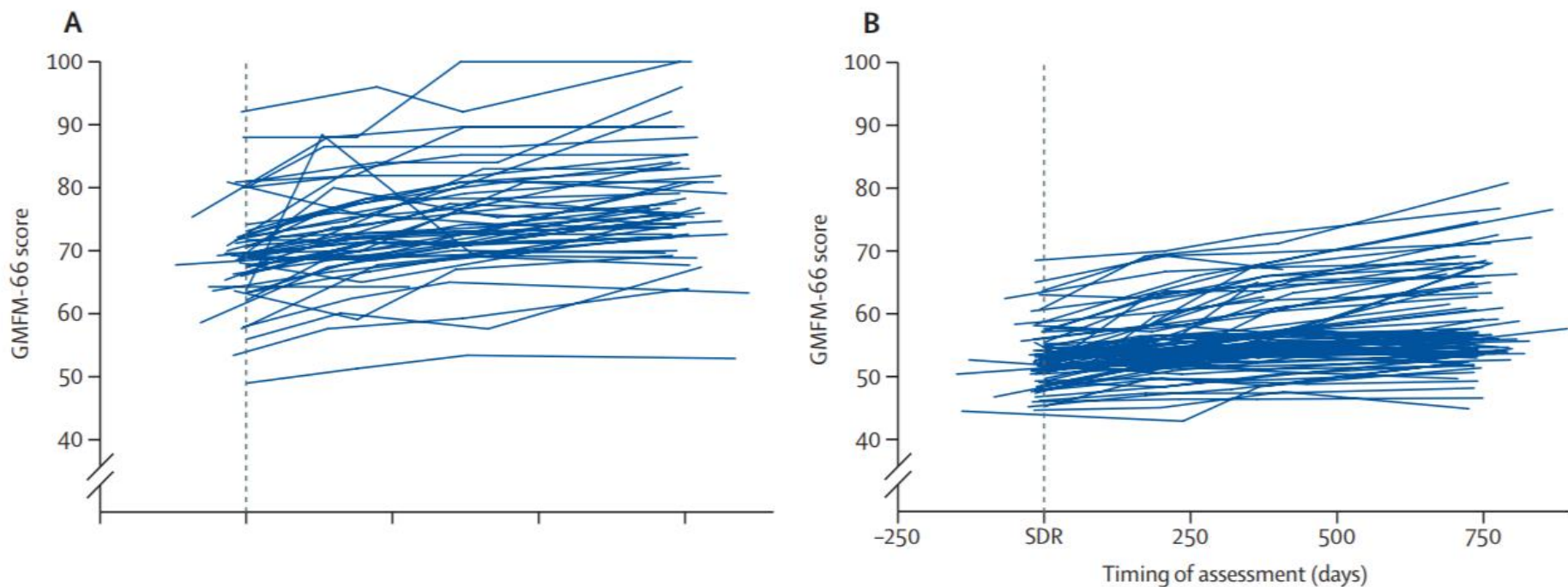
Selective dorsal rhizotomy in ambulant children with cerebral palsy: an observational cohort study



*Jennifer Summers, Bola Coker, Saskia Eddy, Maria Elstad, Catey Bunce, Elli Bourmpaki, Mark Pennington, Kristian Aquilina, Stephanie Cawker, Richard Edwards, John Goodden, Sally Hawes, Kate McCune, Benedetta Pettorini, Jennifer Smith, Christine Sneade, Michael Vloeberghs, Hannah Patrick, Helen Powell, Christopher Verity, Janet L Peacock, for the Selective Dorsal Rhizotomy Steering Committee**



Change in Gross Motor Function



Interpretation SDR improved function and quality of life in the 24 months after surgery in children with cerebral palsy classified as GMFCS levels II and III. On the basis of these findings, an interim national policy decision was made that SDR would be funded for eligible children in England from 2018.

What about children with spasticity without cerebral palsy?

Selective dorsal rhizotomy for spasticity of genetic etiology

Laura-Nanna Lohkamp¹ • Ian Coulter¹ • George M. Ibrahim¹

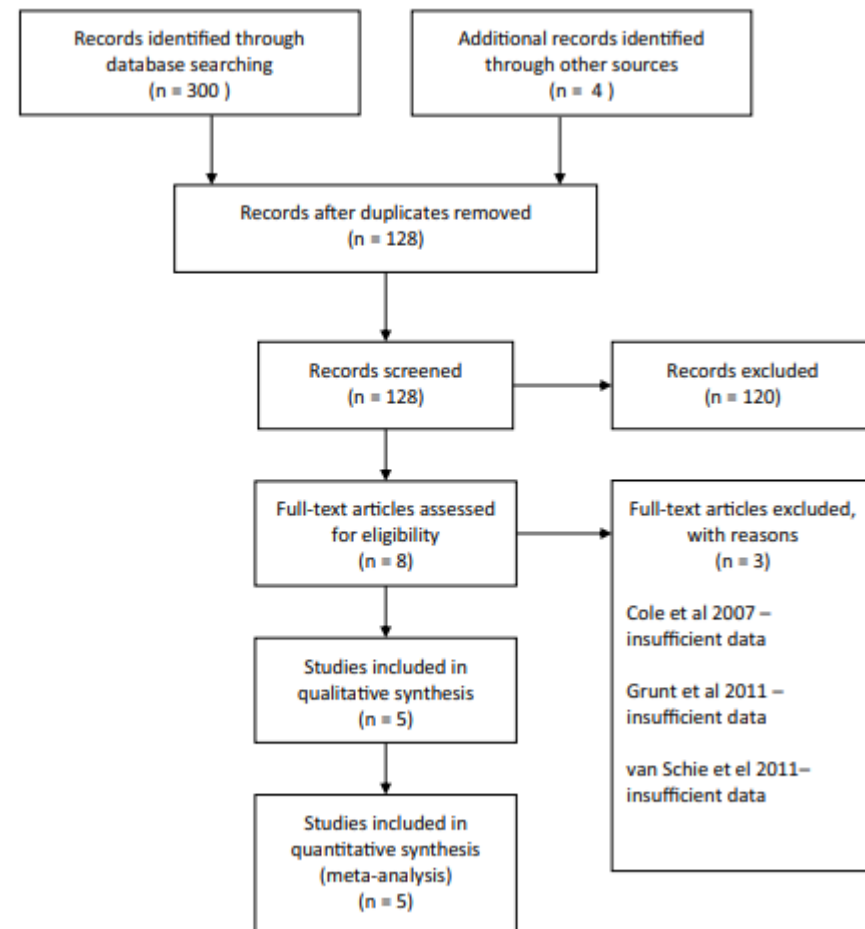
Received: 9 January 2020 / Accepted: 30 March 2020 / Published online: 16 April 2020
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Identification

Screening

Eligibility

Included



What about children who are non-ambulant?

Intrathecal baclofen versus selective dorsal rhizotomy for children with cerebral palsy who are nonambulant: a systematic review

Benjamin Davidson, MD,¹ Nathan Schoen, MD, MPH,² Shaina Sedighim, BSc,² Renée Haldenby, MSc(PT),³ Blythe Dalziel, MSc(PT),³ Sara Breitbart, MSc,⁴ Darcy Fehlings, MD, MSc,³ Golda Milo-Manson, MD, MHSc,³ Unni G. Narayanan, MBBS, MSc, FRCS(C),⁴ James M. Drake, MBBCh, MSc,^{1,5} and George M. Ibrahim, MD, PhD^{1,5}

What about children who are non-ambulant?

SDR

Authors & Year	No. of Pts	Outcomes
Ailon et al., 2015	20	GMFM, MAS, hip, ROM
D'Aquino et al., 2018	54	MAS, urologic outcome, power/ROM, caregiver survey
Arens et al., 1989	19	Change in Peacock grade
Buizer et al., 2017	24	Complications, caregiver survey
Gigante et al., 2013	12	Association between GMFCS & MAS improvement
Heim et al., 1995	32	Radiographic hip subluxation
Hicdonmez et al., 2005	43	Association between GMFCS & hip subluxation
Ingale et al., 2016	10	MAS, caregiver survey
Kim et al., 2001	66	Change in Peacock grade
Kinghorn, 1992	4	Caregiver survey
Nordmark et al., 2008	16	GMFM, PEDI
Park et al., 1994	8	Hip subluxation
Peacock et al., 1987	16	Tone, function

ITB

Authors & Year	No. of Pts	Outcomes
Borowski et al., 2010	161	Complications, caregiver survey
Campbell et al., 2002	21	MAS, GMFM, caregiver survey, complications, PEDI, orthopedic procedures
Hoving et al., 2009	17	GMFM, PEDI, CHQ, unique VAS, caregiver survey, complications
Krach et al., 2005	18	GMFM
Kraus et al., 2017	13	MAS, CPCHILD, KINDL, complications
Motta et al., 2008	10	MAS, upper-limb "percentage score"
Motta et al., 2011	37	GMFM, caregiver survey
Overgård et al., 2015	46	MAS, caregiver survey, complications
Ravindra et al., 2017	50	Requirement for spinal stabilization
Sgouros & Seri, 2002	6	MAS
Shilt et al., 2008	50	Progression of scoliosis
Vles et al., 2013	17	Complications, pt-specific goals, CHQ, requirement for orthopedic procedures
Ward et al., 2009	23	Complications, CHQ, COPM, MAS

What about children who are non-ambulant?

- Relevant Outcomes
 - Spasticity Reduction
 - Gross motor function
 - Subsequent Orthopedic Surgery
 - Subsequent Spinal deformity correction
 - Caregiver satisfaction
 - Effect on care and comfort
 - Complications

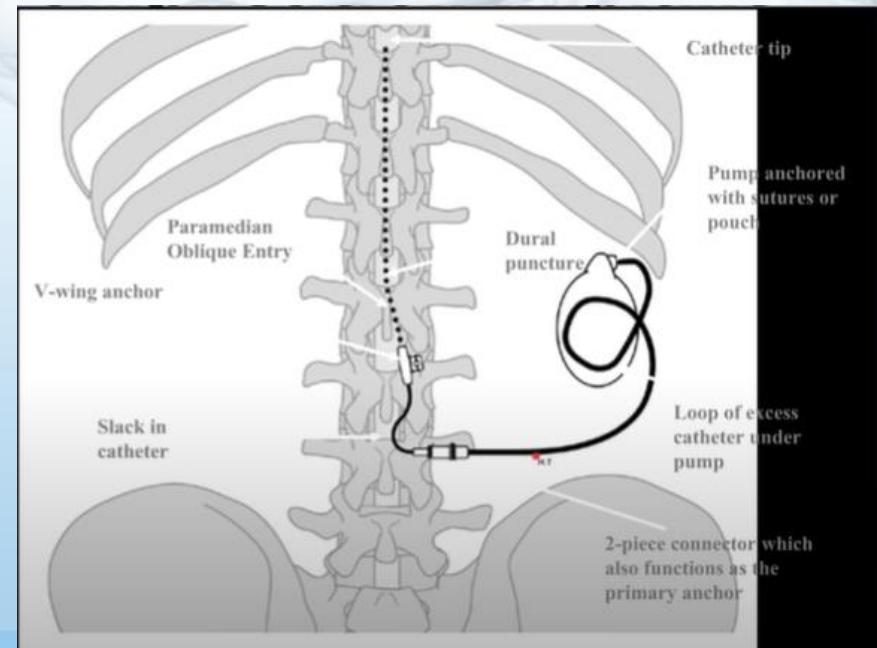


Baclofen Pumps

- Patient factors
 - Size - Minimum weight ~ 25 lbs
 - G-tube (left usually first choice – unless G-tube)
 - Previous surgery
 - Belts/straps/seating
- Pump Volume
 - 40 cc vs. 20 cc
- Pump Position
 - Subfascial (below rectus and external oblique) vs. suprafascial
- Tip Position
 - Spastic diplegia vs. dystonia

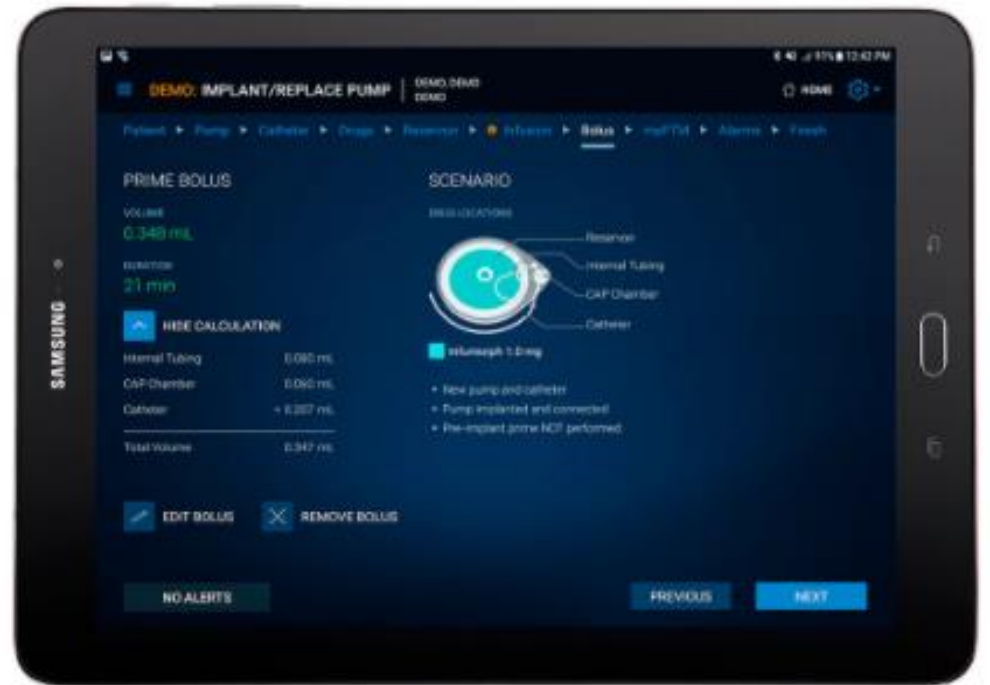
Baclofen Pumps

- Lateral position
- Spinal puncture to insert catheter (L2/3 paramedian oblique insertion with fluoroscopy)
- Between costal margin and iliac crest
- **Measure length of discarded catheter**



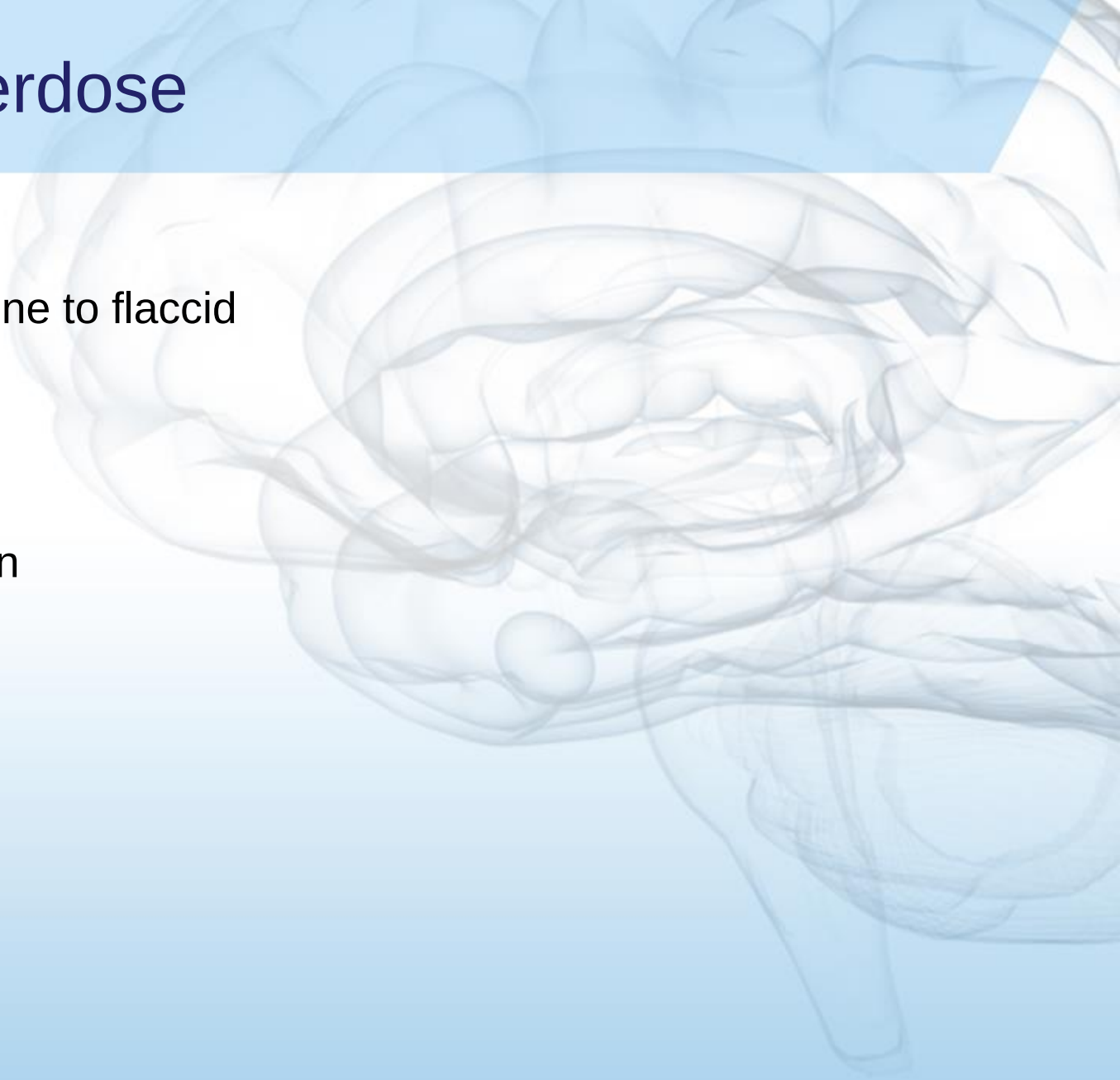
Baclofen Pumps Programming

- Never implant a device you can't program !
 - DBS, VNS, ITB etc
- Continuous dose (double test dose) + bolus (pump tubing and catheter)
- **Always ask:** Where is the drug?
 - Reservoir
 - Internal pump tubing
 - Pump tubing
 - Spine catheter



Baclofen Overdose

- Symptoms
 - Decrease in tone to flaccid
 - Confusion
 - Hypotension
 - Bradycardia
 - Hypoventilation
 - Coma



Baclofen Overdose

- Symptoms

- Decrease in tone to flaccid
- Confusion
- Hypotension
- Bradycardia
- Hypoventilation
- Coma

- Treatment:

- Support:
 - ACLS
 - Volume
 - Airway
 - Ventilation
 - Pressors
- Remove drug
 - Turn off pump (empty pump!)
 - Remove CSF (side port access, LP)

Baclofen Overdose

- Symptoms

- Decrease in tone to flaccid
- Confusion
- Hypotension
- Bradycardia
- Hypoventilation
- Coma

Physostigmine
(cholinesterase inhibitor)

– not really

Increases secretions/bradys

- Treatment:

- Support:

- ACLS
- Volume
- Airway
- Ventilation
- Pressors

- Remove drug

- Turn off pump (empty pump!)
- Remove CSF (side port access, LP)

Baclofen Underdose

- DDX:

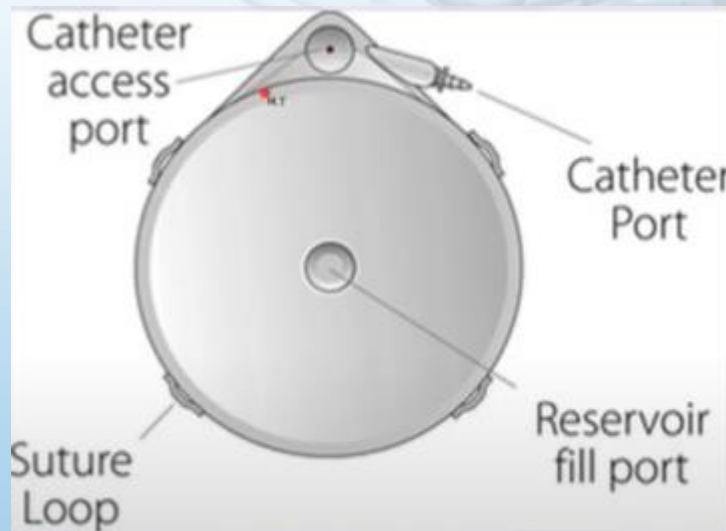
- Programming error
- Empty pump
 - Flow decreases at 1-2mL
- Pain (increased requirement)
 - Constipation
 - UTI
 - Fracture
- Medication change
 - SSRI decrease baclofen efficacy
 - Other antispasmodic
- Pump malfunction
- Catheter malfunction

- Symptoms

- Increased tone
- Itching
- Dysphoria
- Tachycardia
- Fever
- BP lability
- Organ failure
- Seizures
- Hallucinations
- Life-threatening

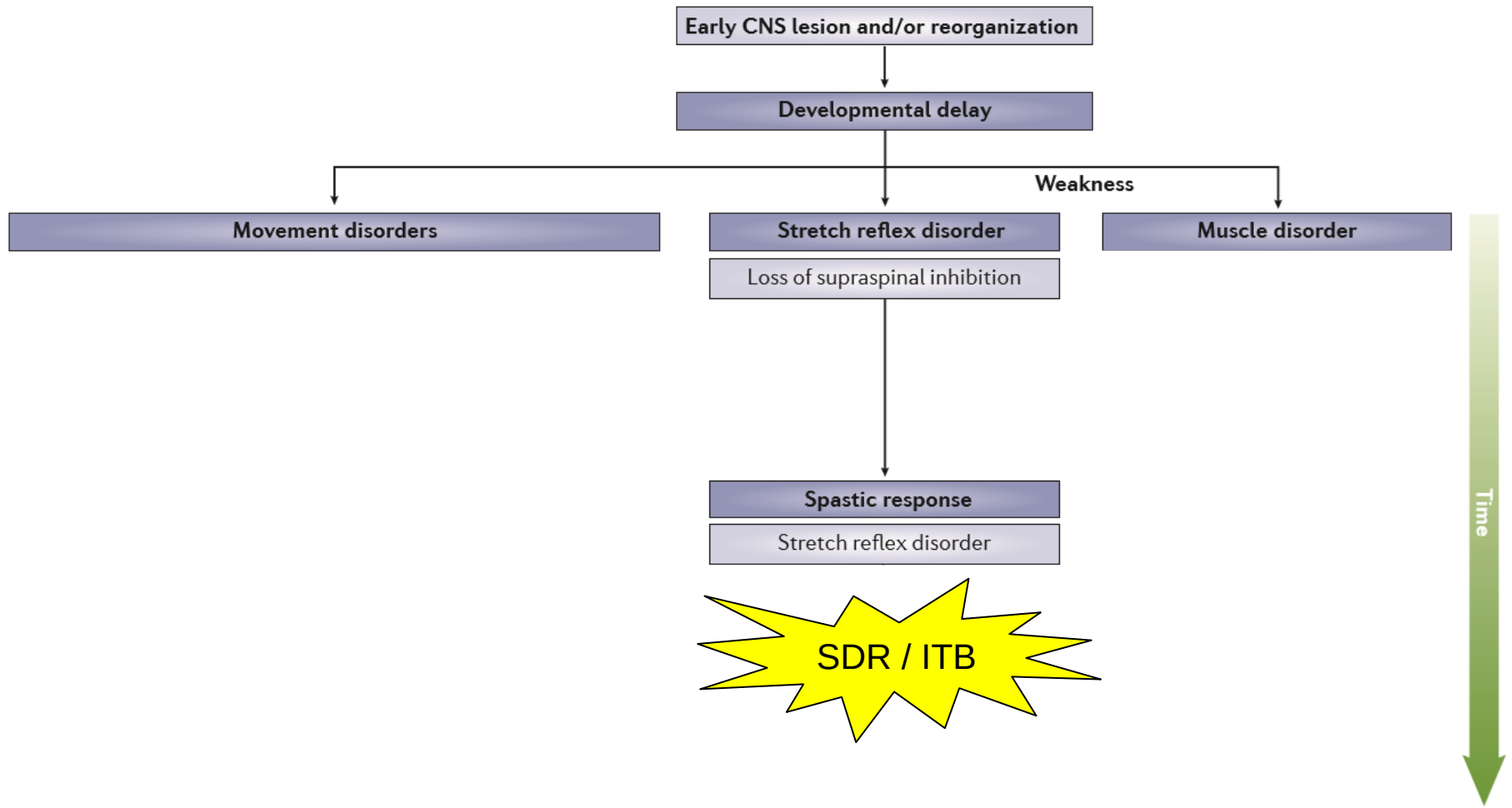
Baclofen Underdose

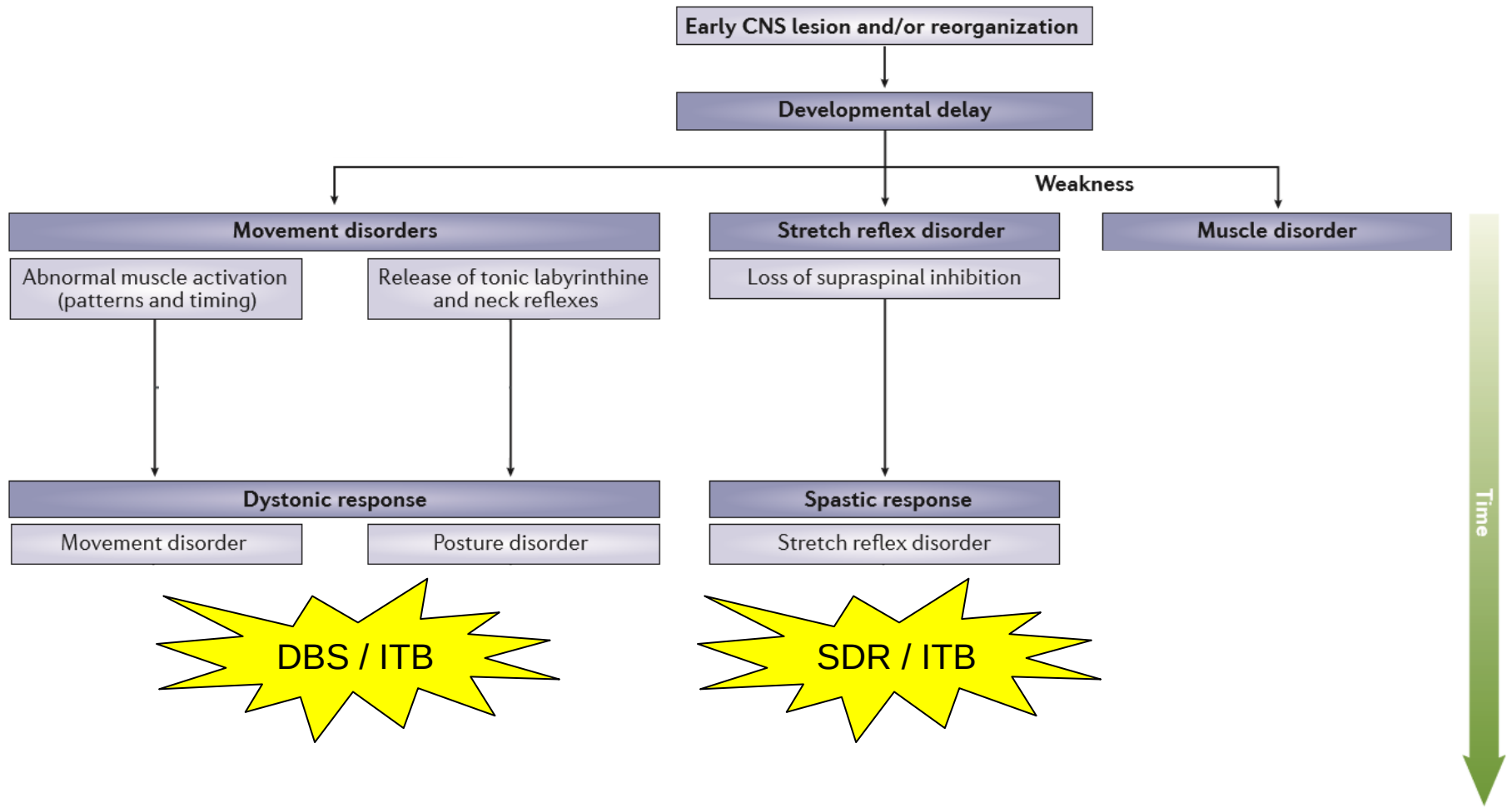
- Management = investigations + treatment
 - Investigate DDx
 - H+P
 - Interrogate pump: alarm date, refill date, elective replacement interval
 - KUB, AP/Lateral lumbar spine
 - Catheter access port aspiration (aspiration 3-4 cc CSF)
 - Myelogram
 - Exploration



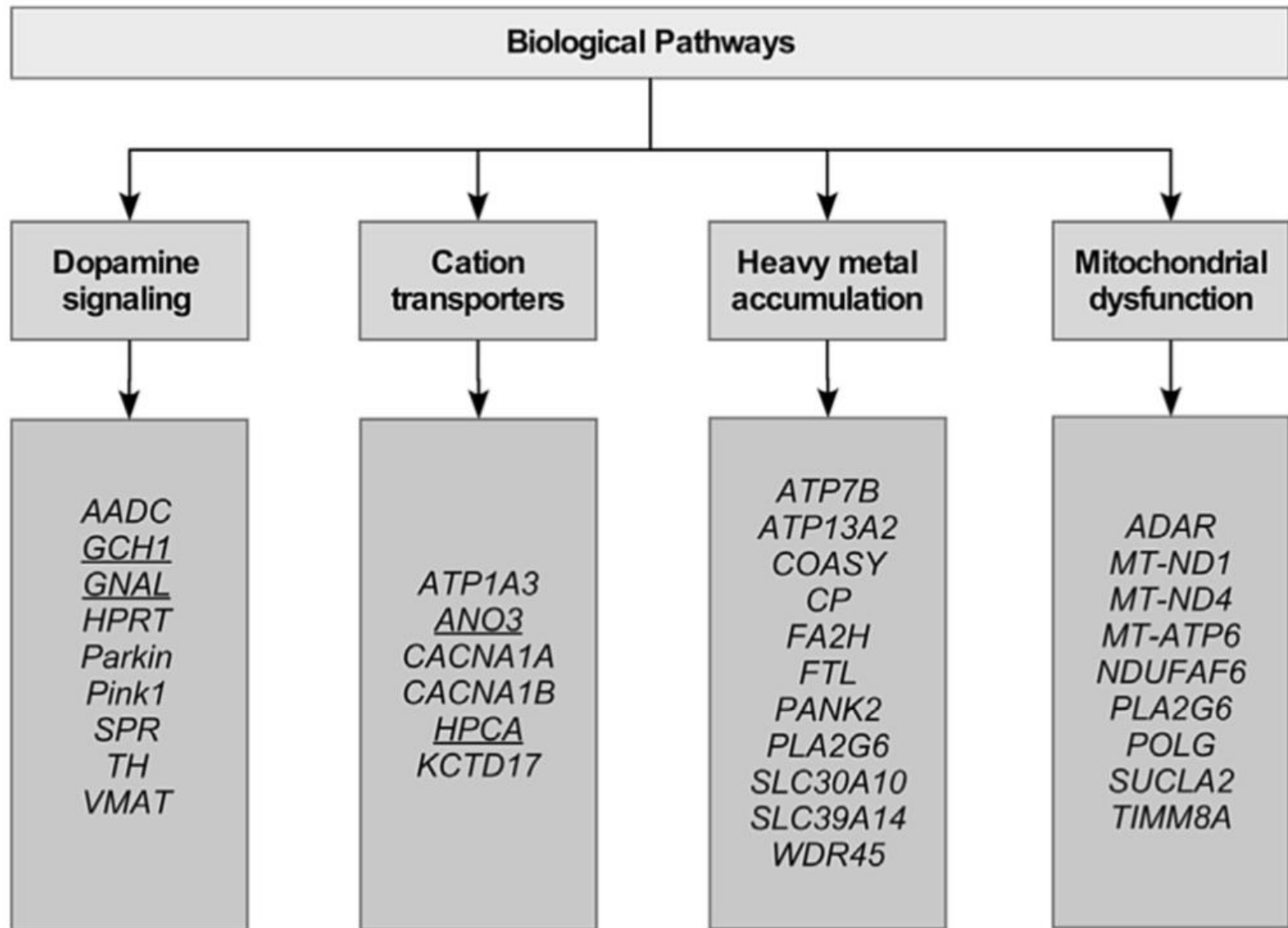
Baclofen Underdose

- Management = investigations + treatment
 - Investigate DDx
 - H+P
 - Interrogate pump: alarm date, refill date, elective replacement interval
 - KUB, AP/Lateral lumbar spine
 - Catheter access port aspiration
- Treatment = baclofen
 - Resume prior dose and route as soon as possible
 - LP and/or lumbar drain
 - Oral baclofen may lessen symptoms
 - Oral/IV benzodiazepenes
 - Supportive care

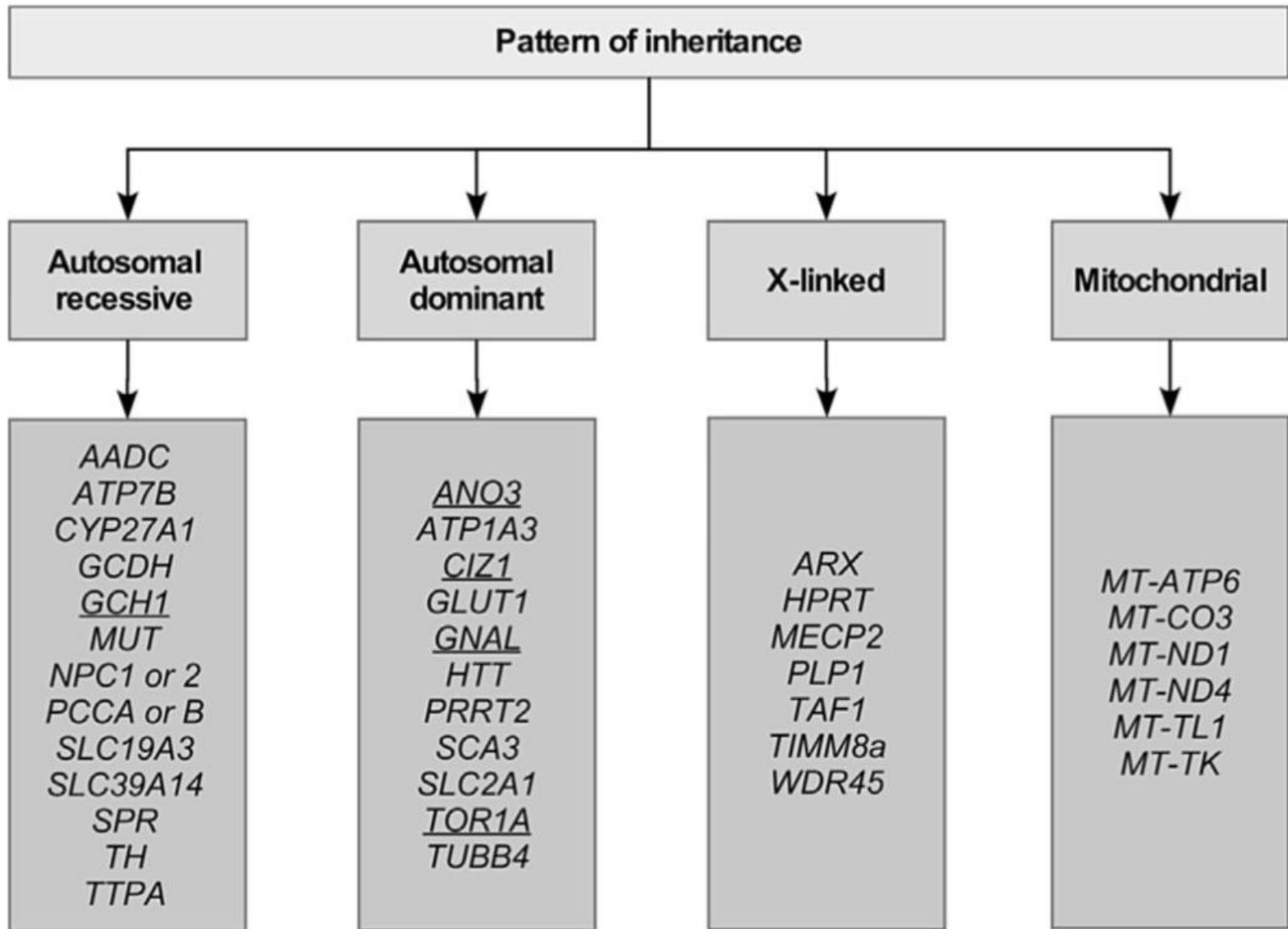




Dystonia: Biological Pathways



Dystonia: Biological Pathways



Dystonia



REVIEW

CME

Phenomenology and Classification of Dystonia: A Consensus Update

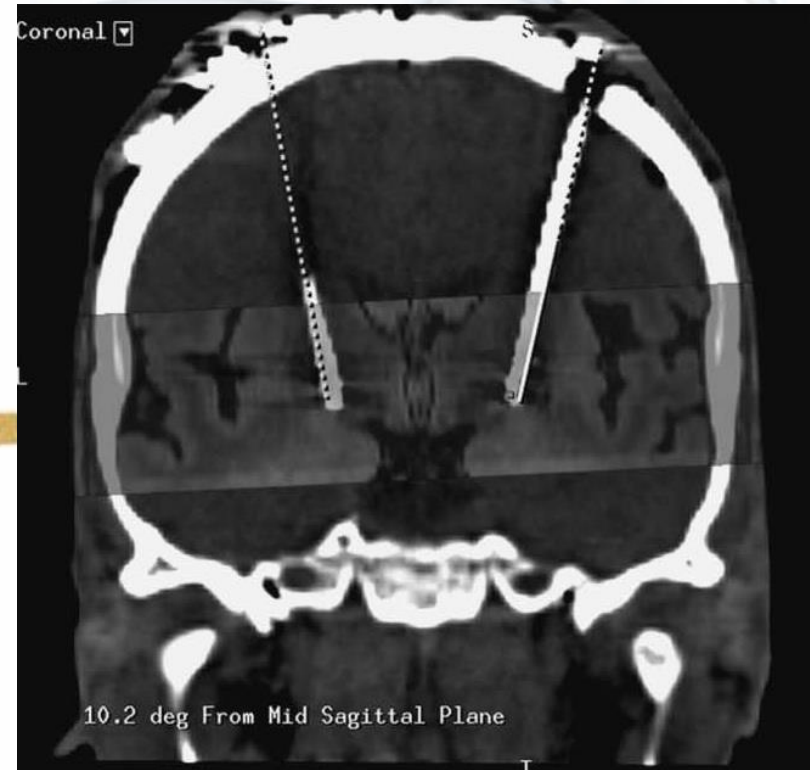
Alberto Albanese, MD,^{1,2*} Kailash Bhatia, MD, FRCP,³ Susan B. Bressman, MD,⁴ Mahlon R. DeLong, MD,⁵ Stanley Fahn, MD,⁶
Victor S.C. Fung, PhD, FRACP,⁷ Mark Hallett, MD,⁸ Joseph Jankovic, MD,⁹ Hyder A. Jinnah, PhD,¹⁰ Christine Klein, MD,¹¹
Anthony E. Lang, MD,¹² Jonathan W. Mink, MD, PhD,¹³ Jan K. Teller, PhD¹⁴

Dystonia

- The second axis of the Albanese classification divides dystonia into four main groups:
 - Inherited dystonia without nervous system pathology,
 - Inherited dystonia with nervous system pathology,
 - Acquired dystonia, and
 - Idiopathic dystonia

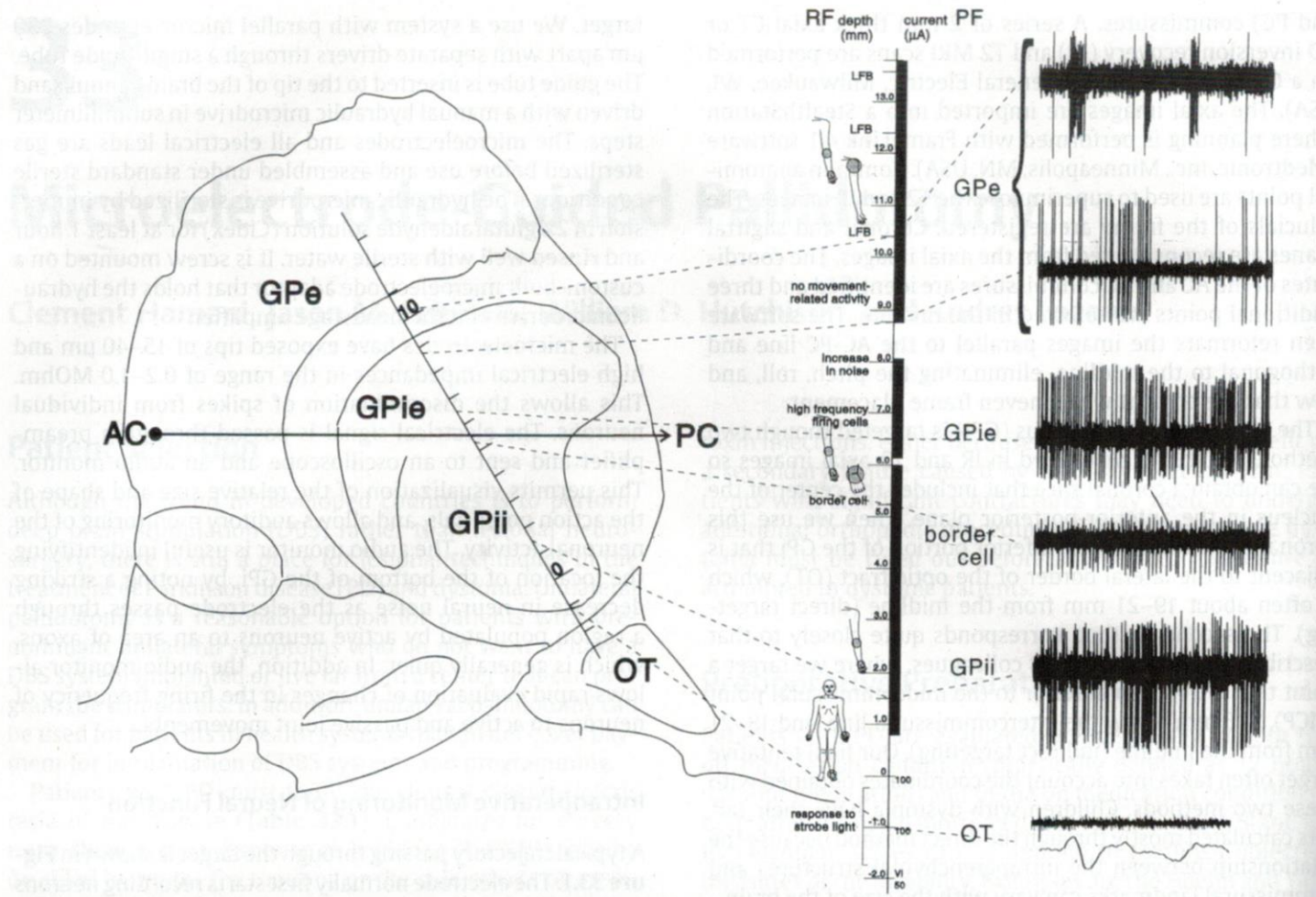
Group	Subgroups Included	Previous Classifications*
Inherited dystonia, without nervous system pathology	DYT1 DYT6 Myoclonus-dystonia	Primary dystonia, dystonia-plus
Inherited dystonia, with nervous system pathology	Pantothenate Kinase-associated Neurodegeneration Lesch-Nyhan disease Glutaric Aciduria Type 1 Mohr-Tranebjaerg syndrome Batten's Disease Methylmalonic acidemia	Secondary dystonia or heredodegenerative dystonia
Acquired dystonia	Cerebral Palsy Kernicterus Stroke Infectious Nigrostriatal Necrosis Metabolic Post-Traumatic	Secondary dystonia
Idiopathic		Mostly primary dystonia

Deep Brain Stimulation



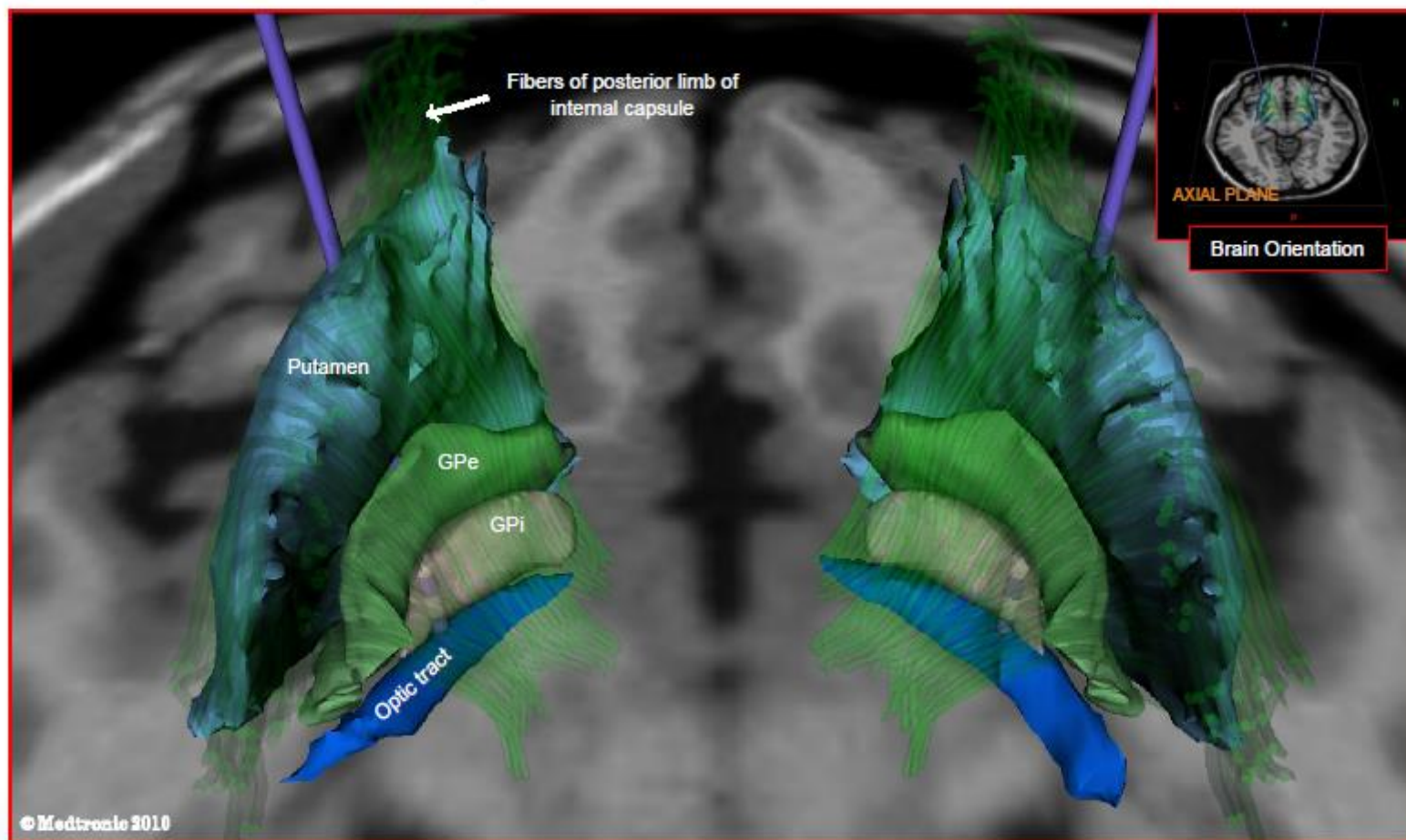
Intraoperative Microelectrode Recordings

212 III Movement Disorders



Optimally Placed GPI-DBS

GPI: Effectively-Placed Bilateral Leads



- Medtronic 3387 DBS leads shown in both hemispheres

Anatomy	Preferred Location for DBS Lead	Observed Effect If Stimulated
GPI	Posteroventral aspect of GPI	Reduction in parkinsonian or dystonia symptoms

Point to consider: The DBS lead may intentionally be placed slightly anterior or lateral in dystonia patients to allow for the use of higher voltages during programming.

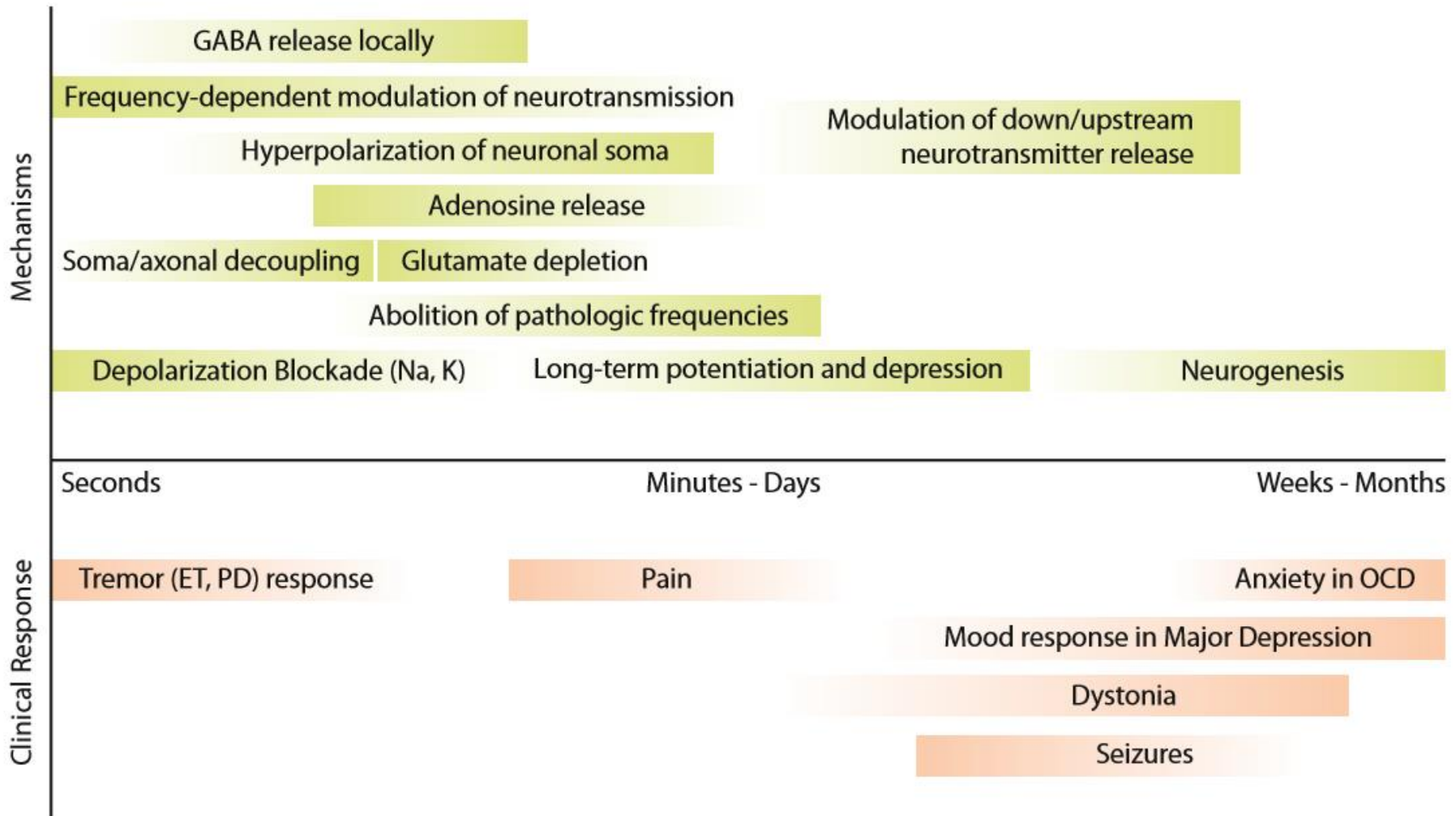
Deep Brain Stimulation

Mechanism	Example Evidence
DBS inhibits the activity of target neurons	
Depolarization blockade (K, Na effects)	Reduction of GPi activity with GPi stimulation (Wu et al., 2001; Dostrovsky et al., 2000); firing suppression in STN neurons after STN stimulation (Tolakis et al., 2012); DBS associated with increased ATP and activation of adenosine A1 receptors (Bekar et al., 2008); frequency-dependent modulation of synaptic function/neurotransmission (Contreras and Llinas, 2001)
Synaptic failure	
Neurotransmitter depletion (glutamate)	
Hyperpolarization of neuronal cell bodies and dendrites	
Release of inhibitory neurotransmitter (GABA; adenosine)	
Synaptic inhibition of afferent projections	
DBS activates target neurons	
Increased glutamate in afferent projection targets	Decreased VL thalamic activity with GPi stimulation (activation)

Many Proposed Mechanisms

DBS both excites and inhibits target neurons	
Decoupling of soma and axons (inhibition of soma, activation of axons) (antidromic, orthodromic)	Computational models showing driving of axonal output by stimulation independent of activity on neuronal cell body (McIntyre et al., 2004); STN DBS leads to LTP and LTD in rat STN neurons (Shen et al., 2003); stimulation leads to plasticity of SNpr neurons in patients with PD (Prescott et al., 2009)
Plasticity (long-term potentiation [LTP], long-term depression [LTD])	
DBS disrupts pathologic oscillatory patterns and generates an “information lesion”	
Replacement of irregular bursting cells with regular high-frequency firing	STN DBS and dopaminergic medication reduce beta-band oscillations in PD (Doyle et al., 2005; Little and Brown, 2012); DBS disrupts and overrides network-wide pathologic activity (McIntyre and Hahn, 2010)
Promotion of “prokinetic” frequencies and abolition of pathologic beta-band frequencies	
Produces/generates “jamming” signal	
DBS has trophic effects, leads to neurotrophin release, and generates new neurons (neurogenesis)	Entorhinal cortex stimulation leads to new DG cells that are functionally incorporated into memory circuits (Stone et al., 2011a, 2011b)
GABA, gamma-aminobutyric acid; ATP, adenosine triphosphate; SNc, substantia nigra compacta; HVA, homovanillic acid; STN, subthalamic nucleus; VL, ventrolateral; DA, dopamine; SNpr, substantia nigra pars reticulata; DG, dentate gyrus.	

Deep Brain Stimulation



Deep brain stimulation for pediatric dystonia: a meta-analysis with individual participant data

LIOR M ELKAIM¹ | NAIF M ALOTAIBI² | ALISSA SIGAL¹ | HAIFA M ALOTAIBI³ | NIR LIPSMAN^{2,4} |
SUNEIL K KALIA^{2,5} | DARCY L FEHLINGS⁶ | ANDRES M LOZANO^{2,5} | GEORGE M IBRAHIM^{2,7,8}  |
NORTH AMERICAN PEDIATRIC DBS COLLABORATION*

1 Faculty of Medicine, Université de Montréal, Montreal; **2** Division of Neurosurgery, Department of Surgery, University of Toronto, Toronto, Canada. **3** College of Medicine, King Saud University, Riyadh, Saudi Arabia. **4** Division of Neurosurgery, Sunnybrook Health Sciences Centre, Toronto; **5** Division of Neurosurgery, Krembil Neuroscience Centre, Toronto Western Hospital, Toronto; **6** Child Development Program, Holland Bloorview Rehabilitation Hospital, University of Toronto, Toronto; **7** Division of Neurosurgery, The Hospital for Sick Children, Program in Neuroscience and Mental Health, The Hospital for Sick Children Research Institute, Toronto; **8** Department of Surgery, Institute of Biomaterials and Biomedical Engineering, Institute of Medical Science, University of Toronto, Toronto, Canada.

Correspondence to George M Ibrahim, The Hospital for Sick Children, 1503G-555 University Ave, Toronto ON, M5G 1X8, Canada. E-mail: george.ibrahim@sickkids.ca

*See Appendix S1 (online supporting information) for names and affiliations of the members of the North American Pediatric DBS Collaboration.

Table II: Independent participant characteristics and univariate analysis for studies reporting BFMDRS ($n=301$)

Variable	Estimate (SE)	<i>t</i> -value
Age at onset	1.7 (0.7)	2.3 ^a
Age of surgery	-0.01 (0.5)	-0.03
Sex (males vs females)	5.2 (4.0)	1.3
Life with dystonia	-1.6 (0.6)	-2.7 ^a
Family history (positive)	-4.3 (8.0)	-0.5
Dystonia type (contrast: idiopathic, no degeneration) ^b		
Idiopathic	-13.7 (5.1)	-2.7 ^a
Inherited, with degeneration ^c	-36.5 (6.1)	-6.0 ^a
Acquired, static lesions ^d	-43.1 (5.7)	-7.6 ^a
Preoperative BFMDRS-motor	-0.13 (0.07)	-1.8
Status dystonicus	3.8 (10.1)	0.4
Generalized (vs focal)	3.3 (17.6)	0.2
Trunk affected	34.1 (9.5)	3.6 ^a
Target (GPi vs non GPi) ^e	-4.5 (10.3)	-0.4
Stimulation frequency (high vs low)	-17.3 (9.6)	-1.8
Follow-up duration (mo)	0.05 (0.09)	0.6
Percentage change in score	—	—
Patients with >20% improvement	—	—

Heritability	Nervous system pathology	Subtype	<i>n</i>	Preoperative severity (mot), median, (25/75 centiles)	BFMDRS-M % improvement, median, (25/75 centiles)	Follow-up (mo), median (25/75 centiles)	Proportion with clinically significant (<20%) Improvement
Inherited	None						
		DYT1 DYT6	102	45.5 (33.6/64.7)	78.1 (53.7/94.0)	15.0 (12.0/42.0)	88%
		Myoclonus-Dystonia	9	18.5 (9.7/31.5)	68.3 (35.8/84.6)	-	100%

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Inherited	Degenerative						
		PKAN	36	83.7 (64.0/96.5)	27.7 (0.6/65.4)	12.0 (7.1/30.1)	57%
		GA1	5	111.2 (64.5/113.6)	6.4 (1.3/13.1)	12.0 (2.9/12.0)	0

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Acquired	Static lesions						
		Cerebral palsy	59	86.2 (66.7/100.3)	11.1 (0.0/21.4)	12.0 (6.0/16.5)	26.9%

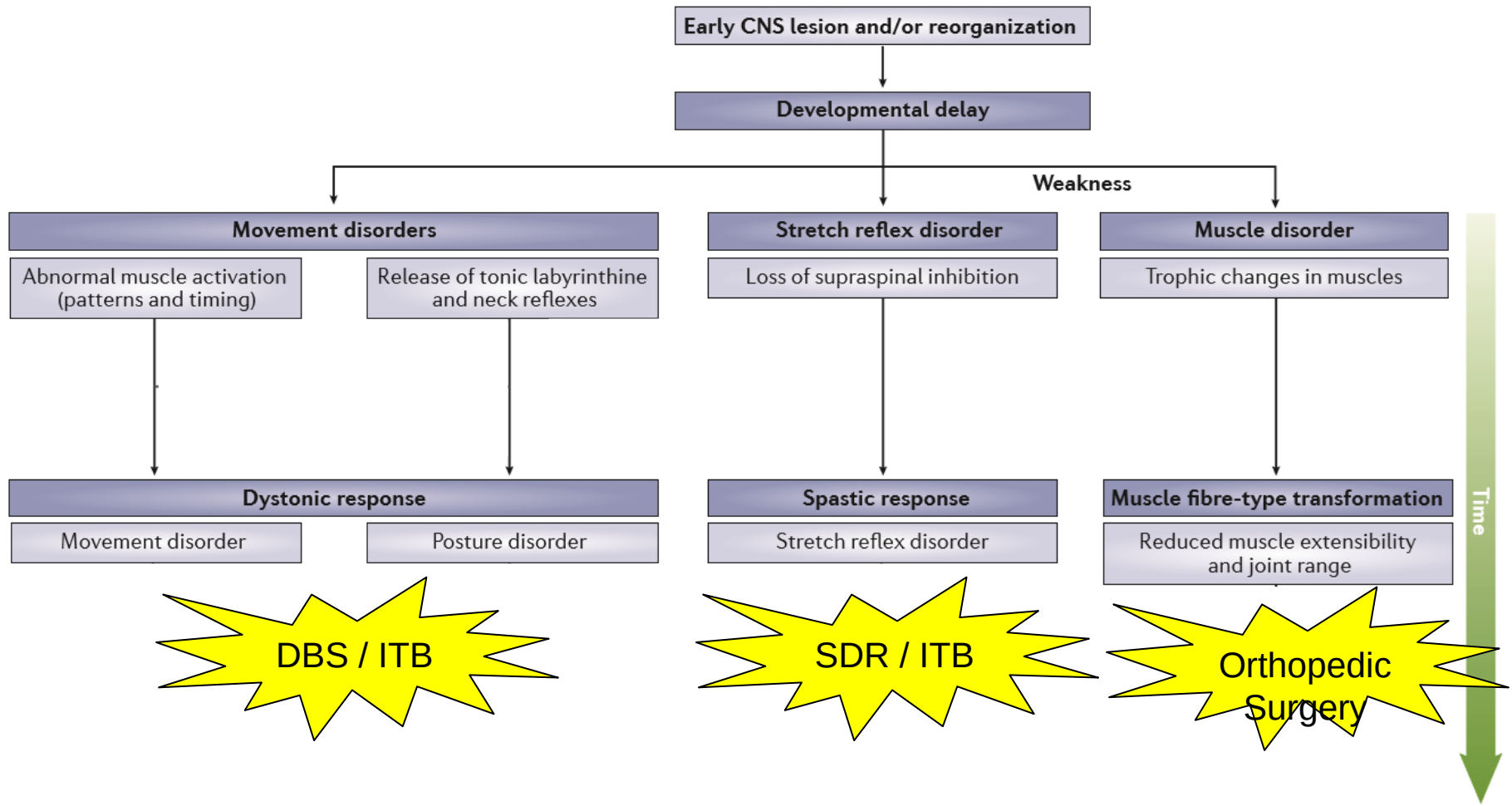
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Acquired	Static lesions						
		Cerebral palsy	59	86.2 (66.7/100.3)	11.1 (0.0/21.4)	12.0 (6.0/16.5)	26.9%
Idiopathic			72	62.0 (42.5/89.2)	50.5 (24.3/73.1)	20.0 (12.0/36.0)	78%



Preop – Status Dystonicus
GA+Midazolam Infusion



Postop



Take Home Message

- Multidisciplinary, highly individualized, child and family-centered approach.
- Goals of surgery & expectations of treatment must align.
- Future :
 - Evidence from comparative studies
 - Relevant outcomes, including quality of life measures
 - Redefine ideal candidacy for treatments given goals, values, priorities and risk tolerance.
 - Better understanding of neural circuitry



Thank you for your attention

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